

Gender norms in agri-food systems; Economic Resilience



INITIATIVE ON
Gender Equality

Cathy Rozel Farnworth with Rebecca Morahan and Anne Rietveld

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Context Annotated Bibliography

This annotated bibliography (AB) provides a research overview on the basis of 77 articles. The annotated bibliography was produced to support an evidence synthesis on gender norms in agri-food systems (Rietveld et al., 2023¹). The aim of the evidence synthesis was to describe how gender norms facilitate, or hamper, women's abilities to achieve economic resilience in the face of climate change challenges. For each article the title and citation are listed and a description is provided. The description covers the research topic, the methodology, the research findings and conclusions of the article. Sometimes the reviewer has added additional comments.

A total of five ABs on 'gender norms in agri-food systems' were produced, each focusing on another theme. Next to this AB on 'Economic Resilience' (#5), ABs on 'Consumer Behaviour' (#2); Food Environment (#3); Adaptive behaviour (#4), and 'Value Chains' (#1) were produced.

The evidence synthesis was conducted as part of Work Package One (WP 1) of the CGIAR initiative on Gender Equality (HER+). WP 1 works with a range of partners to address normative constraints that limit the capacities of women agri-food systems actors to build economic resilience to climate change challenges through developing and using gender transformative approaches.

Article selection

The Web of Science (WoS; <https://clarivate.com>) was used to systematically identify appropriate literature. The search terms in Table 1 were cross-referenced with "gender" and "women".

Vulnerability; resilience climate change; resilience agriculture; climate change; sensitivity, shocks.
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Table 1. Search terms for "Economic Resilience"

The following restrictions were applied: i) geographical focus on LMIC, (ii) agri-food system focus, (iii) analysis of primary data, (iv) written in the English language, (v) articles from 2016 to 2022 only. Deep reading of the thus selected articles resulted in further articles being discarded. Articles were discarded if they showed no or insufficient focus on agrifood systems, or if gender issues were not meaningfully addressed.

Author statement

Methodology and article selection (AR), article review and annotated bibliography (CRF), editing (RM and AR), project supervision (AR).

¹ Rietveld, A.M., Farnworth, C.R., Shijagurumayum, M., Meentzen, A., Voss, R., Morahan, R. and López, D.E. (2023). An evidence synthesis of gender norms in agrifood systems: Pathways towards improved women's economic resilience to climate change. Rome, Italy: Bioversity International. <https://hdl.handle.net/10568/136053>

Annotated bibliography

1. **Abdul-Razak M, Sylvia Kruse (2017) The adaptive capacity of smallholder farmers to climate change in the Northern Region of Ghana, Climate Risk Management, Volume 17, 2017, Pages 104-122, ISSN 2212-0963. <https://doi.org/10.1016/j.crm.2017.06.001>**

Research Topic

The main research questions of this paper are:

1. How can the adaptive capacity of smallholder farmers in the Northern region of Ghana be characterised?
2. What is the capacity level of smallholder farmers in the Northern region of Ghana to adapt to climate change and which are the specific determinants of higher or lower adaptive capacities?

Methodology

The paper attempts to develop a comprehensive and applicable framework for assessing the adaptive capacity of smallholder farmers in Sub-Saharan Africa. It proposes an indicator-based framework for assessing the adaptive capacity of smallholder farmers in two communities in the Northern Region of Ghana along six main determinants of adaptive capacity: economic resources, social capital, awareness and training, technology, infrastructure and institutions. The Northern Region was chosen because it is particularly vulnerable to a number of negative climate change effects (floods, drought, bushfires).

The indicator framework is sector and region specific. It was developed in a deductive and expert-driven way: a heuristic was developed grounded on a systematic review of scientific literature considering both previous approaches assessing generic adaptive capacity, specific adaptive capacity of farmers with particular focus on smallholder farmers as well as approaches focussing particularly on sub-Saharan Africa. Before applying the indicator-based assessment to two communities, the deductive heuristic was validated by expert interviews.

The main focus is on adaptive capacity not economic resilience *per se*. Gender is one aspect examined.

Relevant findings

The findings focus on all six determinants of adaptive capacity.

Statistically significant differences in scores between women and men on six determinants apart from social capital (same level). More than twice as many women (78.05%) as men (38.46%) assigned low adaptive capacity. Moderate adaptive capacity 35.9% men and 9.75% women. 23.08% men high adaptive capacity /no women. It should be noted that even high adaptive capacity is relative.

"Male farmers have better access to economic resources such as higher diversity of incomes and better access to credit, to technological resources such as better knowledge of soil fertility and soil moisture retention techniques as well as awareness and training such as level of literacy, access to climate information and extension services. Further, those with formal education perform significantly better than those farmers without formal education. These results confirm Asante and Amuakwa-Mensah (2014) who based on a literature review expect women to be among the groups that are negatively impacted by climate change and variabilities." (p.115).

Conclusions

The starting points for building resilience should target smallholder farmers with low adaptive capacity, especially women and farmers without formal education. The Northern region shows the most accentuated gender differences in adult literacy, and gender differences in literacy are more significant in rural areas. An extremely high percentage of women and men in the country have not completed primary education, the situation being more critical for women.

Reviewer comment

The authors also provide suggestions for further research including qualitative. Nothing on gender norms really apart from what one can deduce e.g. women are even less likely to go to school than men. The main focus is on adaptive capacity.

2. Ado, A.M., Savadogo, P. & Abdoul-Azize, H.T. Livelihood strategies and household resilience to food insecurity: insight from a farming community in Aguié district of Niger. *Agric Hum Values* 36, 747-761 (2019). <https://doi.org/10.1007/s10460-019-09951-0>

Research Topic

Households' resilience to food insecurity depends on the risks they are exposed to, their capacity to respond to shocks, and their ability to predict future risks. The households' ability to respond to shocks and anticipate risks is highly dependent on their livelihood strategy. Hence, grouping households according to their livelihood strategy is crucial for the design and implementation of targeted adaptation strategies as interventions to strengthen households' resilience to food insecurity.

This study therefore aimed to assess the different livelihood strategies and levels of resilience to food insecurity among households in the Maradi Region of southern Niger—a region with a high level of household vulnerability to food insecurity. The cluster analysis identified six major livelihood strategy groups—pastoralist-extensive agriculturalists (3.80%), farmers (25.6%), agro-pastoralists (46.9%), public service employees (5.5%), entrepreneurs (12.5%) and wage employees (6.3%).

Methodology

Cluster analysis and the principal component analysis were used to identify the different livelihood strategies and to construct a resilience index, respectively. Regression analysis was used to identify the most significant factors determining households' resilience.

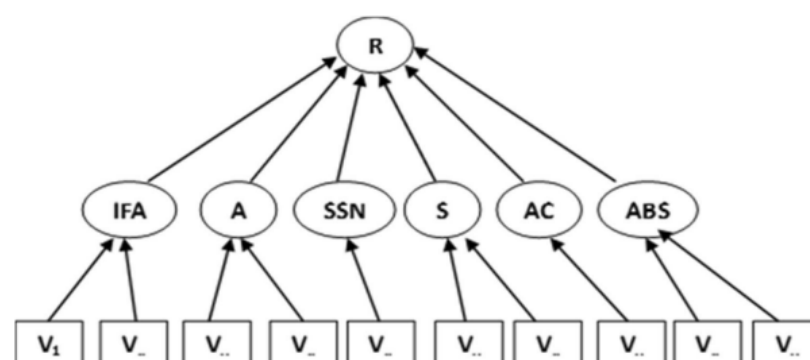
Relevant findings

This article focused establishing food security/insecurity through reference to livelihood strategies which were then cross-referenced by Access to Basic Services, Adaptive Capacity, Social Safety Nets, Assets, Income and Food Assets, and Stability (Fig p750) and p751. Comparison provides information on the distribution of these factors by HH livelihood strategy.

750

A. M. Ado et al.

Fig. 2 Analytical model of household resilience (Alinovi et al. 2009). *R* resilience, *IFA* income and food access, *A* assets, *SSN* social safety nets, *S* stability, *AC* adaptive capacity, *ABS* access to basic service, and *V* variable. The variables in circles represent the latent variables whereas the variables in boxes indicate the observed variables (Grothmann and Patt 2005)



The article does not talk much about gender per se but says, "Regarding gender, the households headed by men are the most resilient (0.017) and have the highest scores for income and food access (0.019), assets (−0.005), adaptive capacity (0.029) and social safety net (0.035), whereas the households headed by women have the highest score for access to basic services (0.117), and stability (−0.054). The main implication of these results is that assets, income and food access, and adaptive capacity are the most important factors determining households' resilience to food insecurity; and social safety net is a great indicator of vulnerability. Therefore, specific attention should be given to small-scale farmers such as wage employees, entrepreneurs and women for resilience capacity building! (p.578)

Overall, "pastoralist-extensive agriculturalists who are the most resilient (0.486), have the highest score for assets (0.987), income and food access (0.988), and adaptive capacity (0.537), but have a very low index for social safety net (− 0.011) and the lowest for access to basic services (− 0.265). The pastoralist extensive agriculturalists group is followed by the public service employees with a higher index for resilience (0.067), determined by higher score of adaptive capacity (0.229), and income and food access (0.489). The wage employees who are the least resilient (− 0.305) has the highest score for social safety net (0.354), whereas the entrepreneurs' group has a negative value for all the resilience components and hold very low level of resilience (− 0.113)." (p.758)

... the group with the highest stability or access to basic service is not necessary the most resilient. The households headed by men have the lowest score of stability and access to basic services yet are the most resilient resulting from higher score of adaptive capacity and income and food access. This implies that the combination of adaptive capacity, access to basic services, assets,

income and food access, stability and social safety net are very important in understanding and determining resilience to food insecurity (p.758)

Conclusions

The development of institutions to improve farmers' income and food access, and adaptive capacity toward food insecurity could help to increase their resilience to food insecurity. This would inevitably lead to a more sustainable use of natural resources and the improvement in households' production system depending on household's livelihood strategy (p.758).

Reviewer comment

This article is valuable because it cross-references livelihood strategies with indicators of resilience. It shows that male-headed households (MHH) score more highly on key indicators than female-headed households (FHH). Improving women's resilience involves focusing on the areas where they score low. This, as with other articles, suggests that external partners are very important - in the article they do mention that women are targeted by UNICEF and others. So potentially these and other partners need to hook up to strengthen FHH in areas where they currently score low. This study does not, of course (similar to so many others) discuss women in MHH.

- 3. Alhassan, S.I., Kuwornu, J.K.M. and Osei-Asare, Y.B. (2019), "Gender dimension of vulnerability to climate change and variability: Empirical evidence of smallholder farming households in Ghana", *International Journal of Climate Change Strategies and Management*, Vol. 11 No. 2, pp. 195-214. <https://doi.org/10.1108/IJCCSM-10-2016-0156>**

Research Topic

Farmers' vulnerability to climate change and variability in the northern region of Ghana.

Methodology

Assessed the vulnerability of male-headed households (MHH) and female-headed households (FHH) farming households to climate change and variability by using the livelihood vulnerability index (LVI) and tested for significant difference in their vulnerability levels by applying independent two-sample student's t-test based on gender by using a sample of 210 smallholder farming households.

Relevant findings

Significant difference in the vulnerability levels of FHH and MHH farming households. FHH were more vulnerable to livelihood strategies, socio-demographic profile, social networks, water and food major components of the LVI, whereas MHH were more vulnerable to health. FHH were more sensitive to the impact of climate change and variability but have the least adaptive capacities.

Livelihoods

About 36 per cent of FHH depend solely on agriculture as a source of income compared with 30 per cent for MHH. Yet, FHH recorded low average agricultural livelihood diversification (2.08) compared to MHH (2.16). This makes FHH significantly more vulnerable (0.575) than MHH (0.548) in terms of agricultural diversification index, as revealed by the results of the two-sample t-test. Computed vulnerability indices showed that FHH (0.380) were more vulnerable in terms of farmland ownership than MHH (0.120), with 38 per cent of FHH households not owning their farms relative to 12 per cent for MHH. FGD shows women were often engaged in non-farm activities such as rice processing, shea business, burning and selling of firewood and charcoal, petty trading and food vending and men are often engaged in fishing, casual labour at sand loading sites and masonry works.

Social capital

Both women and men have low social capital, but FHH were more vulnerable in terms of access to credit and assistance than MHH. FGDs - revealed that women can hardly walk to a traditional or community leader for assistance because of cultural factors. When in need, women can only seek for assistance from such leaders through their husbands or brothers. This limits the social networks of FHH, making them more vulnerable than MHH.

Conclusions

FHH should be given priority in both on-going and new intervention projects in climate change and agriculture by empowering them through financial resource support to venture into other income-generating activities. This would enable them to diversify their sources of livelihoods to boost their resilience to climate change and variability.

Reviewer comment

I read the whole paper but the abstract, cited here, summarises all the key lessons learned. The main focus is adaptive capacity. The actual reasons why women's adaptive capacity is lower is not explored.

4. Ali, Essossinam. (2021). Farm Households' Adoption of Climate-smart Practices in Subsistence Agriculture: Evidence from Northern Togo. *Environmental Management*. 67. 10.1007/s00267-021-01436-3.

Research Topic

Analyses farm households' perception of CC, the determinants of choice of the number of adopted practices, and correlation between the most used climate-smart strategies in subsistence agriculture.

Methodology

Zero-inflated Poisson regression and multivariate analysis are employed using data collected from 704 farm households in Northern Togo.

Relevant findings

Households' minimum consumption needs, gender, land, access to credit and extension services are the main determinants of the choice of the number of adopted strategies. Male Headed Households (MHH) are more likely to adapt to CC compared to females. Men more likely to adopt strategies such as the change of crop varieties (S2) and soil and water conservation practices compared to women. This result might be associated with the lower levels of education or land constraint mostly faced by women (Dhanaraj and Mahambare 2019; Ali et al. 2020; Deressa et al. 2009).

Deressa et al. (2009) found that education would increase farmers' adaptive capacities, while MHH are likely to adapt to CC compared to women. The Ministry of Agriculture, Livestock and Fishery of Togo (MAEP 2014) reports that the deprivation rate of education is more than 30% in the study areas. MAEP (2014) points out that a significant inequality in access to education is observed among gender (43% of illiterate women against 30% in Central region; 53% of women against 37% of men in Kara region; and 63% of women against 46% of men in Savannah region) in the study areas. Most of the time, women do not have access to land resources for agricultural activities and this could contribute to lower agricultural production and increase food insecurity. This is true in rural areas in the northern regions of Togo, where the land is often owned by men according to their culture.

Conclusions

The article shows that farmers are highly aware of climate change and observe changes in the rainy season etc closely. Most farmers are adapting by applying at least one measure (adjustment of sowing time, integration of crop and livestock, changing varieties by using high drought tolerant and yielding, soil and water conservation practices, planting early maturing crops, soil fertilization system, off-farm employment, mono-cropping and intercropping systems). The author suggests these are short term measures - major investments in agro-forestry, irrigation and also farmer's educational levels are needed as is improved access to extension, agricultural credit, and markets.

Reviewer comment

Some confusion I think between MHH and FHH, and women in general /men in general. Gender opportunities and constraints were not specifically studied and the author relies on generalized literature to support their findings that women are less able to adapt. This article is yet another that shows that women and men farmers face serious constraints in their ability to adapt, but women face deeper constraints in regard to every indicator. All this hints at underlying norms but does not discuss them nor if they are changing.

5. Akurugu, C.A., Mathias Mwinlabagna Jatoe, Maximillian Kolbe Domapielle. Empowering rural women for sustainable development through the provision of water infrastructure in north-western Ghana, *World Development Perspectives*, Volume 21, 2021, 100287, ISSN 2452-2929, <https://doi.org/10.1016/j.wdp.2021.100287>.

Research Topic

This article examines the Water for Resilience project implemented in Ghana by HELVETAS and others (p.2) in rural NW Ghana. This is fundamentally about digging bore holes for drinking water and encouraging vegetable gardening in the dry season using a range of agro-ecological methodologies (composting, organic-based pesticides etc). The overall aim is to strengthen poor women's resilience in drought prone areas / experiencing chronic vulnerability.

Methodology

Qualitative in-depth interviews, focused group discussions and observation with women beneficiaries and project staff to gather data in four intervention communities. Total 52 (40 in FGDs).

The authors draw on a range of conceptual frameworks, particularly Batliwala (1994). This FW argues that women's empowerment constitutes three dimensions: economic development, integrated development, and consciousness raising. The economic development element assumes that women's increased income will filter into other aspects of their lives. The integrated element addresses women's practical gender needs. The consciousness raising element focuses on women's strategic gender needs.

Relevant findings

The Water for Resilience approach "has enormous prospects for reducing poverty and eradicating vulnerability in these deprived settings." Household relations often (but not always, due to men's jealousy in some cases as women obtained incomes) improved considerably and women were able to discuss their ideas and issues with each other in the group spaces provided by the vegetable gardens.

The authors "show that this innovative intervention can promote the transformation of unequal gender relations and empower women. Nonetheless, these can only be achieved if the intervention is executed within a framework that incorporates strategic gender interests, including issues of status relative to men, improved capacity for action, equitable access to household productive resources and access to decision-making and influencing power. We conclude that these structural issues, which remain untouched by this instrumentalist intervention require critical attention if women's empowerment is to achieve meaningful results in the male-centric settings of rural northern Ghana and beyond. We explore lessons to emerge from our analysis for the empowerment of women in these and similar resource-scarce and patriarchal contexts for sustainable development" (this is all from the Abstract as it is not possible to copy and paste from this article).

Conclusions

The key finding is fundamentally that "power" must be reinserted into empowerment. The authors argue that the Water for Resilience project remains instrumentalist because it does not raise consciousness, among men as well as women, of male-centric power relations in Ghana. Unequal power relations must be addressed.

They also make the point the current vegetable gardening projects are very small in terms of land size, and insufficient for really empowering women in terms of income.

Reviewer comment

This article usefully uses Batliwala's conceptual model of three dimensions to explain that it is important to address women's practical gender needs and to improve their income. But unless critical consciousness is mobilized for transforming inequal gender norms and male centric power relations then project interventions are going to make little overall difference.

6. Anderson, B., Poeschel, F. & Ruhs, M. Rethinking labour migration: Covid-19, essential work, and systemic resilience. CMS 9, 45 (2021). <https://doi.org/10.1186/s40878-021-00252-2>

Research Topic

Existing research on the effects and regulation of labour migration has largely focused on 'efficiency' (costs and benefits of immigration) and 'distribution' (effects of immigration on different groups of people) as key outcomes of interest without paying significant attention to issues related to systemic resilience (p.16).

Considerations of how migrants may affect the systemic resilience of essential services – in a pandemic or similar crises – are pervasively absent, not only in policymaking but also in research. Drawing on several disciplines, we outline the concept of systemic resilience and develop implications for the analysis and regulation of labour migration. We call for shifting the focus from the role of migrants in specific occupations and sectors in particular countries to transnational systems of production and service provision (p.1).

Methodology

Literature review. To study how migrant workers affect systemic resilience, we propose an agenda for comparative research along three lines: comparing migrants to citizens within the same system, comparing migrants' roles across systems, and comparing strategies for resilience adopted in different systems.

Relevant findings

Systemic resilience

In the broadest sense, systemic resilience can be defined as a system's capacity to continue functioning given external shocks or changes (Martin-Breen & Anderies, 2011). Early use of this notion was made in ecology, where systemic resilience captures the capacity of an ecosystem to recover from a shock, returning to the status quo ante. A similar idea of systemic resilience has also long been used in physics and engineering. In a social science context, systemic resilience has been used with a key modification: after a shock, a resilient system continues to function but might end up in a state that is quite different from the status quo ante (p.5).

Social systems are typically in a process of change also in the absence of shocks (e.g. Robinson & Carson, 2016). In addition, social systems are adaptive: they can self-organise and find new modes of operation in order to continue functioning (Martin-Breen & Anderies, 2011). These considerations lead to defining systemic resilience here as the ability to withstand, recover from, and adapt to unexpected external shocks (p.6).

Key elements of systemic resilience shown in diagram. When a shock hits, systemic performance might initially decline substantially, down to a point ('impact'). Over time, however, the performance of a resilient system bounces back. In the baseline scenario (solid dark line), performance fully recovers, eventually proceeding as if the shock had never occurred. In another scenario, the recovery is only partial. In a third scenario, the adaptation by the system not only leads to recovery but also creates lasting benefits, so that performance bounces forward.

In order to investigate systemic resilience, systems need to be the unit of reference. A holistic approach to entire systems has to go beyond the common practice of investigating the impact and policies of labour migration in terms of specific skill levels or occupations. This narrow focus ignores the systems that occupations are embedded in, meaning that effects on the functioning and the resilience of the systems go unnoticed (p.8).

If resilience strategies in destination countries lead to a greater demand for migrant labour, this can threaten resilience in origin countries. Global care chains, e.g. nurses from Europe going to the US while nurses from the Philippines go to Europe, are an example of how emigration can create shortages in migrants' countries of origin (Yeates, 2010). By seeking to increase resilience of their essential services, destination countries risk 'exporting' vulnerabilities to origin countries (p.9).

Conclusions

"Towards a new research agenda"

... to help understand the link between migrants and systemic resilience, requires a comparative approach at several levels, to disentangle the effects of migrant workers on systemic resilience from the effects of policies or institutions. We need comparative analysis of how migrants and citizens affect systemic resilience within given systems, and of how and why resilience varies across

systems and across countries characterised by different institutional and policy frameworks and, therefore, different degrees of reliance on and roles for migrant labour. The comparative institutional analysis also needs to pay attention to politics and systemic changes over time, considering whether, how, and why different systems change, including but not limited to their use of migrant labour (p.17).

Reviewer comment

This paper is not immediately relevant to our work, and it does not discuss gender per se. And I do not cover all of the interesting observations in my summary above. However, it is a very powerful paper because it suggests pathways for us to think about how gender inequalities affect systemic resilience in agrifood systems at multiple levels. It could be used to help us develop, or recommend developing, a systemic resilience ToC for HER+.

7. Asmamaw M, Mereta ST, Ambelu A (2020) The role of local knowledge in enhancing the resilience of Dinki watershed social-ecological system, central highlands of Ethiopia. PLoS ONE 15(9): e0238460. <https://doi.org/10.1371/journal.pone.0238460>

Research Topic

Local knowledge (LK) = “cumulative body of knowledge, practice, and belief, evolving by adaptive processes and handed down through generations by cultural transmission, about the relationship of living beings (including humans) with one another and with their environment” (Berkes, Colding, and Folke)

Aim: to explore the resilience enhancing roles of LK, and determinant factors for the acquisition, and transmission of traditional knowledge, and practices in the Dinki watershed social-ecological system, central highlands of Ethiopia. Research Questions: (i) Are there local knowledge systems that are applied to manage local-level disturbances in this locality? (ii) How is the management and current utilization trends of local knowledge systems in this locality? (iii) Do local knowledge systems have adaptive roles in managing environmental disturbances? (iv) What are the social mechanisms that sustain local knowledge systems? (v) What factors determine the acquisition of local knowledge systems?

Methodology

Key informant interviews, focus group discussions, and a household survey, and climate knowledge and agro-ecology etc from relevant govt departments. Chi-square statistics were computed to explore the effects of a set of predictors on the knowledge level and participation status of practical and socioeconomic domains. The effect of age on the acquisition of the eco-cognitive domain was analyzed through one-way Analysis of Variance (ANOVA). Gender-based variation in acquisition of the eco-cognitive domain was computed using the t-test.

Relevant findings

Men over 60 express much more LK than younger men (see esp p.8). The authors then say that:

“The gender of the respondents was found to have a positive and significant correlation with farmers’ knowledge acquisition at 1% significant level. Accordingly, for each one male added in the community, the odds ratio will raise 12, 10, and 10 times to acquire eco-cognitive, practical, and sociocultural domains, respectively (Table 5). ... The t-test result showed that females were inferior to males in identifying components of eco-cognitive knowledge. Similarly, the chi-square statistics result revealed a statistically significant difference between males and females in their knowledge and participation in practical domains (Fig 3A) and most normative (Fig 3B) domains. This suggests that these cultural and religious ceremonies are celebrated equally in both males and females in the Dinki watershed socio-ecological system.

It might be the effect of cultural influence where the females in most rural Ethiopia are restricted to in-door tasks, the knowledge and participation of females were inferior to males. In rural Ethiopia, the majority of outdoor activities, including natural resources management practices are predominantly performed by males. This finding agrees with the previous study in Southern Ethiopia that states females’ knowledge and participation in agroforestry practices were minimal than males [26]. Likewise, a study in Northern Ghana showed that females’ awareness of the roles of traditional ecological knowledge in ecosystem management were limited than males [11].” (p.14).

Conclusions

This paper shows that LK is being eroded and this is associated with schooling etc. The paper also notes strong environmental degradation and suggests that LK should be taken seriously as a means to mitigate this.

Reviewer comment

It may be worth ploughing through the statistical methodology and analysis to see in how far the authors are correct in assigning women poor LK. As it stands, the authors suggest that women know less because they do not occupy decision-making roles in relation to environmental management. Yet the fact that the authors suggest women are restricted to indoor tasks is puzzling. Women are known to be prominent in farming tasks across the country; this suggests that the authors worked with a stereotyped Gender Division of Labour (GDL) and other concepts. Even so, it may well be true that if women are denied roles in Natural Resource Management, then their knowledge will be affected. The issue here is that some of those roles may have been missed by the research team.

8. Azong, Matilda, Clare J. Kelso, and Kammila Naidoo. 2018. "Vulnerability and Resilience of Female Farmers in Oku, Cameroon, to Climate Change." *African Sociological Review / Revue Africaine de Sociologie* 22 (1): 31-53. <https://www.jstor.org/stable/90023845>

Research Topic

Gender is linked to vulnerability through a number of factors. These include access to and control over land, division of labour, marriage relationships, access to education and responsibility for dependents. Participants' life histories show how vulnerability in the region develops over time and is both complex and non-linear. The authors also show that women develop adaptive strategies.

Methodology

This study makes use of a life history methodology in order to examine the particular nature of the vulnerability experienced by rural women in Oku in the Bamenda Highlands region of Cameroon (10 women).

Relevant findings

Participants expressed how they used their agency, both individual and collective, in coping with vulnerability. They narrate different adaptation strategies employed including livelihood diversification, and changing farming practices.

Vulnerability is defined as "the state of susceptibility to harm from exposure to stresses associated with environmental and social change and from the absence of capacity to adapt" (Adger, 2006, 268). Vulnerability includes physical risk to a person or property as well as the social, economic and political context that results in people experiencing worse impacts from a hazard (p.33).

Adaptation is the term used to define change that takes place over the long term and leaves people better able to adjust to climate change in the future (references). The term 'coping' is used to identify ways in which people make small changes in the short term, just to get by. Coping strategies often leave people worse off in the future and can involve a spiral of worsening circumstance (p.33).

Gender, from a feminist analytical perspective, does not define essential differences between men and women, but rather "highlights engendered practices and the differentiated performance of rights and entitlements stemming from unequal power relations (Butler and Gambetti, 2013)". Gender differences in the "experienced impacts of climate change are filtered through gendered behaviours and lived realities within a particular society (Jost et al., 2016)". Gendered vulnerability should be understood as a complex, nuanced and layered experience (p.34).

Women's agency in promoting adaptation should also be acknowledged (references. Tschakert and Machado (2012, 276) argue that "lessons from adaptation illustrate that simplistic notions of women as vulnerable victims conceal deep-rooted inequalities, patterns of marginalization and unequal power structures." Researchers should not portray women exclusively as victims, as this fails to acknowledge their potential as agents of change and actors in complex networks of adaptation (ibid.). Women should be understood as both vulnerable and resilient, they constitute adaptive agents both collectively and individually and often act to change their circumstances and those of their dependents (Butler and Gambetti, 2013 (p.35).

Weaken resilience: Through the life histories it emerged that particular cultural and socio-economic practices worsen women's vulnerability in Oku. The factors that emerged included: a lack of secure land tenure (access not ownership), leaving school

before completion of studies, having an unplanned early pregnancy, being responsible for the care of ill adult dependants, arranged marriages and polygamous relationships in which resource allocation can be problematic (p.40).

Increase resilience: These included improving educational levels (especially of children to later benefit from remittances and also to help them become more resilient themselves), diversifying livelihoods, gaining access to collective organization and funding through loan schemes (p.44).

Specific response to climate change: changes in farming practices, livelihood diversification, multi-cropping, increase in the use of inputs such as insecticides, pesticides, fertilizers, manure, to curb pests and diseases and enhance soil fertility. They also apply erosion control measures such as contour farming, planting of trees, vetiver grass and construction of ridges with blocks and windows to control the flow of water. A few women have diversified their livelihoods and are rearing domestic animals such as goats, pigs and chickens. There has been a growing participation of women in coffee production which was, in the past, done by men (p.45). There was some concern that men may become more engaged in crop agriculture and take over the income.

Conclusions

Although women are improving their economic conditions in the midst of adverse effects of climatic changes, their efforts are limited by challenges such as insufficient funds for agricultural inputs, insecurity of tenure and lack of adequate training. Promotion of women's resilience to climate change can be enhanced by supporting existing collective organizations. By promoting women's agency and empowerment, agricultural productivity and opportunities for livelihood diversification will be enhanced better enabling them to adapt sustainably to climate change (p.48).

Reviewer comment

This article is useful because it provides good empirical examples of how women tried to adapt and cope with difficulties in their lives. The life course approach shows that barriers to women's resilience are inculcated very early in their lives, with schooling being denied or very limited due to (usually) fathers not permitting them to go to school. Other structural barriers include lack of land ownership and complete voicelessness if, say, a brother wants to sell your land even in the middle of a cropping season. Typical interventions like VSLA, engaging women in cooperatives etc are making a difference. I think that the article shows that women are always trying to make their lives better and to survive, and to build resilience into their children by funding their education.

9. Benítez B, Nelson E, Romero Sarduy MI, Ortíz Pérez R, Crespo Morales A, Casanova Rodríguez C, Campos Gómez M, Méndez Bordón A, Martínez Massip A, Hernández Beltrán Y and Daniels J (2020) Empowering Women and Building Sustainable Food Systems: A Case Study of Cuba's Local Agricultural Innovation Project. *Front. Sustain. Food Syst.* 4:554414. doi: 10.3389/fsufs.2020.554414

Research Topic

Case study of Cuba's Local Agricultural Innovation Project (PIAL). PIAL currently operates in 75 municipalities across 12 provinces and in the Isle of Youth special municipality, and the model has evolved from its initial focus on participatory plant-breeding to include more holistic programming aimed at increasing food system sustainability and community resilience. In this paper we highlight how the gender-specific aspects of the PIAL model facilitate female participation and leadership and how this contributes to positive economic, ecological and sociocultural changes in farming households and communities.

Methodology

Research conducted according to the principles of Participatory Action Research, which Greenwood and Levin (2005, p. 54) explain "aims to solve pertinent problems in a given context through democratic inquiry in which professional researchers collaborate with local stakeholders to seek and enact solutions to problems of major importance to the stakeholders." This approach involves rethinking the role of the researcher through processes of collective knowledge construction and recovery that break with the subject-object binary typical of traditional research (Salazar, 1991). While such work is sometimes criticized for lacking rigor [see Mosavel et al. (2011), Levin (2012)], others argue that it can be just as rigorous—and in some ways even more so—than more conventional research approaches [see Davies and Dodd (2002), Brydon-Miller et al. (2003), Greenwood and Levin

(2005)], in part because “it must demonstrate its credibility to a broader audience that brings a more diverse set of questions and standpoints to bear” (Warren et al., 2018, p. 448) (p.7).

Relevant findings

P.6-7 describe seven strategies in depth that PIAL implements to empower women. “Coupled with a re-affirmation of the value of traditional campesina activities, PIAL has involved explicitly challenging traditionally entrenched gender norms in rural households and communities. The directness of this challenge sets the PIAL methodology apart from many other farmer-led research, PPB and agroecological development initiatives, which demonstrate significant gender-related impacts [see Classen et al. (2008), Rosset et al. (2011), Humphries et al. (2012), Bezner Kerr et al. (2019)] without so directly confronting patriarchal values. In the case of PIAL, as the methodology evolved, it actively expanded beyond its initial plant breeding focus to foster innovation in everything from flower arranging to vinegar production, and now also includes workshops on gender norms, gender-based violence, self-esteem, and associated issues that further serve to pull at tightly woven knots of male bias.” The article goes on to suggest that it is harder to measure such changes than e.g. reduction in pesticide use. (p.12).

“This considerable expansion of what constitutes “agricultural innovation” not only provides more space for women’s involvement in PIAL, it also centers women’s knowledge and expertise, their capacity to innovate, and their role as leaders and decision-makers in community development. In so doing, PIAL’s equity orientation confronts head-on some of the deeply gendered power imbalances that construct men as the primary generators, holders, transmitters and users of agricultural knowledge and innovation [see Classen et al. (2008), Cockburn (2015), Pimbert (2018), Bezner Kerr et al. (2019)]. As the research results demonstrate, this confrontation has resulted in concrete changes to the gendered division of household labor, as well as to broader conceptions of gender roles at the community scale. Such results are consistent with findings by Schwendler and Thompson (2017, p. 111), based upon work with Brazil’s Landless Peasant Movement, that when agroecological education is coupled explicitly with “gender-oriented pedagogy” the results empower “women and men to disrupt the traditional sexual division of labor in rural communities...”” (p.12).

The discussion then highlights work to include youth and says (p.13) this “suite of inter-generational efforts, insights and impacts from PIAL are being actively incorporated into curricula in Agronomy as well as Socio-Cultural departments at a variety of universities across Cuba. The ways in which Cuban youth are learning and reproducing new attitudes about agriculture and society through involvement with PIAL is reflective of the need to challenge power dynamics, engage traditionally marginalized actors, and build relationships across a diversity of groups if social learning for agroecology is to be effective.”

The paper then discusses food sovereignty (much discussed in this paper) and adds a quote from Pimbert (2018, p. 263) “Instead of being seen as passive beneficiaries of trickle-down scientific development or technology transfer, farmers and other citizens are viewed as knowledgeable and active actors who can be centrally involved in both the “upstream” choice and design of scientific innovations, and their “downstream” implementation, spread and regulation. In this context, science and the construction of knowledge are seen as part of a bottom-up, participatory process in which citizens take center stage in decisions on what knowledge is produced, why, how and for whom”. The authors then argue that Pimbert’s arguments is “especially relevant in the case of women farmers, who face additional layers of marginalization. This is where the PIAL methodology offers perhaps its most useful lessons, as it represents a means by which the production and reproduction of agricultural knowledge and innovation can truly be democratized in a way that is inclusive and empowering for, not just small-scale farmers, but specifically the women in their ranks”. (p.13-14).

Conclusions

The paper concludes by saying “Beyond the ecological and economic impacts of the work, PIAL has catalyzed transformations in gender relations in participating households and communities, as its gender-specific activities have created new spaces for women’s participation and leadership in agricultural innovation processes. This work is not easy, and it is certainly not complete. However, through PIAL, both women and men are learning to take on new roles and identities, women’s knowledge and labor is rendered visible and valuable, traditional campesina activities are foregrounded in economic and community development plans, and rural women gain a sense of agency and empowerment, thus enhancing their overall well-being”. (p.15).

Reviewer comment

This is a keynote paper for us and stands in sharp contrast to the previous paper on Ethiopia which essentially marginalized women’s knowledge (the conceptual framework and methodology itself was insufficient). The Cuban paper is extremely detailed and would need to be read again for details on how it understands food sovereignty, the specific gender-transformative steps in PIAL, etc. Its main value for us is that it shows that male-led agricultural and knowledge systems can be transformed to empower women whilst offering men many benefits (there is a long section using quotations etc from women and men on various learning points like this).

10. Beyer, Linda & Chaudhuri, Jay & Keino, Barbara. (2016). Kenya's focus on urban vulnerability and resilience in the midst of urban transitions in Nairobi. Development Southern Africa. 33. 3-22. 10.1080/0376835X.2015.1115739.

Research Topic

This paper focuses on understanding indicators of urban vulnerability and resilience.

Methodology

Analyses results of five rounds of the Indicator Development for Surveillance of Urban Emergencies surveys conducted in three informal settlements of Nairobi.

Relevant findings

Results show a significant deterioration in food security and household hunger in marginalised urban populations, with other deprivations including insecurity, negative coping behaviour and inadequate access to water and sanitation. Significant gender disparities have been shown in lowest income quintiles, with female breadwinners earning 62% compared with male breadwinners ($p < 0.05$). Also, alarming rate of severe household hunger reported in households of Korogocho with female breadwinners – double the rates of those reported in the two other locations (p.1, p.14).

P.5 says that “Resilience is the capacity of individuals, communities and systems to survive, adapt, and grow in the face of stress and shocks, and even transform when conditions require it. Building resilience is about making people, communities and systems better prepared to withstand catastrophic events – both natural and manmade – and able to bounce back more quickly and emerge stronger from these shocks and stresses”(The Rockefeller Foundation, 2015). ...Resilience analysis has provided a relevant and plausible framework for understanding the regional or local capacity of individuals/households and a communities’ ability to withstand future shocks and stresses, their ability to adapt and the potential for enhanced future capacity (Béné et al., 2012).”

P.5 adds that “Measuring the multi-dimensional nature of what puts these urban populations at risk as well as the unique dimensions of how they cope will be the focus of this analysis. These measures of resilience are grounded in the conceptual understanding of risk, vulnerability, chronic and transient poverty, insecurity and complex interactions between shocks and stressors at various scales as well as adaptive capacities in the face of shocks and stressors (Maxwell et al., 2013; Barrett & Heady, 2014). Measuring resilience involves a higher frequency of analysis to ensure sensitivity to seasonal variation and dynamic changes in welfare indicators and other shocks that are context specific (Barrett & Heady, 2014). It requires measuring the ability to withstand or bounce back from these shocks. This DEVELOPMENT SOUTHERN AFRICA 5 Downloaded by [Radcliffe Infirmary] at 17:06 11 April 2016 involves using a variety of qualitative and quantitative methods including a focus on local economies and ecologies (Barrett & Heady, 2014). These aspects of resilience analysis have helped shape both the formation and approach underpinning IDSUE efforts. These include basic measures of well-being as well as absorptive, adaptive and transformative capacity (Frankenberger et al., 2013).”

Conclusions

The authors make a number of recommendations and say (p.18) “Gendered dimensions of social capital, social exclusion, coping, decision-making and vulnerability would strengthen some of the initial work in this area. Social capital would focus on qualitative elements underlying social organisations including networks, norms and social trust that can serve as assets for communities to engage and benefit from collective action (Bernier & Menzen-Dick, 2014).”

Reviewer comment

As with most papers surveyed so far there are no insights into why women /Female Headed Households are more vulnerable. This is simply observed in the descriptive statistics section.

The value of the paper lies in its range of indicators to measure vulnerability and resilience. Personally, I got less insight into what creates resilience. Also, the authors argue that urban and rural vulnerability and resilience are not the same, but I am not clear why.

One thing that strikes me (which is why I cited the Rockefeller definition of resilience) is that a resilient population (or gender) is supposed to bounce back. This actually seems to be a common understanding, but I just don’t think this is helpful as in my view systems change all the time and one cannot bounce back to the previous one. ... if we are to make this case (bouncing back), we need to discuss very carefully which system (household? Farming system? Extension system etc) we are talking about. And I still think that changes in one system will create changes in another. I prefer the tone of the Bene et al., statement which talks about the potential for enhanced future capacity.

11. Bryan, E., Quinn Bernier, Marcia Espinal & Claudia Ringler (2018) Making climate change adaptation programmes in sub-Saharan Africa more gender responsive: insights from implementing organizations on the barriers and opportunities, Climate and Development, 10:5, 417-431, DOI: 10.1080/17565529.2017.1301870

Research Topic

Examines extent to which climate change adaptation programmes carried out by various NGOs, government agencies, and others in SSA are gender-sensitive and the extent to which research can support implementation of more gender-responsive programming (p.418).

Methodology

A knowledge, attitudes, and practices survey and key informant interviews targeted at government agencies, local and international NGOs, and other practitioners (sample size 76). On p.419 is a detailed description of KAP survey and topics.

Relevant findings

The impact of climate and other shocks on men's and women's assets depends greatly on the socio-cultural context in which these shocks occur (i.e. social norms and gender roles), as well as the type of shock experienced (p.417).

In Niger, men were more likely to sell their assets in response to climate shocks given a cultural norm that men are responsible for ensuring the food security of the household (Tumer 1999 in Kristjanson et al., 2010) (p.418).

Women are more constrained in responding to climate change and climate shocks. In part, this is due to the fact that men and women perceive climate change and climate shocks differently, with women often less likely to perceive the impacts of climate change (many references) [...] differences in men's and women's perceptions can have profound effects on whether or not they adapt and, if so, which strategies they choose. Differences in perceptions and responses also relate to the fact that, women are less likely to have access to climate information as well as information about options for responding to climate change (many references). As a result, women tend to be less aware of options for responding to climate change (Bernier et al., 2015). However, when women are aware of climate-smart practices, they are just as likely as men to adopt those practices (Bernier et al., 2015; Mittal, 2016). (p.418).

Factors constraining women's resilience (p.418):

- Women often have less mobility, which limits their ability to seek off-farm sources of income (many references).
- Gender differences in resources for adaptation, including assets, inputs, and decision-making authority, also constrain women's responses to climate challenges and influence the degree to which women benefit from changes at the household level, such as the adoption of new agricultural technologies (many references).
- Women have different preferences and needs for adaptation responses given their gender roles within the household that are often not taken into consideration when technologies are designed or introduced to agricultural households (references).
- Men and women interact with development agencies differently, with women less likely to be engaged with external agencies working to build adaptive capacity in rural communities (Cramer, Forch, Mutie, & Thornton, 2016). At the same time, these agencies may be less responsive to the needs and preferences of women (Cramer et al., 2016)
- Barriers to women's participation in adaptation to climate change are very context-specific and there are no one-size-fits-all solutions to increasing women's adaptive capacity (Gonda, 2016). Even within the same communities, there are important differences between different groups of women, which influence their adaptive capacity (Fischer & Carr, 2015).
- Women's voices are often missing from policy decisions and in leadership roles within implementing organizations which further prevents women's needs from being address through policies and programmes (references)

KAP Survey Modules

Knowledge (p.420); Attitudes (p.421); Practices (p.426)

Although the potential linkages between gender and climate change (e.g. barriers and outcomes) can be gleaned from the available case studies, such research does not provide the answers required for particular adaptation programmes to successfully integrate gender. Addressing this research gap will require programmes and projects to integrate more rigorous, academic-style research into these programmes, either through stronger partnerships with research organizations or through the application of research tools and approaches, for example in their scoping and M&E work.

Local NGOs felt that information was not well shared among the network of organizations working in the area of climate change adaptation. This finding suggests that there is large potential to improve the quality of programmes dealing with climate change adaptation in a gender-sensitive manner, simply through better communication among different stakeholders ... Government agencies, in particular, may benefit from greater information-sharing given that they were less likely to have access to information and research on gender and climate change. (p.427).

There is tension in terms of how much emphasis to place on gender. While gender has become a buzzword in international circles there is some resistance to emphasizing gender during project implementation, with participants reporting that projects targeting women tend to be less successful than projects that emphasize community benefits. While the way in which communities are approached is important to get community buy-in, it is also important not to lose sight of key gender considerations during project design, targeting, implementation and M&E. Communities must be sensitized to understand that gender-sensitive projects do entail a focus on both women and men and that this joint focus can improve overall outcomes. At the same time, it is important that culture is not used as an excuse to justify gender inequality. (p.428).

Conclusions

Although organizations have access to research on climate change from various sources, more is needed to identify entry points for gender integration into climate change adaptation programmes across a range of local contexts. Lack of staff capacity on gender, lack of funding to support gender integration, and socio-cultural constraints were identified as key barriers to gender integration by many respondents, particularly from government agencies. Enabling organizations to pay greater attention to the gender dimensions of their programmes is possible through greater collaboration across different types of organizations in order to share knowledge and best practices and strengthen the integration of research into adaptation programmes.

Reviewer comment

This is the standout article so far. It needs to be reread when writing the synthesis. It does not have details per se on norms, but it summarizes the key constraints women farmers face. However, its main contribution is in analysing a wide range of institutions from governments to NGOs regarding the knowledge of gender and capacity in gender (project design, M&E etc) and capacity to integrate research into their work.

This article is especially valuable for the CGIAR as it provides clear entry points for research organisations and how they may best partner with different types of organisations.

12. Call, M. and Samuel Sellers (2019) How does gendered vulnerability shape the adoption and impact of sustainable livelihood interventions in an era of global climate change? *Environ. Res. Lett.* 14 083005

Research Topic

Acknowledges the gendered unevenness of the global landscape of vulnerability, exposure, and adaptive capacity to environmental stressors and shocks. Historically, policies and interventions addressing natural resource-based livelihoods have rarely considered underlying gender dynamics despite the global pervasiveness of gendered disparities in both economic opportunities and welfare outcomes. This article tries to summarize what is known across agriculture, fisheries and forestry.

Methodology

Literature review (methodology explained in detail, could be useful for us to base our write up of our AB on this).

Relevant findings

Gender has significant impacts on the adoption of natural resource-based livelihoods interventions, affecting the ability of households to adapt to the effects of climate change. In all three sectors explored (agriculture, fisheries, forestry), women or female-headed households generally have lower rates of adoption or participation (p.10). Interventions can inadvertently propagate vulnerability and inequity in adaptive capacity rather than reduce it. Example (there are many in main text) Men or Male Headed Households (MHH) are much more likely to adopt improved seeds, which are largely promoted to both increase household crop productivity as well as to help households cultivate climate-resilient crops. Women and men also often engage in different forest harvesting activities - when men are often the main voices in forest management programs like REDD+ this can impact women's vulnerability by reducing their capacity to benefit from forest resources. The most successful interventions and policies are those which are based on a sound understanding of how individuals use natural resources, and if gender differences do exist, are tailored to accommodate gendered practices.

Gender often correlates with disparities in resources and information, and effective natural resource-based livelihoods programs recognize and adapt to these disparities to yield beneficial social outcomes for both women and men. When natural resource-based livelihoods interventions are sensitive to the power dynamics shaping gender relations in communities, they have the potential to yield broadly beneficial social outcomes (sources). In agriculture, a limitation for many women may be access to information, for aquaculture, the barrier to entry is often the cost of establishing an aquaculture pond. Research from India and Nepal has demonstrated that interventions that subsidize adoption of aquaculture improve income and employment gains for women, reducing inequity (sources). As gender is often correlated with other disparities, successful interventions and policies often use creative strategies for identifying and addressing multiple deficits, including gaps in knowledge and resources. When these disparities are adequately addressed, women and men are able and interested in adopting new natural resource management practices, and these practices are more likely to yield gender-equitable outcomes.

Gender may affect environmental outcomes of natural resource-based livelihoods interventions, but the literature is scant. Across agriculture, forests, and fisheries, natural resource-based livelihoods interventions infrequently evaluate the environmental outcomes of policies that are ostensibly designed to improve both environmental and social conditions. Regarding forests, studies from India suggest female forest user committees can improve ecological outcomes (Agarwal 2009, Das 2012), but this may not be a universal phenomenon. More evidence is needed to understand the mechanisms through which some women's user groups have successfully improved forest quality outcomes (Mwangi et al 2011, Sun et al 2011). Similarly, scant evidence exists in the agriculture and fisheries/ aquaculture sectors. As such, although there is a small body of evidence suggesting that women's engagement in natural resource-based livelihoods interventions has the potential to be environmentally beneficial, we caution against broad generalizations or assumptions regarding more sustainable environmental outcomes associated with greater involvement of women in these activities.

Other points made elsewhere in paper (p.2) N.B. Importance of intersectional approaches to examining gender issues, noting that the effect of gender can be modified by power structures, discrimination, poverty, geographical, political, and historical legacies, among other factors (Nagel 2012, Moosa and Tuana 2014). As a result of cultural gender roles, rules, and norms, women and girls may be exposed or vulnerable to environmental stressors or shocks in ways that differ from the challenges faced by men and boys (Ellis 1998, Gladwin et al 2001, Cleaver 2005). Likewise, women and men may differ in their access to the resources and skills necessary for adaptation, or adjustments that improve an individual's capacity to cope with external stressors (Smit and Wandel 2006). Though women and girls are often portrayed as being especially vulnerable, men and boys often face significant, though sometimes distinct, challenges as a result of environmental stressors and shocks (Omolo 2010, Alston and Whittenbury 2013, Keshavarz et al 2013).

Conclusions

The authors conclude this paper by noting how scant the literature really is, though it is building up. They make a list of recommendations for future work, particularly focusing on the need for rigorous (and intersectional) work. They note geographic disparities in research across agriculture, fisheries and forestry.

Reviewer comment

As with the Bryan article, this is a powerful literature review that highlights what is known about gender (as a binary) in relation to adapting to climate change. As with Bryan they point to the importance of institutions effectively targeting women and addressing gender disparities. This seems to be emerging as a key area affecting women's economic resilience (the norms in institutions - organisations).

13. Chingombe, W., & Musarandega, H. (2021). Understanding the Logic of Climate Change Adaptation: Unpacking Barriers to Climate Change Adaptation by Smallholder Farmers in Chimanimani District, Zimbabwe. *Sustainability*, 13(7), 3773. <https://doi.org/10.3390/su13073773>

Research Topic

This study focuses on the barriers to farmers adapting to climate change. It suggests that farmers are confronted with an "assortment of barrier dynamics related to climate, finance, infrastructure, generational change, water resources, inefficient bureaucracy, gender inequality, and health barriers ... barrier components ... act together to cripple smallholder farmers' efforts to raise their adaptive capacity. Adaptation barriers are complex and, therefore, cannot be addressed using policies that are fragmented. The paper recommends a multidimensional approach by policy makers to analyse adaptive barriers to build more climate resilience within smallholder farmer communities.

Methodology

Emic approach (via FGDs but the whole conceptual framework and analysis is built around farmer understandings).

An inductive approach with much focus on mixed sampling was used to select respondents in Manicaland Province (close to Mozambique). Several FGDs held based on purposive sampling. Women happened to be more common in some FGDs than others but there were no sex-disaggregated FGDs. The authors say they used AGRITEX to help solicit key gender issues.

Relevant findings

(IPCC) Climate change adaptation means “adjustment to actual or expected climate and its effects”. Adaptation aims to “moderate or avoid harm or exploit beneficial opportunities”. It generally means a shift in livelihood practices meant to cope with newly emerging climatic conditions. It is an ongoing process with no specific endpoint [source]. Climate change adaptation is also viewed as a way of reducing vulnerability, increasing resilience, moderating the risk of climate impacts on lives and livelihoods, and taking advantage of the opportunities posed by actual or expected climate change [4]. Thus, smallholder farmers’ productivity depends on how effectively they adapt to their physical environment in the wake of climate change. (p.1)

Productivity Commission - barriers to climate change adaptation refer to circumstances that “restrict people’s ability to identify, evaluate or manage risks in a way that delivers the highest level of community wellbeing”. Eisenack et al. define adaptation barriers as those impediments to specific actions needed to fulfil the goals of particular actors. (p.1-2)

To date, there seems to be no agreed framework to understand the proximate causes and complexity of climate change adaptation barriers because of the subtleness of the boundaries between suggested categories (sources). More practical and evidence-based studies need to be done to establish their proximate drivers and the connectedness of the commonly recognised barriers. It remains unclear as to how *adaptation barriers relate to one another* to create a complex network of barriers that derail local and external intervention initiatives towards improving the adaptive capacity of the farmers. (p.2)

Although women were the majority of FGD respondents, little gender specific data was obtained. A general point was that “male household heads reportedly held the responsibility to purchase seed and other inputs, leaving the task of planting and tending of crops to women and children whilst they engaged in off-farm activities. As such, female respondents widely bemoaned their lack of equal involvement in choosing the practices and specifying which inputs to use for each season” (p.10)

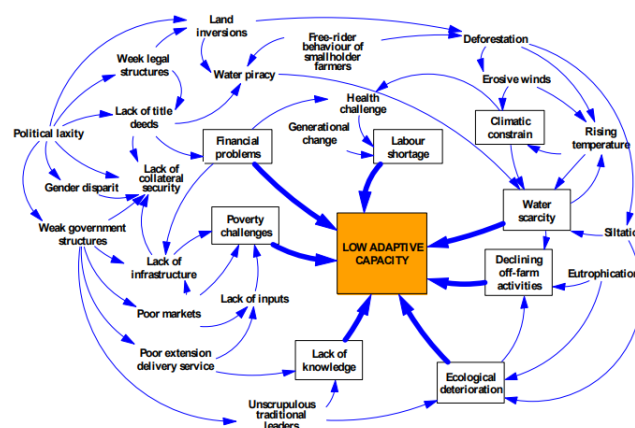


Figure 4. Causal loop diagram showing the connectivity of climate change adaptation barriers in Chimanimani District.

This figure on p.12 is a very interesting attempt to link all the barriers together using emic perspectives. The authors comment “As farmers struggle with one challenge it translates into a whole new form of challenge, together forming a complex network of barriers that draw back the efforts put in place by the farmers themselves. ... For example, farmers felled thorn branches to fence crop fields, which in turn translated to increased deforestation and more ecological challenges.” (this is not a fully direct citation and there are other examples).

The authors note that farmers in the past managed climatic change but now “climate change has continued to bring more extreme weather scenarios beyond the farmers’ ability to understand and adjust to accordingly. Gitay et al. [31], in this case, argued for the mobilization of institutional support from state universities and civil society organizations to assist farmers with climate-smart strategies. This is the component that many smallholder farmers are lacking, hence, their increased failure to adapt to climate change. Otherwise, effective agriculture production requires the proactive provision of relevant inputs at a timeous moment [7,32] when there are government institutions directly responsible for all these mandates.” (p.13-14)

Conclusions

The authors recommend that:

- Communities are further educated for them to understand that they can build more climate change resilience beginning with the little resources available to them.
- Local traditional authorities can also harmonize the sharing of land and water resources amid climate change adaptation initiatives if they remain transparent and honest to their subjects through their practices.

- An all-inclusive and nonlinear approach is required in addressing barriers to climate change since they affect smallholder farmers in an interconnected manner.
- Institutions working with smallholder farmers should be well-coordinated in their intervention to fight common climate change adaptation challenges.

Reviewer comment

This paper hardly addresses gender per se, but women were the majority of informants. This case study is fascinating in how it weaves together emic perspectives to show how barriers link together to create an almost impenetrable mess, with farmers attempting to overcome one barrier therefore creating new barriers / running into new ones. We could certainly think about developing a similar kind of diagram to show how gender constraints link and associate and reinforce each other. Then we would need to think about how to break these links.

The authors again point (not cited here but in article) to considerable institutional deficits, lack of will, etc in trying to solve problems. This is the first article so far which also points to considerable community tensions around key resources like water.

14. Calderón, C.I., Claudia Jerónimo, Alexandra Praun, Jaime Reyna, Ivan Dimitri Santos Castillo, Raquel León, Rose Hogan & José Pablo Prado Córdova (2018) Agroecology-based farming provides grounds for more resilient livelihoods among smallholders in Western Guatemala, *Agroecology and Sustainable Food Systems*, 42:10, 1128-1169, DOI: [10.1080/21683565.2018.1489933](https://doi.org/10.1080/21683565.2018.1489933)

Research Topic

"Livelihood insecurity emerges from the interaction of factors and pressures affecting people's ability to cope with change. Yet the effects of interactive stressors on the dynamic nature of livelihoods and their underlying gendered relationships are still unexplored. We evaluated a case study in Maria la Baja, a municipality in the Colombian Caribbean severely affected by the Colombian armed conflict and oil palm expansion. With the pacification of the conflict in the late 2000s, traditional inhabitants aimed to reconstruct their livelihoods amidst a context of reduced access to resources. We assessed how local livelihood strategies have been affected by the interplay of socio-political distress, agrarian change, agrobiodiversity loss, seasonality, and unequal access to resources for men and women. We conducted surveys and interviews to 1) characterize local livelihoods and their transformations; 2) evaluate current livelihood insecurity and pressures; and 3) assess whether disruptions in local livelihoods have equally affected men and women and influenced their mechanisms to cope with change. Current livelihoods in Maria la Baja have been largely characterized by months with water, crop and income scarcity. These effects have been shaped by the reduced access to resources and social control that resulted from the armed conflict and oil palm expansion. Such pressures also induced the spatial segregation of women. Diversification of livelihood activities was a coping strategy to overcome the effects of resource scarcity. Despite segregation by gender, diversification was greatly conducted by women. Women-led diversification may become more important in contexts of increasing socio-ecological change."

Methodology

Surveys and interviews. Focus is on Afro-descendent women and men. Uses Sustainable Livelihood FW and also Livelihoods as Intimate Government FW (LIG) by Carr, 2013. This aims to make sense of internal forms of governance shaping roles, values and relations through which households pursue material and social goals (p0). Surveys conducted in five communities using a wide range of questions.

Relevant findings

Paramilitary conflict had driven farmers from the area. They returned (2010s) to find oil palm plantations and had to reconfigure their lives. Previous mechanisms for managing change were no longer effective and the new situation demanded new strategies (p.9).

Men forbade women from returning to the fields, where they had previously worked, for their safety (women had suffered violence during the war) and some men felt women should stay at home. Women were restricted to activities around their home, kitchen gardens, and nearby areas of villages. The authors see some benefits - women fight change with change - p.9 but the comment overall is that women's lack of access to open areas inhibits their ability to access biodiversity (important for food security and nutrition) and leads to a massive care and housework burden. Men's livelihood strategies reproduce former strategies (mono-cash crops) and are not effective in the current situation (p.9).

"Men and women engaged in different activities with contrasting economic goals (Table 3, Fig. 5). Men continued specializing in a few activities that included growing bastimento crops, waged agricultural jobs, and construction. Men were the decision-makers in bastimento parcels grown for commercial circuits, which were frequently mono-cultures managed with agrochemicals. Women developed a diverse array of activities in domestic spaces including the sale of groceries, transformed foods and cellphone calls; paid laundry; husbandry of pigs and poultry; a broad range of domestic care activities; and the establishment of patios productivos or productive food gardens. In those gardens, women were the main keepers and producers of fruits, vegetables and medicinal or aromatic plants. One woman sold vegetables and fruits from her home garden in the village's grocery store. Collectively, productive home gardens and yards hosted more than 70 of the plant types grown or tended by families participating in the study, including bastimento crops grown for household consumption. Gardens were more diverse when both men and women participated in garden maintenance. Women signaled some difficulties in garden management such as damages made by free-range animals affecting crops, lack of water and seeds, and lack of space to grow food."

Conclusions

The authors argue that women have experienced forced displacement, sexual violence and spatial confinement. They are fighting against material scarcity. Yet overall, Afro-descendent communities have experienced displacement through armed conflict and land-grabs. Women have often been at the forefront of resistance, resilience and re-existence in these communities. Yet the ability of these specific communities to manage is challenged by simultaneous pressures limiting gender equity and economic autonomy. They need access to resources (water, land, and other productive resources) and assistance with economic revitalization. It is important too to help preserve agro-biodiversity as well as people's wellbeing and stability.

Reviewer comment

This paper contains a wealth of information, and it is impossible to include summary snippets here. It is worth reading p.9 especially. Another keynote paper. Focuses on a specific intersectionality – Afro-descendent/ Guatemala. Important because the reality is that many locations are now facing war and other forms of disruption. This article shows the gendered ways in which people try to cope. It shows gender relations in transition, and how this effects resilience, very clearly.

15. Cely-Santos,M., Olga Lucía Hernández-Manrique, Fighting change: Interactive pressures, gender, and livelihood transformations in a contested region of the Colombian Caribbean, Geoforum, Volume 125, 2021, Pages 9-24, ISSN 0016-7185, <https://doi.org/10.1016/j.geoforum.2021.06.014>.

Research Topic

"A set of sustainability attributes was estimated in Western Guatemala by characterizing 20 small-scale farming families with the aim of exploring food security- and climate resilience-related conditions. Solidarity-based economies spawn a stronger social network among agroecological farms. Gender roles in agroecology-adopting families seem to be moving to a more balanced scenario. Differences between agroecology- based and semi-conventional farmers suggest that shares of commercialized produce, gross agricultural income, and plant diversity are significantly higher and thus more resilient in the former. Challenges to agroecological adoption, however, include limited public infrastructure, dearth of supporting policies, and external threats posed by utility-inspired economic agents." (Abstract).

Methodology

Focus group discussions, soil sampling and many other methods were used to compare agro-ecology adopting families with semi-conventional farming (clear distinctions cannot be drawn since some practices are shared).

Relevant findings

There are a vast number of findings which suggest that agro-ecology farmers are more resilient. The findings on gender are very limited and show almost no difference, apart from slightly higher levels of schooling in agro-ecology farming (pasted text p.1161).

"Gender differences were also observed in regard to schooling. Agroecological families seem to distribute schooling opportunities more evenly (15 girls and 15 boys) than their semi-conventional peers (15 girls and 23 boys). Agroecological groups have had more chances of being trained by a number of organizations which seems to have deepened their gender-

related sensitivity. Even so, agroecological female producers still face major challenges as to health, nutrition, education, access to credits, access to water, work, migration, and organization.”

Conclusions

Limited.

Reviewer comment

This is a great comparative paper which examines a vast range of indicators, both physical and social, across both farm systems. Gender findings are limited, though. It is possible that women and men may have had different conceptualisations of what agro-ecological farming /conventional means, but this was not explored (a major issue in northern America literature on sustainable agriculture). As ever there is some suggestion that participation in organisations raises consciousness on gender issues.

16. Dah-gbeto, A.P., Villamor, G.B. Gender-specific responses to climate variability in a semi-arid ecosystem in northern Benin. *Ambio* 45 (Suppl 3), 297-308 (2016).

<https://doi.org/10.1007/s13280-016-0830-5>

Research Topic

This study explores gender-specific responses to climate variability and related coping strategies in the context of agrarian Benin (p.297). It uses an anticipatory learning conceptual framework.

Methodology

This research explores gender-specific responses to climate variability in the context of agrarian northern Benin through a household survey (n = 260) and an experimental gaming exercise among a subset of the survey respondents, to contribute to the overall understanding of resilience among subsistence agricultural systems in semi-arid ecosystems:

- (1) How do male and female farmers perceive and react to climate variability and extreme weather conditions?
- (2) Do male and female perspectives differ in terms of land-use preferences and adaptation to climate variability?
- (3) What determines gender-specific decisions under conditions of climate uncertainty?

Improved knowledge of gender-differentiated exposure and response to shocks contributes to anticipatory learning, which is the key in helping communities to become more resilient to risks and uncertainty associated with global climate change (Nuttall 2010; Kumar and Quisumbing 2014).

Relevant findings

Anticipation is a critical component for building resilience (Boyd et al. 2015). By being proactive, anticipation focuses more on foresight drawn from predictive capabilities, knowledge, experience, and skills (Nuttall 2010). Botkin et al. (1979) distinguished anticipatory learning from adaptive learning, with adaptation being a reactive adjustment to change (or external pressure), and anticipation implying preparation for possible contingencies and consideration of long-term alternatives. In a non-anticipatory situation, people may simply wait for problems to worsen before seeking remedies or may react and search for answers until it is too late to implement effective solutions (Botkin et al. 1979). In this sense, adaptive learning poses a limitation, and anticipation is required to adapt decisions and behaviors to a dynamic environment.

According to Rhea (2005), anticipatory learning has four defining dimensions: *foresight* (thinking ahead about how trends, issues, and developments observed in the present are likely to shape alternative futures), *identity* (how we act in the present, and what is important for us to create in the future), *direction setting* (forging the learning from foresight and identity into wiser decisions about what to create in the future and how to do it), and *innovation* (identifying strategies to avoid a feared future) (p.298).

Of the four defining dimensions of anticipatory learning, only evidence of forecast and identity was found (p.306) though the authors partly explain this due to constraints of the game selected and they suggest follow-up work. Yet they add “Nevertheless, it is important to reflect on the anticipatory capacity of local people, as most adaptation studies draw attention to technologies, regulation, policies, and practices that enable society to adapt to change (O’Brien 2012). In this study, an important social variable that affects resilience with a slow rate of change—identity—is found to be strong among women, and this is often overlooked and yet essential to understanding their resilience (Folke et al. 2010)” (p.306).

Conclusions

Although perceptions and adaptation measures related to climate change are quite similar between men and women in the study area, the means, capabilities, and motivations vary by gender. Thus, their approaches to risk and uncertainty are also different.

While men remain the primary decision makers in Benin households when they are around, women were found to respond in more active, dynamic, and innovative ways (in terms of diversifying income sources) when dealing with rainfall variability. Men continue to engage in seasonal migration or permanent relocation for work purposes, which is a common response to economic hardship. Although resulting migration or relocation may increase household resilience for both those who remain and those who emigrate, women are most likely to remain and will continue cultivating crops for household subsistence, thus bearing much of the impacts of climate change. Identity, as one of the defining dimensions of anticipatory capacity, was found to be strong among women in the study site and is essential for understanding their resilience to climate change.

Reviewer comment

This article is another key article. The concept of *anticipatory learning* is novel (so far) and it is very usefully contrasted with *adaptive capacity* here. It is interesting that the farmers only exhibited two forms of anticipatory learning (noting the provisos in the article) and that women's and men's anticipatory capacity differed.

A new story is emerging, namely that men's strategies for resilience (see Calderón et al., above) may not be particularly powerful contributions to resilience. Also, the sense of women being left behind to manage is strong, but they are key to food security and nutrition. In other words, it is essential they succeed.

A key point is that the packages offered by external agencies need to match up with people's anticipatory capacity.

17. Darabant, A., Habermann, B., Sisay, K. et al. Correction to: Farmers' perceptions and matching climate records jointly explain adaptation responses in four communities around Lake Tana, Ethiopia. Climatic Change 163, 1119–1120 (2020). <https://doi.org/10.1007/s10584-020-02911-2>

Research Topic

This study distinguishes between climate change and climate variability and provides literature indicating why they need to be differentiated. It compares women's and men's understanding on these topics and their adaptation strategies. Mixed methods approach, climate modelling plus local perceptions.

Research objectives (i) complementarities and differences between local perceptions and climate data, (ii) perceived impacts and (iii) climate adaptation strategies of local people as a function of perceptions.

Methodology

FGD and PRA tools to collect data for addressing all three research objectives. Social data collected involving members of local Participatory Research and Innovation Clusters (PARICs) formed by researchers to engage local communities into applied research focusing on adaptation. PRA tools - timelines, land use history, trend analysis and pairwise ranking to assign relative importance to (i) perceived climate change parameters and (ii) adaptation strategies. (p.483).

Local climate data spanning 30 years (1981 to 2010), consisting of daily minimum temperature, maximum temperature and precipitation, were modelled for the four study sites using regional climate data and meteorological stations (much detail in text). (p.484).

Relevant findings

Adaptation takes place in specific contexts: perceptions and traditional ecological knowledge are the basis for local adaptive action by vulnerable stakeholders (refs) even in cases where climate change leads to unprecedented conditions (refs). Contextual knowledge and its variability are necessary to explain whether experiences of climate and its impacts influence farming practices and their environment (refs). Inadequate context knowledge may lead to maladaptation resulting from distorted perceptions of evident long-term trends (climate change) at the local level (refs), and the lack of understanding of local perceptions in the design of adaptation strategies at the policy level (refs). Correlating the variability of farmers' climate perceptions and their adaptive actions with meteorological data assists to develop a wider framework of policy responses to climate change and variability (refs).

Understanding local perceptions helps to explain whether farmers adapt to long-term changes or merely to increased variability (p.484).

On climate (p.491). It is difficult to disentangle climate change and climate variability as both have local impacts (refs) and have increased considerably during recent decades (refs). In line with this, it became evident from the narratives that farmers do not consciously differentiate between these two phenomena. However, our results confirm the findings of Smit et al. (2000) that farmers' perceptions are rather shaped by short-term variability of climate parameters and the frequency of extreme events than slow long-term changes in the average conditions.

Gender differences in rankings of adaptation strategies can be partially explained by gender differences in agricultural tasks and in perceived impacts. Farming in the Amhara Highlands is primarily a male domain and men are responsible for the selection of crop varieties (Abay et al. 2008), ploughing and livestock rearing. This explains higher adaptation priority rankings assigned to these and to pest and disease management by male as compared with female PARIC members, which corresponded well with the findings of Deressa et al. (2009) and Mersha and Van Laerhoven (2016). (p.493).

Females perceived forest degradation, decline in agricultural productivity and differences in stream discharge and thus preferred adaptation options addressing these impacts, such as soil and water conservation, water harvesting and forest restoration. The reporting on improved varieties, mineral fertilizers, terracing, water harvesting ponds and afforestation was balanced between gender groups, indicative that these adaptation practices do not stem from traditional gender roles but result from an externally introduced agenda. Migration irrespective of destination was almost exclusively reported by females, as opposed to the study of Mersha and Van Laerhoven (2016), where males were more likely to engage in local, whereas females in long-distance migration.

Gender differences in the ranking of non-land-based adaptation options cannot be explained by differences in perceived impacts, which indicates that their drivers are more complex and rooted in rigid gender norms and conceptions of the patriarchal society of the Amhara Highlands (Mersha and Van Laerhoven 2016).

Conclusions

Ultimately, scientific climate information is of little use for an important target group—farmers responsible for implementing adaptation strategies. Based on the perceived impacts, farmers implement adaptation options that vary by region, even across small scales, and by gender. To be effective, adaptation options need to be based on local perceptions and specifically tailored to meet local environmental, socio-economic and cultural conditions, including traditional gender roles. Our findings corroborate other studies that call for a meaningful combination of scientific and local knowledge systems to allow for better adaptation (p.494).

Reviewer comment

A strongly empirical paper that says that farmers note climate variability but not necessarily climate change. Adaptation strategies vary by gender with women's ability to respond limited by cultural norms that allow men to decide on key responses e.g., crop choice and variety.

18. Dossou-Yovo, Elliott R., Sander J. Zwart, Amadou Kouyaté, Ibrahima Ouédraogo, and Oladele Bakare. 2019. "Predictors of Drought in Inland Valley Landscapes and Enabling Factors for Rice Farmers' Mitigation Measures in the Sudan-Sahel Zone" *Sustainability* 11, no. 1: 79. <https://doi.org/10.3390/su11010079>

Research Topic

Drought is a noteworthy cause of low agricultural profitability and of crop production vulnerability, yet in numerous countries of Africa little to no consideration has been paid to readiness for drought calamity, particularly to spatial evaluation and indicators of drought occurrence. The study intervened in three West African countries located in the Sudan-Sahel zone, viz. Burkina Faso, Mali and Nigeria.

Methodology

Biophysical and socio-economic data, farmers' community surveys and secondary data from remote sensing on soil characteristics and water demand were used to evaluate the predictors of drought in inland valley rice-based production systems and the factors affecting farmers' mitigation measures.

Relevant findings

Numerous mitigation measures were used by farmers against drought in inland valley rice-based production systems. They encompassed crops diversification, farming practices and land use measures. Land ownership and the gender of farmers influenced farmers' commitment to use any mitigation measure against drought (Table 4). Farmers were not willing to implement any measure when their land was not secure or when they did not have full rights on the aforesaid land. Likewise, when property rights are not secure, there is a probability of expropriation which acts as a disincentive to invest. Indeed, a farmer is more motivated to invest when he inherits or purchases the land. However, he is less willing to invest in the land when he knows that the land can be sold. Therefore, policies to secure property rights could promote farmers' investment in labour, input and other resources to mitigate drought and sustain productivity.

Women were more likely to use mitigation measures against drought than men. This could be explained by the fact that in the study area, men were more interested in investing in cash crops such as cocoa, cotton and sesame than in rice [42]. Women lacked land and capital to invest in cash crops [43] and focused on rice production to improve the livelihood of their households. Recognizing the greater ability of women to mitigate drought in inland valleys rice-based production systems than their male counterparts could be particularly important in focusing on women's association for advancing drought mitigation measures.

Conclusions

Land ownership and gender affected the commitment of rice farmers to use any mitigation measure against drought. Drought studies in inland valleys should include climatic water balance and groundwater data. Securing property rights and focusing on women's associations would improve farmers' resilience and advance drought mitigation measures.

Reviewer comment

This study is highly statistical, and it was not clear to me how they established the gendered findings. However, the story that men focus on cash crops meaning that women focus on food for the household, is continuing over from earlier studies in this review. The suggestion is again being made that women rather than men are more interested and receptive to mitigation measures. However, to be able to implement they need property rights as farmers do not want to invest in land, they do not own which can be expropriated. It is unclear where the idea for women's associations comes from, but it is probably conceptualized as a counterweight to gender norms that restrict women's agency.

19. Druza, K., Valentina Pever (2018) Literature on gendered agriculture in Pakistan: Neglect of women's contributions, Women's Studies International Forum, Volume 69, 2018, Pages 180-189, ISSN 0277-5395, <https://doi.org/10.1016/j.wsif.2018.02.007>.

Research Topic

The role of Pakistani women in producing wheat and the way wheat products are used, valued and shared by them is grossly under-researched. The initial search around the basic key words 'rural women', 'agriculture', and 'wheat' did not produce any significant outcome, identifying not so much a paucity of current published literature pertaining to gender in wheat producing regions, but rather a more fundamental and widespread inconsistency about the connection between women and grain crops (p.179).

Methodology

Feminist standpoint theory is applied in analyzing 73 documents on women's roles in agriculture from 1990 until 2016 to answer the following questions: How does the world look, and operate, for males and females in wheat growing households? What do we know about social relationships and mediating processes (i.e., social factors mediating men's and women's access to resources and activities) that exist in the prime wheat growing regions in Pakistan?

The authors endorse the hypothesis that an exclusive focus on agricultural production would make women fade away into invisibility/illegibility, and they therefore aim at providing alternative entry points into the lives, feelings, concerns and unofficial and informal networks of rural Pakistani women (p.181) Feminist standpoint theory involves standpoint analysis, which involves (Hawkesworth, 2006):

- Step 1 - Construct alternative standpoints by collecting and synthesizing as many competing views of the subject under investigation as time allows;

- Step 2 - Acknowledge the partiality and contentiousness of comparing competing claims and methods. In line with step 2, and Haraway (1988:193) who argues that 'unlocatable' knowledge claims are irresponsible, the situated knowledge of the reviewers is acknowledged. (the authors provide details about themselves here).
- Step 3 - Compare theoretical assumptions and empirical claims in conflicting accounts;
- Step 4 - Present the alternative views fairly while avoiding caricatures

The article describes feminist standpoint theory in more depth as well.

Relevant findings

Despite the dogmatic insistence of the strictness of purdah (the code of honor and modesty resulting in the seclusion of women) in public fora, women and men's actual participation in various activities points to the flexibility of the concept. It provides substantial evidence that the spheres of men and women do overlap much more than they are likely to admit. A meaningful relationship emerges from the literature linking women's energies, goals, and interests with home gardens (harvesting vegetables), storage of cereals (post-harvest activities) and natural resources management, especially with livestock tending and animal production. (p.180).

The agricultural literature reviewed tends to accept cultural norms and gender roles, rather than question their persistence or attempt to examine them. The binary thinking which simplistically identifies men with technology and farming, and women with tradition and home, accompanies much gender-blind work and is strongly reasserted in agricultural literature. Conversely, the anthropological literature reviewed questions the traditional view of gender and the nature of purdah. There is extreme variation across parts of Pakistan that is reflected in the literature and yet frequently downplayed by agricultural literature which focuses on yield, crops or climate, rather than people, history and culture.

Conclusions

This review sheds light on the dynamics that are rarely tackled in both policymaking and the development agenda (p.187).

Firstly, it foregrounds the inversely proportional relation between the legibility of grain crops and the legibility of rural women.

Secondly, Pakistani women use alternative scripts to cope with the dominant paradigm, be it inside the domestic walls or in open fields, talking around rather than contradicting it head on. There is scope for exploring how women's assertion of their independence may well cohabit with their desire to be modest, self-sacrificial, subservient, and humble and the way this plays out in the agriculture sector.

Thirdly, women's work mainly and meaningfully stands in the realm of subsistence; yet often development institutions, in their attempt to commercialize agricultural activities, competes with the subsistence sector for natural resources and women's labor. Women's role in providing food security for the family and managing farms when their spouses engage in non-farm income elsewhere should play a more prominent role in the literature.

And finally, women's relationship with livestock has to be brought into clearer focus in its emotional as well as practical components if a full understanding of the important role women play in agriculture and in intra-household decision-making is to be identified.

Reviewer comment

This paper puts forward many complex arguments which cannot be captured effectively here. A key point is that focusing on crops alone excludes women because they are not necessarily active in these spaces, but they are agentic and active in other spaces. Also, women may be resisting in multiple ways which may not be immediately obvious. Simplifying debates around purdah and cultural norms is unhelpful not only due to variation around the country, but because so much else is going on. A lot more complexity and nuance are needed. Finally, this paper returns again to the emerging theme over the previous papers that women are strongly active in subsistence, providing food security, livestock and actually managing farms when husbands away, but this is neglected in research.

20. Drury O'Neill, E., B. Crona, A. J. G. Ferrer, R. Pomeroy, and N. S. Jiddawi. (2018). Who benefits from seafood trade? A comparison of social and market structures in small-scale fisheries. *Ecology and Society* 23(3):12. <https://doi.org/10.5751/ES-10331-230312>

Research Topic

How benefits from a fishery and the associated trade are ultimately distributed, specifically how market structures and relations, and the combined dynamics of the local fishing society, can mediate these flows. The paper compares two sites, in Zanzibar and the Philippines, and makes a comparative analysis of market types, assistance networks and income inequality, using a value chain framework.

Methodology

Structured interviews with fishers and trading agents in Unguja Island, Zanzibar, and Iloilo Province, in the municipality of Concepcion in the Philippines. 194 fisher and 163 traders were interviewed in Unguja and 280 fishers and 52 agents in Concepcion.

Relevant findings

Results – Examining Market Structures and Relationships (p.4-5): It was found that women's and men's roles were different in the two contexts. In Concepcion, men and women play similar roles, and sometimes work in partnership, e.g., in husband-wife teams, while in Unguja, men and women have distinct roles. In both contexts however, women's income was lower than men's on average. (p.6).

Discussion – Seafood Market structure and participation (p.10): In the Philippines, women occupy non-traditional roles in the value chain. However, at the production (fishing) level, women end up with comparatively fewer benefits than men. The authors attribute this to women's domestic responsibilities which limit the time they can dedicate to fishing, as well as to the physical demands of the role:

As explained to the authors, rather than what they feel are lived cultural constraints or social norms, they spend less time out fishing because of the harder physical labor demanded by this activity and necessary domestic duties, thus they can end up with less daily income. Data on hours spent fishing per day from structured interviews supports this argument.

In Unguja, there are also women who deviate from the expected roles and earn more income than average. The authors note that women seem to be gaining ground in fishing, but it is not clear if these are exceptions to the rule, or if they point to future trends in VC participation for women.

They also highlight the importance of intersectional analysis to understand this better:

"Nevertheless, the stories of women's participation in fisheries that are increasingly connected to global seafood trade are multifaceted and shaped not just by their gender but also by factors such as household assets, size and composition, education and skills (Tindall and Holvoet 2008). Gender must be placed in the broader social processes at play around the fisheries, rather than being singled out, if researchers are to better understand women's long-term VC positions (Wosu 2017)."

Conclusions

The study draws similar conclusions to the article reviewed above (Drury O'Neill, N., Crona, B. (2017) *Assistance networks in seafood trade – A means to assess benefit distribution in small-scale fisheries*), that social and economic relations are both social insurance mechanisms, but also can be inflexible, due to the nature of social ties/obligations, and that their inflexibility might perpetuate unsustainable resource extraction. It also concludes that "the relations do appear to play a critical role in structuring the marketplace and conduct, yet they are generally never explicitly considered in fisheries related policy or governance." (p.11)

It finishes by raising questions about how small-scale fishers and traders can access growth from global market integration and upscaling of the fishing industry:

"The year 2018 marks major development in the fishery VCs of Unguja and Concepcion as the market and port expansions get underway. Neoclassical economic theory suggests global market integration is a viable strategy to reduce poverty. However, it is difficult to understand how exactly fishers and trading agents in the current cases can access the growth in wealth projected to emerge from this integration."

Reviewer comment

This article builds another one reviewed here by two of the same authors- Drury O'Neill, N., Crona, B. (2017) *Assistance networks in seafood trade – A means to assess benefit distribution in small-scale fisheries*. This article provides more detail on the value chain structures and maps the exchanges of assistance more clearly between different actors. It also has added richness in its comparison between the Philippines and Zanzibar contexts. It does not deal directly with climate change but does contain some

nuanced analysis of women's roles in fishing, and of the constraints they face, highlighting that their incomes are limited not so much by their traditional roles within fishing but more by their time commitments relating to their other roles in the home.

21. Drury O'Neill, E. and B. Crona. (2017). Assistance networks in seafood trade-A means to assess benefit distribution in small-scale fisheries. *Marine Policy* 78, 196-205.

<https://www.sciencedirect.com/science/article/abs/pii/S0308597X16306443>

Research Topic

This paper aims to broaden understanding of market structures and trading relations in tropical marine small-scale fishery and how these contribute to food and livelihood security for all actors involved. In particular it looks beyond purely economic transactions between actors to include non-financial reciprocity such as exchange of assistance, recognising that fishers and traders are embedded in a broader web of social interactions. Research was carried out on Unguja Island, the largest island of Zanzibar, Tanzania.

Methodology

Rapid assessment; followed by semi structured interviews, informal conversations and observation to map the value chains. Fieldwork was then conducted over 5 months using surveys and participant observation in 8 sites. Only 5 women fishers completed surveys, of a total of 168, because of a lack of female fishers available and willing to take part. 49 female and 89 male traders also took part. At each site, an average of 9.5% fishers and 46.7% traders (of the total population) took part.

Relevant findings

Fishing is central to the Zanzibar economy: The rural fishing sector supports through livelihoods 25% of the Zanzibari population through fishing directly (34,268 fishers in 2007) and secondary fishery services (auctioneers, traders, retailers, boat builders etc.) (2.3)

There is the potential for fishing to be affected by climate change and/or environmental degradation though this is not the focus of this paper: the fishery of Unguja Island, like most of coastal East Africa, is highly dependent on the Monsoon cycles which are characterized by Kusi (Southeast) and Kaskazi (Northeast). The coastal environment of Unguja, and the rest of Zanzibar is dominated by coral reef, seagrass beds, sandy beach and mangrove ecosystems.

Women and men interact in different ways with the fishing sector:

Female traders typically deal with smaller lower-value species, often frying them and selling them for local consumption, while male traders will not traditionally fry the larger species that they can export or sell to tourism. (2.2)

Female traders appear to have very few predetermined buyers, with only 15% (n=47) of women traders (largely rural and all frying the products) reporting arrangements primarily to local consumers. (3.2)

Women serve principally the local consumers, and are largely disconnected from the tourism industry, with the exception of the five fisher women selling to one neighbouring resort. (3.1)

The relationships of reciprocal support were shown to benefit men more frequently than women. The exception to this was mutual support between women traders.

Fewer women traders receive support from fishers than male traders do – 40% of women traders in rural areas, and none in urban areas, versus 67% male traders in rural areas and 37% in urban areas. Support includes fish for home consumption, deferred payment arrangements and discounts. his paper argues that this support is vital for traders' food security and to help them to survive on narrow margins.

Women fishers do not receive assistance from traders, while men in rural areas report receiving support in the form of money and products for home consumption.

Urban female traders do not generally give fishers help (only 4.8% do this), but rural female traders are more likely to (56%). Overall, the main type of help traders provide fishers is money for general use, gear, fuel where relevant, food and to fix gears or vessels. Secondly they provide fishers with seafood for home and, to some fishers, bait and fuel.

Women and men traders support each other (usually women-women and men-men) by buying together pooling products to sell, lending or borrowing money and selling products on each other's behalf when necessary.

Conclusions

The paper doesn't look at the impact of climate change *per se* but does highlight the need for value chain mapping to include reciprocal exchanges of assistance rather than only formal economic relations, because the former are important for resilience in the face of any economic shocks.

The findings confirm previously researched differences in the positions women and men take in tropical marine small-scale fishery. The paper does offer insight into the differences in reciprocal relationships between women and men, and urban and rural traders and fishers. However, it does not explore in depth the reasons for these differences (e.g. social norms), beyond acknowledging that social ties are a factor which determine relationships between fishers and traders. The findings on women fishers are limited by the very low number of women fishers included in the research.

With regards to trader-fisher relations, the paper challenges the perception that traders and fishers are in an asymmetrical and exploitative relationships (with traders having more power) and indicate that the relationships are more of mutual assistance. The paper warns against programs which aim to bypass local level traders or middlemen and highlights the "inappropriateness of management strategies that consider these fisheries as isolated from non-financial social relations (e.g. assistance), sectors and resources" (Section 5, p.203).

Reviewer comment

The interest of this paper is in the inclusion of informal exchanges of assistance as a strategy for resilience in the face of shocks. The conclusions about women's roles in fishing are in themselves not new. The paper also doesn't delve into gender norms that affect women's benefit from reciprocal relationships between traders and fishers.

22. Duyen - Tran Nhat Lam, Roberto F. Rañola, Bjoern Ole Sander, Reiner Wassmann, Nguyen Dinh Tien & Nong Nguyen Khanh Ngoc (2021) A comparative analysis of gender and youth issues in rice production in North, Central, and South Vietnam, *Climate and Development*, 13:2, 115-127, DOI: 10.1080/17565529.2020.1734771

Research Topic

How rice farmers at all gender and ages perceive climate change and adopt Climate Smart Agriculture (CSA) technologies to enhance resilience and adaptation in three sub-regions of Vietnam. Explores gendered perceptions of climate change, barriers, and attitudes to CSA technologies.

Methodology

Survey data from HH survey questionnaire from 579 randomly sampled households in North, South and Central Vietnam.

Relevant findings

The study shows different patterns of outmigration in the three sub-regions and that the gender division of labour (GDL) differs. There are no constraints on women's mobility.

Based on the labour inputs (person days/hectare) in rice production, the relative share of men is generally higher than women in Bac Lieu province in contrast to Ha Tinh and Thai Binh province where women have higher level of engagement in rice production than men. Men contributed 70% of total labour inputs per hectare while women contributed only 30% in Bac Lieu province. In contrast, women have higher labour contribution in rice cropping activities than men in Ha Tinh (46% of men and 54% of women) and Thai Binh (45.4% of men and 54.6% of women). The actual tasks conducted by women and men vary by subregion. Male outmigration is higher in Ha Tinh and Thai Binh because the contribution of rice to family income is low. Women remain to conduct more of the work. In Bac Lieu men do not feel an incentive to out-migrate or take on other off-farm work since rice provides the bulk of family income. (p.119)

Gender differences in decision-making are discussed on p.121 but are not surprising. There is stronger participation in provinces where women participate more in rice, and lower in Bac Lieu, where women are also not reached by extension.

The respondents were asked whether the proposed CSA technologies are suitable for women given the nature of farm work? (p.121). A large range of CSA technologies were discussed and on balance it seems like respondents agreed overall that they were suitable for women as well as men. However, there were regional differences., for example:

Fewer men than women in three provinces reported that the water-saving techniques were suitable for women. This is probably because men dominated women in irrigation operation. Although men are dominant in irrigation activities, women could be trained on water-saving techniques because a high percentage of men and women in these provinces agreed that water-saving techniques were appropriate for women.

With regards to improved stress-tolerant varieties and crop diversification, the study found the difference in response of men and women in Bac Lieu and Thai Binh provinces. In particular, there were many women than men who reported that these CSA technologies were suitable for women. In general, high proportions of male and female respondents of three provinces reported that these CSA technologies were appropriate for women. The finding implied that women should be provided opportunities to attend training activities and field demonstrations on new rice varieties and crop diversification.

Participants were asked about barriers to CSA adoption. Barriers were classified into: (A) technical factors, i.e. CSA technologies are too complex to apply and require costly inputs; (B) government policy, i.e. lack of government policies to support CSA technologies, lack of accessible services (new varieties, agricultural machines); (C) human factors, i.e. shortage of labour, CSA technologies are knowledge-intensive and require skills training; (D) financial factors, i.e. lack of capital to purchase inputs, lack of access to agricultural credit; (E) economic factors: low market demand, low market price of rice; (F) bio-physical factors, i.e. topography (fragmented lands, slope of plots), soil and water condition (infertile soil, water shortage, poor water quality). (p.122).

In Bac Lieu province, there were gender differences in women's and men's contribution, control of decision-making, and experience in rice production. More women than men reported that CSA technologies are too complex to apply (35% men, 51% women) and the technologies to knowledge-intensive (64% men, 76% women) were constraints to adoption. This is because men dominated women in rice production activities. Thus, women tend to avoid complex and knowledge-intensive CSA technologies because they have limited experiences in rice cultivation. In contrast, fewer women than men reported that lack of access to agricultural credit as well as economic factors such as low market demand and low market price of rice are barriers to CSA adoption.

There was no gender difference in responses on barriers to adoption of CSA technologies in Ha Tinh province while a higher proportion of women than men cited that shortage of labour, lack of capital to purchase inputs and low market price of rice were the constraint to adopting CSA technologies in Thai Binh province. This is plausible since women in Thai Binh province dominated men in rice farming activities. This can be attributed to a higher rate of male out-migration due to the small contribution of rice in total household income and the extreme weather for farming activities.

Conclusions

The paper concludes with recommendations for addressing women and youth in rice, basically through targeted extension.

Reviewer comment

This paper usefully points out that there are strong regional differences in women's overall participation in agriculture, and specifically in paddy. It also suggests that women's greater or lesser propensity to find CSA technologies suitable is in large part a consequence of the existing GDL. This is important.

This said I felt throughout the paper that this was a hypothesis pre-held by the authors ... if women do not do X, then they will not be interested in Y. This was admittedly never stated but the analysis is glib and straightforward. It needs more complexity to help understand women (and men's) agency. Even so, it is interesting that overall respondents did not (appear to) signal that women could not do something, rather that the technology was not associated with women's labour.

I did not review the youth data largely because it was clear that the data relied on adult perceptions of youth needs.

23. Estrada-Carmona, N.; Attwood, S.; Cole, S.M.; Remans, R.; DeClerck, F. (2020) A gendered ecosystem services approach to identify novel and locally-relevant strategies for jointly improving food security, nutrition, and conservation in the Barotse Floodplain. International Journal of Agricultural Sustainability 25 p. ISSN: 1473-5903

Research Topic

Multiple lines of evidence call for the use of locally relevant strategies to guide and support sustainable agricultural intensification while improving development and conservation outcomes. The goal of this study was to identify the ecosystem services from natural and agricultural systems to achieve this aim in the Barotse Floodplain of Zambia.

Methodology

Gender-sensitive ecosystem services approach, whereby local knowledge from women and men was harnessed to understand which services and their sources are important. Identified the various constraints and options people encounter for developing sustainable and nutritious agriculture while achieving conservation outcomes.

Some of the gaps for guiding sustainable and inclusive agricultural development can be filled by: 1) emphasizing nutritional and agricultural diversity (rather than simply closing yield gaps) for food and nutrition security (McKenzie & Williams, 2015); 2) highlighting the benefits or ecosystem services delivered by natural resources essential for ecosystem functioning (TEEB, 2015) and for the wellbeing of the people who depend on them (Suich et al., 2015); and 3) recognizing and incorporating local knowledge and context-dependent constraints for agriculture (refs) (p.151).

Three communities were studied.

Evaluated 17 ecosystem services jointly discussed by and selected before the activity with community facilitators and traditional leaders. The selected ecosystem services included nine provisioning, seven regulating and one cultural type aiming to cover the wide use and benefits that women and men obtain from the natural and managed ecotypes in the floodplain (p.153).

Relevant findings

The floodplain provides a broad range of ecosystem services, which are important for securing local livelihoods and wellbeing. The forests in the uplands and the grasslands in the plains are the primary sources of the 17 provisioning and regulating ecosystem services assessed. Both are often converted to agriculture due to their high soil fertility. We identified opportunities and challenges for sustainable agricultural intensification and development in areas with lower conservation concerns. We discussed the constraints and limitations for promoting sustainable and inclusive agriculture in those areas.

In all three communities studied women and men cultivated the same or different ecotypes and to use different areas to fish.

Example. In Lealui, three eco-types (cultivated grasslands, home gardens, islands) seem to be the most commonly cultivated ones during both seasons among women and men participants' fields (>25%). Nonetheless, a small proportion of women from this community also mentioned cultivating different eco-types during both seasons (i.e. wet sandy fields, riverbanks) and some eco-types exclusively during the dry season (cultivated forests, woodlands, dry sandy fields). In Nalitoya, very few participants reported having fields on the same eco-type during both seasons. The fields of more than half of the women and men participants were located on wet sandy fields during the wet season and cultivated forest during the dry season. Women and men from Nalitoya seem to cultivate different eco-types. Women's fields were commonly on seepage (dry season) and islands (wet season) whereas men's fields were on dry sandy fields (wet season), cultivated forest (wet season) and cultivated grasslands (dry season). (p.361-362)

Women fished in shallower waters than men and caught smaller species (p.358)

Women participants (mainly from upland communities) listed to plant more crops than men participants, including additional food groups and neglected and underutilized species (NUS). For example, in Mapungu, women listed ten additional crops (four NUS), which added three food groups (fruits, other vegetables, and sweets). Similarly, women in Nalitoya listed eight additional crops (one NUS), which added three food groups (sweets, vitamin A-rich fruits and vegetables and tubers). Whereas in the plain (Lealui), women only listed two different crops (including one NUS) and men listed one different crops. (p.363)

Conclusions

The paper makes many valuable suggestions based on the trade-offs identified between maintaining ecosystem services (essential for long-term) and pressures to convert various ecosystems to farming. They note the many stakeholders involved, way beyond the farmers. They suggest a number of ways forward including agro-ecological intensification (CA/multiple cropping) and agro-forestry among others. Farmers much more likely to perceive provisioning than regulatory ecosystem services (p.368). In the context of the dynamic Barotse Floodplain, efforts fostering sustainable development through agriculture should also consider the challenges farmers face such as long travel distances to their fields, the gender differences in use and knowledge of the floodplain and limited labour availability, particularly during the dry season. Besides, tapping into the use of local crop diversity through multiple cropping systems, could lead to a sustainable floodplain that provides nutritious food and other multiple ecosystem services year-round.

Gender points (p.367/8):

Differences in how women and men use and understand the floodplain are overall statistically non-significant (potentially linked to the small sample size). Nonetheless, some specific and identified differences are relevant for developing strategies or programmes to support sustainable agriculture development that are gender sensitive.

Women and men access the same eco-types for provisioning ecosystem services although they tend to access them in different parts of the landscape, resulting in a considerable disparity between the two groups and implications for income generation and nutrition (refs). One such example is particularly marked in fishing activities. Distances to fishing resources are similar, yet each group accesses different products (e.g. small vs. large fish, by women and men respectively).

Crops (including diverse NUS) and fields cultivated by women potentially complement household diets with essential nutrients such as vitamin A. Ignoring the importance of women's crops on household nutrition security and focusing exclusively on cash crops could exacerbate intra-household power dynamics and lead to increased food insecurity (refs).

Other specific aspects that could challenge the adoption of sustainable agricultural practices in this region includes the 1) reduced labour availability during the dry season particularly experienced by women in the upland, and 2) the distances covered by farmers to carry out activities and to monitor the multiple planted and scattered fields. Labour scarcity is likely stronger in the upland communities since men are more prone to dry season migration for fishing, confirmed by women's more accurate knowledge about eco-type drought patterns than their male counterparts.

Diversifying livelihoods by planting various fields on different eco-types is a common strategy in the study area to reduce the risk of food insecurity due to drought or floods. Nonetheless, this strategy renders constant management and monitoring of crop growth difficult given the distances that farmers need to travel among fields and activities (e.g. fishing and cropping areas). For instance, the dynamic floodplain requires improved access to appropriate low-cost labour-saving technologies (e.g. through access to oxen and ploughs), while considering the potential labour demand that alternative practices such as conservation agriculture can pose to farmers (refs). This is particularly relevant for women since, in the Western Province, the percentage of female-headed households is greater than the national average (refs). Careful attention is required as the evidence suggests that female-headed households tend to have greater difficulty in adopting practices and crops that require more resources (e.g. cattle for manure, or money for agricultural inputs or labour).

Reviewer comment

This is a key paper with very detailed assessment of gendered differences in how women and men in different communities perceive and access ecosystem services. It shows the need for fine granular work on understanding how women and men live within and use a landscape. It suggests that agrifood systems need to focus as much on nutrition as on food security, and that women's use of NUS crops (neglected and underutilized) is important here.

24. Farnworth, C.R., Clare M. Stirling, Amon Chinyophiro, Andrew Namakhoma, Rebecca Morahan, Exploring the potential of household methodologies to strengthen gender equality and improve smallholder livelihoods: Research in Malawi in maize-based systems, Journal of Arid Environments, Volume 149, 2018, Pages 53-61, ISSN 0140-1963, <https://doi.org/10.1016/j.jaridenv.2017.10.009>.

Research Topic

Findings of a research study into GALS in Malawi where the National Smallholder Farmers' Association of Malawi (NASFAM) has been implementing GALS since 2013 with 4274 farmers (2821 women and 1453 men to May 2016).

Methodology

Sex-disaggregated FGDs with 40 GALS households and 40 non-GALS households, all NASFAM members. Community profiles and a matrix activity focusing on task allocation, asset distribution, and expenditures by gender with 125 non-GALS and 135 GALS respondents were also conducted.

Initiatives which foster effective partnership between women and men, based on fostering more equal gender relations, are more likely to result in higher productivity and other gains. We do not agree with the apparent assumption behind FAO's and broader work on women's economic empowerment that male productivity will remain unchanged whilst female productivity will increase if women are supported effectively. Rather, - jointness is likely to have synergetic effects contributing to a number of benefits across the farm and within the household. (We also hypothesize that gender inequalities contribute to low male productivity in smallholder systems though this has not been researched to our knowledge.) (p.54)

Improvements in female productivity on smallholder farms in SSA to the extent envisaged by FAO is not achievable unless there are changes on an enormous scale in gender relations. Achieving this means shifting away from understanding gender as a characteristic of individuals which can somehow be strengthened, to understanding gender as an iterative dynamic process in which gender is constantly being 'remade'. Shifts and reconfigurations which strengthen women's gender interests and women's voice are unlikely to succeed unless men consider themselves partners and beneficiaries of this process. In our view, too much gender analysis has historically been constructed around explicit and implicit dichotomies - his assets, her assets - thus failing to pick up sufficiently on collaborative decision-making processes around assets.

Studies showing jointness in practice, then - these imply that jointness in intra-household decision-making has the potential to strengthen input use, improve adoption of climate-smart technologies, and to underpin more equitable distribution of benefits within the household, including better food and nutrition security. Rather than work within dichotomies, improving synergies appears to be a useful way forward.

Women more than men are involved in a 'zero-sum game', a closed system in which time or energy devoted to any new effort must be diverted from another activity (refs). Women's labour becomes fragmented to handle existing and new work, often resulting in reduced efficiency and effectiveness across productive as well as care work. (p.55)

Relevant findings

Many women involved in GALS still prefer to concentrate on building assets that indisputably belong to them. It can, therefore, be argued that lack of jointness is continuing due to the weak position of women in these patrilineal, patrilocal communities. Improved jointness in intra-household decision-making is difficult to achieve when community institutions do not support it. At the same time married women are demonstrably more able to articulate and realise their own visions and plans, and they expect their spouses and children to support them. They, likewise, support their partners. At this level, jointness is improved.

Taken together, two conflicting forces seem to be in operation. The process of developing visions strengthens the individual agency of both men and women while at the same time it is having a transformative effect on household relationships, increasing co-operation, optimism and resilience. This is contributing towards assets being managed together.

Moves towards dissolving the gender division of labour in farm tasks, and in the designation of crops as women's or men's, are taking place. Respondents trace causal links between jointness and improved yields. Adult men, and boys, are taking on domestic tasks. This is in a context where it is culturally almost unheard of for men to do so.

The GALS methodology allows household members to 'practice jointness' in the company of other GALS households. Safe spaces in which to debate, to identify and challenge social constraints and vulnerabilities and model alternative behaviours, visions, and trade-offs help to anchor behavioural change over the longer term.

Conclusions

Our research shows that the GALS is indeed a methodology which promotes the processes by which people who have been denied the ability to make choices acquire such an ability.

The visioning process and accompanying tools facilitate women (and men) to define choices that can be life-changing, first-order choices (houses, land are first order choices in the study communities) or choices which are lodged somewhere along the way between first and second order choices (such as livestock - particularly larger livestock). Findings suggest that resources as conventionally understood - particularly productive assets - take a secondary rather than primary function. The GALS enables poor people, including very poor people, to change their lives in significant ways, and in so doing stimulate changes in how they are viewed in wider society. This in turn provokes positive feedback loops with a variety of effects.

We can therefore modify Kabeer's simple linear model with a more complex version (Fig. 2). Fig. 2 prioritizes agency as the primary condition for change and suggests that even in circumstances of extremely low resource allocation women's agency can be a very powerful force for change. The two-way arrows symbolize what will become increasingly systemic iterations between agency - resource - achievement as feedback loops are set in motion. The findings draw attention to the extraordinary intrinsic power poor people can have to change their lives. Household methodologies represent a mechanism whereby poor people gain control over empowerment and defines what it means to them in their particular life and particular situation. In so doing, they are very much following Sen's (1990:44) conceptualization of empowerment as "replacing the domination of circumstances and chance by the domination of individuals over chance and circumstances (p.60).

Reviewer comment

This article provides comparative data between a GALS and non-GALS community. It shows that women and men build much stronger asset portfolios in GALS households. This is achieved through the visioning tool and its parts (milestones etc) and it also suggests that household methodologies are key in mobilising agency. This is more important than access to resources per se as in Kabeer's original conceptualisation of meaningful choice. Agency releases resources etc.

25. Feola, G. (2017) Adaptive Institutions? Peasant institutions and natural models facing climatic and economic changes in the Colombian Andes. Journal of Rural Studies, 49. pp. 117- 127. ISSN 0743-0167 doi: <https://doi.org/10.1016/j.jrurstud.2016.10.007>

Research Topic

In the Colombian Andes, peasants have co-evolved with their environment for centuries, but it is uncertain whether traditional informal institutions and natural models are adapting to current and possibly unprecedented economic and climatic disturbances.

This study investigated institutional adaptation and the social mechanisms of institutional change or continuity among peasants in the Eastern Andean Cordillera.

Methodology

The research was informed by evolutionary theories of institutional change and based on a qualitative approach: FGDs, KII, oral histories, observations.

For the purposes of this paper, institutions will be defined as the "humanly devised constraints that structure political, economic and social interaction" (North, 1990:97). Institutions constrain and enable action by assigning power, rights, obligations, and giving a reason for action, based on collectively shared systems of meaning (Searle, 2005, Aoki, 2007).

Research Questions:

- Whether and which informal institutions and connected natural model were changing or persisting in response to climatic and economic disturbances.
- What processes have led to institutional change, or continuity in the face of external (i.e. contextual) economic and climatic changes.

Relevant findings

This study suggests that reciprocal work exchanges, festivities and gender-based divisions of roles have been disused or changed due to economic pressures, but that most informal institutions have persisted due to selective outmigration, conformist intergenerational transmission, and practices of everyday resistance. The natural model of vital energy and the traditional peasant ethos represents a 'social attractor' that has influenced institutional continuity. This study highlights tensions between resilience, cultural diversity, and transformation that are important in many other marginal rural locations in the Andes. Future research should further explore first, under what conditions institutional adaptation is observed and when it is related to increased resilience, and second, how transformability, social-ecological resilience and cultural diversity are related.

Conclusions

The model of vital energy suggests that energy circulates between people and soil, plants, animals etc. (p.120)

"The household is the fundamental unit of social organization of Colombian peasantry, which is similar to what is observed in the broader Andean region. Most traditional informal institutions are defined around the function of the household, and structure resource access and distribution, organization of production, and labour mobilization around that (Fals-Borda, 1955; Comerci, 2012; Forero, 2013).

According to Gudeman and Rivera (1990) and Gudeman (2012), the house is a model of the economy, and, like a physical house, is a closed unit that needs to be maintained. Similar to what is observed in other Andean countries (Sarmiento et al., 1993 in Venezuela), peasants form the base of the house, its foundations. They act as much more than its physical foundations, however and in fact form the wealth or strength of the house. The strength (*fuerza*), or vital energy, is the energy that people use in daily life, and that crops, land, animals, and objects embody. The vital energy flows cyclically from crops, to animals, to and between people through labour and through labour back to the physical environment-the vital energy is the current of the economy and of life (Gudeman, 2012).

In this natural model, the vital energy cannot be created, but is conveyed by people, who secure it, transform it and remake it. Peasants "have a conservation, organization, and entropic notion of force or vital energy. For example, according to their model, the strength in soil, water, and wind is preserved as it passes to crops and then to humans, who assemble and use it up in living and, by planting new crops, help to compose more strength" (Gudeman, 2012, 5). Peasants "see wealth in terms of a cycle that is naturally limited and must be carefully managed. They occupy a humble position in their economic model that assumes limits to accumulation" (Gudeman, 2012, 5).

The household's goals are to maintain itself and to increase the base; due to the cyclical nature of the vital energy flow, these goals are never really reached."

Reviewer comment

Changes in the GDL are mentioned and arise primarily due to the outmigration of young people, leading to older people reorganising their labour.

This article otherwise does not focus on gender much but suggests that peasants strongly follow and create social institutions, which are informed by the life force. Whereas outsiders have long cast peasants as primitive and lacking interest in innovation, for the farmers themselves this is not a lack of interest but a belief in having enough, and not wanting more.

Beyond this, the literature review would support articles on institutions.

26. Forsyth, Tim & McDermott, Constance L. & Dhakal, Rabindra, 2022. "What is equitable about equitable resilience? Dynamic risks and subjectivities in Nepal," *World Development*, Elsevier, vol. 159(C). <https://doi.org/10.1016/j.worlddev.2022.10602>

Research Topic

Equitable resilience is an increasing focus of development policy, but there is still insufficient attention to how the framings of equity itself shape what, and who, is targeted through development efforts. Universalistic assumptions about climate risk or social marginalization can define equity in ways that hide dynamic and intersectional influences on what constitutes risk to whom under different circumstances. This paper investigates the implications of two different equity framings for resilience in Jumla District, western Nepal.

Methodology

One hundred household surveys plus in-depth qualitative interviews in six villages.

Relevant findings

State-led efforts to present post-civil war development as the “equal distribution” of roads and infrastructure, agricultural commercialization, and protection against systemic climate risk fail to reflect local experiences of risk, which are often expressed in terms of social exclusion rather than vulnerability to climate change.

Simultaneously, other efforts at building resilience that use caste and gender as indicators of social marginalization overlook how transitions in livelihoods and individual agency have changed vulnerability contexts for many people, or the increasing vulnerability to climate change of more landed farmers. Excerpts from conclusion (p.11).

“Debates about equitable resilience need to pay more attention to the implications of tacit framings of equity itself. This statement applies to political actors such as the nationalist government of Nepal that employs “equity” as a means to achieve its imagination of post-civil war identity. Yet, it can also apply to analysts of equitable resilience themselves. There is a common assumption that equitable resilience means empowering marginalized groups in order to achieve transformative social change (Ensor et al., 2021; Manuel-Navarrete & Pelling, 2015; Matin et al., 2018). But these positive intentions need to be examined for how far they adopt fixed subjectivities or assumptions of shared physical risks that can conceal the dynamic nature of risks, and the changing constitution of who experiences them.”

“Plus, how does describing these factors in terms of systems hide the specific ways in which norms of equity have been made or applied to social groups for different outcomes? An important challenge here is that the word “subjectivity” is often used interchangeably with agency without acknowledging the social science understanding of the term as projected assumptions about a social group (a point also made by Garcia et al., 2021, p. 191). Seeking to achieve equitable resilience based on predefined ideas of “what” is needed by “whom” can therefore risk imposing subjectivities of its own and avoiding the interdependent ways in which what and who are linked.

Discussions about equitable resilience, therefore, need to include greater capabilities for how far affected people can influence ideas of resilience without being restricted by predefined ideas of risks or social identities. Some influential writings within environmental justice have called for vulnerable people to have “political capabilities” as a way to influence adaptation and climate change policies (Holland, 2017; Schlosberg, 2012). Yet, the debate about capabilities needs to acknowledge the relational nature of risks and identities further (Nightingale et al., 2019; Nightingale et al., 2021). This does not mean disregarding the significant marginalization of social groups such as people of low caste or between genders in many locations including Nepal. Rather, it means understanding how representing people’s risks and agency in terms of these social categories alone might hide dynamic changes in those risks and agencies. Moreover, it means considering how external framings about transformative change or social justice might employ or exacerbate those blindspots (Beck & Forsyth, 2020).”

“The analysis of Jumla in western Nepal in this paper illustrates these challenges for understanding equitable resilience. The paper contrasted two different framings of equitable resilience: the government’s vision of equal distribution without reference to social difference; and approaches that use caste and gender to indicate socially marginalized groups. Evidence confirms widespread findings that Dalit and women workers in Jumla face social segregation and restricted access to livelihood opportunities. Government approaches to resilience through roads, infrastructure, and commercialization fail to overcome these barriers. But relying on caste and gender to determine “who” should get “what” also fails to capture the dynamic, intersectional, and contextual agencies that are applied by different people, or the risks arising faced by higher caste farmers who are exposed to direct climate change impacts because of their generally higher access to land. As Regmi et al (2016, p. 545), note, “vulnerable people do not always conform to popular ideas of vulnerability, such as women or people of low caste.” Acknowledging the dynamism by which risks and identities change and being aware of how normative assumptions might hide these changes is necessary to make resilience equitable.”

Conclusions

The paper urges more critical attention to how normative framings of equity shape what, and for whom is considered equitable resilience, including assumptions about transformative change from analysts themselves. “Representing risks and vulnerability in terms of socially marginalized groups alone might deny the dynamic, intersectional, and contextual interconnection of risks and social agency; and might impose unhelpful subjectivities of their own.”

Reviewer comment

This is a keynote paper. It reviews a lot of philosophical literature on equity. The key point is that blanket interventions do not improve the lot of marginalized people, but assuming that specific identities are marginalized e.g. women x low caste – and then focusing on them – is also a mistake (and can potentially essentialize). Everyone in a community regardless of caste status can

experience vulnerability to climate change, and sometimes climate change affects higher caste more than lower caste. It is vital to not to use simplified understandings of agency, and not to lock people into specific identities. Rather, it is important to pay attention to how identities, risks and agency interconnect in dynamic processes.

27. Friedman, R., Mark A. Hiron & Emily Boyd (2019) Vulnerability of Ghanaian women cocoa farmers to climate change: a typology, *Climate and Development*, 11:5, 446-458, DOI: 10.1080/17565529.2018.1442806 <https://doi.org/10.1080/17565529.2018.1442806>

Research Topic

This paper applies a vulnerability framework to a case study of cocoa farming in the Central Region of Ghana, depicting not only the variety of factors contributing to climate change vulnerability but also *different narratives on vulnerability that emerge based on a woman's relation to cocoa production itself*. Current scholarship argues that treatment of women as a homogeneous group and difficulties integrating biophysical and social components of vulnerability persists in development research and practice. This paper aims to address these shortcomings by applying a novel typology to respond to the question: How does vulnerability vary among women within cocoa farming communities?

Methodology

The case study, located in the Assin South District around the northern and eastern perimeter of Kakum National Park in the Central Region of Ghana, consisted of six cocoa-producing communities, with a high level of heterogeneity in ethnic composition, tenure systems, and social dynamics. The typology of women farmers emerged from initial interviews with community leaders, broadly defined as direct and indirect cocoa farmers (see next paragraph). The case study used semi structured interviews (102 – 49 men and 53 women) and FGDs (6 male and 6 female). Research participants were purposively selected, using referrals from interviewees wherever possible.

Under the typology, a woman was considered a 'Direct Cocoa Farmer' (29/53 interviewees) if she operated her own cocoa land that was inherited, purchased, or sharecropped, although she might have also participated in cocoa farming activities (e.g. harvesting or weeding) on a spouse's farm. Conversely, an 'Indirect Cocoa Farmer' (22/53 interviewees) did not operate her own cocoa farm but primarily was engaged in cocoa as a labourer on a spouse's land in addition to growing food crops, either on a spouse's land or her own land.

Relevant findings

The research analyses the exposure, sensitivity and adaptive capacity of women direct cocoa farmers and indirect cocoa farmers and finds that:

The income that direct cocoa farmers obtain from cocoa enables them to more easily diversify into off-farm income generating activities, such as sale of prepared foods, which strengthens their economic resilience. The autonomy of decision making that female direct cocoa farmers experience is a factor which increases their economic resilience to climate change challenges, as does the opportunity to collaborate with their spouse (if they are married) and support each other on their respective farms. Both of these factors contributed to what the researchers observed as *purposefulness* and *future orientation* among direct cocoa farmers.

Female indirect cocoa farmers have a greater dependence on crop diversification (rather than off farm) which can also be affected by climate change (e.g. changes in rainfall patterns). Many of them lack financial resources to start a business beyond selling vegetables in a sometimes-saturated local market. Female indirect cocoa farmers had varying degrees of participation in decision-making on land use and cocoa income. Interestingly, wives of men who inherited land seemed to have less say in the management of that land or control over outcomes. Women whose husbands did not enlist their help in cocoa farming (rather hiring labour) felt particularly powerless.

"One indirect cocoa farmer explained: 'I do not have much faith in the future of cocoa; it varies by year. In five years it could be good or dead. But there is no another option besides farming.' This attitude of resignation and without a vision for future opportunities depicted by many indirect cocoa farmers suggests that the prospects for cultivating adaptive capacity are limited. It also suggests a relationship between lack of agency and lack of control over cocoa farming." (p.452).

Conclusions

Diversification is key but type of diversification (e.g., crop diversification versus off-farm business) affects women's vulnerability to climate change.

The primary area of structural gender norms identified in relation to economic resilience are customary norms around land inheritance, affecting women's opportunity to operate their own cocoa farms. Women direct cocoa farmers were shown to have greater economic resilience because of their control over cocoa income.

"The interview responses also indicated that it is not gender per se that forms weather and climate perceptions and exposure, but the types of livelihood strategies and land tenure that shapes their concerns and responses. In this instance, the extent of women's involvement in cocoa cultivation, rather than gender itself, suggests in what ways they might be exposed to climate change." (p.453, 5.1).

"There is clearly an interplay between land and income. For instance, food crops may have the advantage over cocoa in terms of flexibility of (shorter) cropping cycles, but cocoa production serves as an important precursor to obtaining funds or credit to invest in off-farm income sources and makes branching into more ventures economically viable." (p.453, 5.2)

The paper calls for programs such as the National Climate Change Adaptation Strategy to recognise the dynamics of gender differentiated vulnerability it outlines. It also suggests that further understanding of the drivers of intra-household decision making would be valuable.

Reviewer comment

The typology of female direct and indirect cocoa farmers is a helpful frame (more so than a focus on male and female headed households for example as it recognises that female farm operators are not always female household heads). The paper is interesting in that it draws a direct link between women's opportunity to farm cocoa directly and control income from cocoa, and their economic resilience to climate change impacts. The analysis seems a little binary in places - for example, it is assumed that climate change impacts on cocoa only affect the time and effort of direct cocoa farmers, perhaps underestimating the role that indirect farmers play in cocoa production (4.1.1)

28. Friedman, R.S., Ellis Mackenzie, Tom Sloan & Nicole Sweaney (2022): Networking for gender equitable climate-smart agriculture, Climate and Development, DOI: 10.1080/17565529.2022.2076645 To link to this article: <https://doi.org/10.1080/17565529.2022.2076645>

Research Topic

This paper looks at climate smart information sharing networks of women and men farmers, and their perceived barriers to information access and use, in three provinces in Papua New Guinea, recognising the role of climate information in adaptation to climate threats in Pacific Island nations.

Methodology

Surveys targeting participants who identified as farmers, were over the age of 18 years old, engaged in cropping-based agriculture, and who resided in the study area. Study areas were identified in consultation with the National Agriculture Research Institute (NARI). Within each study area, village locations were randomly identified and then confirmed as appropriate with NARI. A stratified random sampling approach was used to identify respondents, as well as sampling quotas to ensure a representative spread by age and gender. P.5 has more details of the methods of analysis.

Relevant findings

Men and women access and provide information through different networks and perceive different sources of information as more or less reliable, useful and accessible:

"Women indicated they gave as well as received information among the same groups, suggesting that women's interactions embodied a 'sharing' model with close relationships. Church groups, family and friends, and to a lesser extent leaders and women's groups, were key actors in women's information exchange networks. We found that informal interactions were more important for women as compared to men, relying most heavily on information received from and provided to family and friends

and to church groups. Women did not generally receive information from media, community groups, or agricultural and government groups, and they more frequently reported providing information to farmers' groups, than receiving it. Finally, women's networks tended to be more highly centralized than men's networks, which suggests that women relied on fewer key organizations for two-way transmission of information." (p.6).

Conclusions

A differentiated approach is needed in the provision of weather and climate-related information services to women and men farmers:

"A one-size-fits-all information product is not likely to meet the information needs of a whole community, nor provide adequate guidance for adapting to climate change. Cases of climate information services in other countries have found that targeting of services can be achieved by focusing on specific channels, creating gender-specific information products (Tall et al., 2014), or tailoring the timing of this information (Gumucio et al., 2018; Jost et al., 2016). Our study showed that sourcing information through formal channels, like agricultural extension, the national weather service, and media was almost entirely seated within the domain of men. This has implications for whether the most up-to-date climate and weather forecasts are accessible to women. Although farmers can often rely on informal exchange of traditional knowledge and forecasting to inform their farming decisions, the reliability of these indicators could shift or may not be able to account for significant shifts or deviances from the norm due to climate change (Chambers, et al. 2019)" (p.9)

The research does not analyse the reasons for the differences in women's and men's networks, but it builds on and affirms previous research into the ways in which women can be marginalised in extension services, and also understanding of some of the ways in which women can be reached through existing networks, such as VSLA groups and women's networks. The patterns of information access, and barriers are noted to be similar in the PNG context to others in SSA for example, e.g., lack of female extension officers, logistical constraints preventing women accessing meetings, lack of ownership of cell phones and radios.

Reviewer comment

The paper indicates that women are currently not being reached through formal information channels, and that men having access to a more diverse range of information sources, are better placed to understand and implement adaptation measures such as climate smart agricultural practices. It does not present new findings on the social norms which cause these barriers for women, but rather point to findings from other research to explain these barriers. I think its main interest is that it looks at a context which is less frequently researched (PNG) and it finds similarities between the PNG context, and the context in SSA which is more frequently the focus of research on this topic.

29. Gartrell, Alexandra, Emma Calgaro, George Goddard, and Ngin Saorath. "Disaster Experiences of Women with Disabilities: Barriers and Opportunities for Disability Inclusive Disaster Risk Reduction in Cambodia." *Global Environmental Change* 64 (September 2020). doi.org/10.1016/j.gloenvcha.2020.102134

Research Topic

Despite increased international commitment to disability-inclusive disaster risk reduction (DiDRR) people with disabilities remain largely unseen, unheard and unaccounted for in DRR processes and planning. This is most marked amongst women with disabilities who experience specific gender, disability and poverty-based disadvantages, which disasters exacerbate. This paper unpacks the root causes of women with disabilities' marginalisation in disaster contexts, many of which are extensions of exclusionary processes that play out in their daily lives. We also present steps to position women at the centre of DRR discourse, which will benefit all.

Methodology

Survey, KIs, FGDs (page 4)

1. What do women with disabilities know about natural hazard risk and preparedness?
2. What challenges do women with disabilities face when natural hazards occur? And
3. What types of support (resources) do women with disabilities need and want when hazards strike, in what form and from who?

Relevant findings

Women with disabilities are disproportionately impacted by disasters and are the least able to access institutional support across the preparedness, response and recovery phases of disaster events. Furthermore, the increased threat of violence following disasters heightens their risk of additional harm. In the absence of formal supports women with disabilities have few choices but to rely upon the social capital of their households and neighbours for assistance. They 'recover' in whatever ways they can – through short-term loans, reduced food consumption and/or migration – each carry significant costs to their longer-term resilience.

"An estimated 10 percent of Cambodia's 15 million population (aged five years or older) have one or more disabilities, but data accuracy is questionable and estimates differ significantly depending on the source (NIS, 2015; National Institute of Statistics and Ministry of Agriculture, 2014; Bailey and Nguon, 2014). The most common disabilities are visual (34.8 per cent) and those related to mobility (33.4 per cent) (NIS, 2013). People with disabilities in Cambodia face multiple, intersection marginalizations rooted in negative socio-cultural perceptions and stereotypes that equate disability with inability and lack of capacity to contribute to household productive and reproductive economies and political spaces (Gartrell, 2010). Households with members with disabilities have double the rate of poverty – even before gender is taken into account – and are more likely to be headed by persons that are older and female, unmarried and uneducated than households without disability (Palmer et al., 2019). Women and girls with disabilities experience stigma, limited access to healthcare, education and employment (Bailey and Nguon, 2014). Women with disabilities are also at greater risk from family, non-partner physical, emotional and sexual violence (Bailey and Nguon, 2014, Astbury and Walji, 2013). These every-day gender and disability specific disadvantages pre-determine the risks, vulnerabilities and impacts of disasters on women and young girls with disabilities."

P.3 also has a discussion of institutional barriers. Disability Action Council weak at implementing govt strategies on disability.

"The majority of women reported significant barriers to access non-agricultural employment. Wage-based factory work and informal sector work including construction work and begging, are a common source of supplementary income for farmers when crop cycles are interrupted by hazards and is a key resilience strategy (Parsons and Lawreniuk, 2016). However, migration to urban areas for wage-based factory work is not open to many women with disabilities due to: employers' disability and age-based discrimination; failed entrance tests due to women's low levels of education and confidence; inability to pay common entrance bribes due to low incomes; and difficulties in meeting the physical demands of work. In some cases, family members – rather than women themselves migrated to support their households. These structural factors predetermine women's precarity, particularly during disasters.

Women's inability to secure factory work leaves them more dependent on agricultural livelihoods than their non-disabled counterparts and reduces their incomes. Some received remittances from family members who had migrated to supplement agricultural earnings, but they were in the minority. Fewer financial resources limits women with disabilities ability to accumulate reserves of food and money ahead of disasters. Moreover, they struggle to afford the inflated cost of food and health services during disasters, and find extra funds needed to re-establish livelihoods. Women reported two short-term strategies to overcome these financial barriers. First, women with disabilities and their immediate households shared food between extended family (FGD, Kompong Cham, March 2017)."

Conclusions

Women with disabilities are disproportionately impacted by disasters.

Exclusionary social processes cause heightened risk exposure and reduce resilience.

Inclusive disaster risk reduction must deliver support to those most at risk.

Disaster risk reduction practices can contribute to greater social equity.

Reviewer comment

This is a very comprehensive article which lists many factors making it harder for disabled women to cope with floods, drought, etc. Some are included here. One not discussed is their heavy reliance on social networks, and it is hard for them to keep tapping these. A clear point is that they are heavily reliant on agricultural livelihoods and cannot access other forms of work.

30. Gichure, J.N., Njeru, S.K. & Mathi, P.M. Sustainable livelihood approach for assessing the impacts of slaughterhouses on livelihood strategies among pastoralists in Kenya. *Pastoralism* 10, 26 (2020). <https://doi.org/10.1186/s13570-020-00184-z>

Research Topic

The research appraises how social, physical, financial, natural, and human assets within slaughterhouses impact livelihood strategies to mitigate against vulnerability. Kenya.

Methodology

Qualitative data was collected from four counties in Kenya's ASALs predominantly inhabited by pastoralists, using focus group discussions, and narratives. Observations were analysed using the RQDA package within the R statistical computing environment.

Relevant findings

Pastoral communities are increasingly diversifying their economic and social activities through value addition of meat and meat by-products. Gendered norms were found to influence the extent of diversification and the division of roles and responsibilities within a slaughterhouse. Mostly, men were involved in seemingly prestigious operations such as slaughtering (60%), trade in live animals (90%) and retail of meat (65%), while women, alongside youth are involved in seemingly less prestigious jobs such as scalding (85%), roasting (85%), boiling (60%), stuffing by-products (60%), sale of by-products (75%), sale of firewood (55%), water (55%), charcoal (55%), and managing the nearby makeshift shops (70%). Men oversaw livestock marketing and slaughter operations while women are mostly in charge of by-product value addition. While men gain higher income than women, slaughtering have given women opportunities for diversification and improved diets.

Conclusions

The results suggest that value addition of slaughter by-products, carried out mostly by women provided affordable, safe, high-quality protein diets, with majority of the pastoralists having diversified their diets. Value addition of by-products considerably reduced material and economic losses and increased diet range thereby positively impacting natural resources management.

Reviewer comment

Gender norms influence diversification opportunities. It is interesting to notice that while this pattern provides women with access to less remunerative activities, value-addition of meat by-products offer a variety of income-generating opportunities along with a diversified diet. It confirms the importance of value-addition as a key diversification strategy for women.

31. Gmür, D., (2020) Not Affected the Same Way: Gendered Outcomes for Commons and Resilience Grabbing by Large-Scale Forest Investors in Tanzania. *Land* 9, no. 4: 122. <https://doi.org/10.3390/land9040122>

Research Topic

The gendered institutional changes and gendered impacts on common pool resources, due to large-scale-land-acquisition (LSLAs) in the context of the New Forests Company (NFC) in Kilolo district, in the Iringa region of Tanzania.

Methodology

Participatory research including participatory observation, 40 semi-structured and 15 narrative interviews/biographies, 8 FGDs, value chain analysis, and 38 household questionnaires.

Relevant findings

Findings

Large scale land acquisition provokes land grabbing at different levels and women lose out within all of them:

"I argue that the LSLA provokes different types of land and commons grabbing on different social scales, specifically: (a) pure commons grabbing by the company of community and clan land; (b) internal grabbing within clans common land, meaning between households and lineages; and (c) grabbing processes within the households between husband and wife or brother and sisters." (p.2).

Legal pluralism of land governance means that those who have more power (e.g., investors, men and government actors) are able to 'institution shop' and manipulate the different institutional provisions to buy or sell land. This is intensified when investment money flows in.

"In current legal pluralistic contexts, such as in Tanzania, men and women use both different versions of customary and statutory systems to support their claims to land [34]. This can limit women's access to land and land-related resources. In the context of changing relative prices of land, this legal pluralistic setting provides a huge challenge because men, as well as government actors, are trying to sell land, because investors are becoming interested; thus, these men try to use and manipulate the different institutional provisions to their advantage." (p.6).

Climate change is affecting the region, but people are adapting to this, in part through the management of common pool resources:

"long-term climate records for the Southern Highlands of Tanzania show that the climate of the region has been changing with a steady increase of temperatures over the last 40 to 50 years, with a higher frequency of droughts during the last few decades, as well as a large inter-annual variability in rainfall [37,38]. However, the people in the area, as I will show later, have adapted to these conditions, such as CPR (common pool resources) institutional management in order to be more resilient." (p.6, 4)

"In the context of climate change, people are increasingly relying on NTFPs (non timber forest products), such as wild mushrooms, wild fruits, and wild vegetables for food, health and income security but the planted fruit trees also produce NTFPs." (p.6-7, 4)

However, LSLAs have resulted in losses of livelihoods for women, because of the loss of access to these common pool resources:

P.10, Sections 6 & 7 - The New Forests Company has prohibited villagers from using land in fertile bottom valleys (vinyungu land) which are of particular importance due to unpredictable weather patterns. They are the more productive areas where beans are usually planted, which is the most important cash crop for women. Loss of income from bean sales has a direct impact on women's food security:

"Many people depended on these vinyungu, as I experienced during my field stays and my observations living within these households. Their loss means the loss of possibilities to earn extra income to pay school fees, to buy extra food in cases of food scarcity when harvests on subsistence plots are bad, and to buy fertilisers. Due to the loss of the valley bottom land, food security is now said to be low. Even though this food shortage impacts the entire family, it mostly affects the mothers and the wives, who starve, as they give their food to their children, especially between November and December." (p.11)

Women have also lost access to fruit trees that were growing on communal land that has been sold. They have also lost income from losing access to two grasses on communal land, used for basket and mat making. Women are more dependent on their husbands because of this loss of income. Men have more options for income-generation than women, and women have more care responsibilities.

There is a lack of transparency in payout schemes designed to compensate people for loss of income, and women are often not informed about what has been received by clan heads or elders, or men within their families/household. Women have not benefited from CSR schemes promising employment and alternative IGAs, which in general have not lived up to expectations.

Conclusions

The LSLIs have led to land scarcity, and higher land rents. This has led to women and men finding land in other villages, which increases their workload due to the time it takes to walk there, and also poses a security risk for women if they walk back at night. Maize production has lowered because of lower land access, and this has driven up local maize prices. (p.11)

This situation has also led women to intensify work on existing plots, as well as work as labourers on other people's plots. (p.14)

"CSR programmes and compensation do not reach women and are, for them, an anti-politics machine, hiding the grabbing processes, which hit the poorest of the poor, while the company uses a development discourse to legitimise its activities" (p.14).

"It remains to be seen if women react in a more organized way to these grabbings on several levels in the future. Development policies should consider these voices before taking action and recognise that poverty of women is being created, rather than occurring as a natural phenomenon to be addressed by development programmes." (p.15).

Reviewer comment

This is an excellent piece on the gendered impact of LSLAs on communities. It details the history of land institutions in Tanzania (section 3, pages 4&5) which provides helpful context. It shows clearly that the resource of communal land is a vital economic resource for women in the face of climate change, and that sale of this land to investors has further marginalised women, and intensified conflict and land grabs within families and communities. Women's workload, income and food security has been directly impacted. There have also been backwards steps in terms of gender relations, due to women's increased financial dependence on their husbands. This is a helpful, context-setting piece for discussions of economic resilience to climate change and is of relevance to other contexts where LSLAs are taking place.

32. Gramzow, A., Haroon Sseguya, Victor Afari-Sefa, Mateete Bekunda & Philipo J. Lukumay (2018): Taking agricultural technologies to scale: experiences from a vegetable technology dissemination initiative in Tanzania, International Journal of Agricultural Sustainability, DOI: 10.1080/14735903.2018.1473103

Research Topic

The purpose of this paper is to elucidate a scaling approach implemented for vegetable technology dissemination in five regions of Tanzania as part of a United States Agency for International Development (USAID)-funded program, 'Africa Research in Sustainable Intensification for the Next Generation (Africa RISING)', focusing on the key elements necessary for results related to: gender analysis; private sector engagement; social capital; improvement and adaptation; and program performance monitoring and evaluation.

Methodology

Gender Analysis: quantitative baseline survey augmented with help of focus group discussions conducted to verify the baseline survey results that covered 360 smallholder farm households from nine pilot villages.

Relevant findings

Gender Analysis: Only one woman out of 295 MHHs had a documented titled plot of land, whereas in FHHs at least 28% of the female respondents had at least one. Only 23.4% of the women in MHHs cultivated at least one small plot, whereas 90.1% of women living in FHHs cultivated at least one plot of land. Women in MHHs also had less access to extension training opportunities. Similar results were found regarding access to finance.

Based on these results, the scaling component team decided to place emphasis on women particularly those in MHHs. To enable women to benefit from the project interventions, the gender-specific distribution of tasks during vegetable production and the allocation of earnings from vegetable sales were analyzed. The research component, therefore, included improved leafy vegetable varieties to improve women's income in general and in MHHs specifically. Targeting the participation of women, especially those from MHHs, was also a priority for the scaling component. Given the low participation levels of women in the training activities due to restrictive social norms, the scaling component team started to explicitly invite married couples to participate in the training activities to motivate female and male farmers to participate together in the training. This led to an increase of women participation in the training from less than 15% to more than 30% at the end of the first project year. In the further course of the project the scaling component team intensified its collaboration with national and international NGOs who focus on nutrition interventions and women's empowerment. As a consequence, the total share of women in the trainings increased from 41.1% in 2015, to 56.1% in 2016, and 58.2% in 2017.

Conclusions

Gender Analysis: During an outcome evaluation study conducted in January 2017 in the initial six pilot villages of the scaling component, 104 randomly selected project participants from a sampling were asked whether they have adopted all, one or none of the technologies scaled out during training sessions. One third of the male beneficiaries had adopted all scaled out technologies; for female beneficiaries, it was only one fifth. In contrast, the share of beneficiaries who adopted at least one of the technologies was significantly higher for female beneficiaries. However, 11% of the female beneficiaries in MHHs reported not

having adopted any of the disseminated technologies. This was found to be lower than for FHHs. The fact that almost 89% of all female beneficiaries in MHHs adopted all or at least one of the technologies scaled out shows that the gender approach that was applied by the scaling component team seems to have resulted in some success.

Reviewer comment

Gender-specific constraints (restrictive social norms that limit their participation in training) were identified in this study as an important aspect for successfully taking technologies to scale.

33. Griffiths Michael (2019) Networks of Reciprocity: Precarity and Community Social Organisations in Rural Myanmar, *Journal of Contemporary Asia*, 49:4, 602-625, DOI: [10.1080/00472336.2018.1542450](https://doi.org/10.1080/00472336.2018.1542450)

Research Topic

"In the face of precarious conditions, resilience, arising from adaptive behaviors aimed at mitigating, circumventing or overcoming the effects of precarious living, is a critical factor in maintaining household social and economic well-being (Raven-Ellison 2015, 273-277). Whilst some definitions of resilience collapse it into passive coping actions, there is evidence of emerging co-operative arrangements which enable people to survive and thrive in the face of employment uncertainty and the absence of adequate welfare..." p.603

Methodology

Large surveys and qualitative KIs.

Relevant findings

"Growing precarity amongst rural households in Myanmar is characterised by increasingly debt-fuelled agriculture, decreasing sufficiency and sustainability of rural livelihoods and an absence of social safety nets. This constrains the capacity for viable livelihoods, as risk-averse coping strategies undermine long-term economic sustainability. In this context, informal support networks may be expected to decline or collapse. However, recent evidence demonstrates the widespread emergence of community-based social organisations in rural communities, formed along traditional principles of reciprocity. Analysis of large-scale rural household surveys demonstrates that such organisations are found in nearly 40% of communities in rural Myanmar. These organisations collect and redistribute funds to help with healthcare, education, funerals and other social needs. The presence of such organisations is linked to higher levels of household resilience, achieved through reducing the inequalities linked to gender, disability and poverty, and through providing enabling environments for effective income diversification. Communities with higher levels of migration are more likely to have community-based social organisations, possibly a result of economic and social remittances. In the absence of effective formalised social protection, these social organisations provide most of the social assistance in rural communities, representing new networks of reciprocity in the face of increasing precarity." (Abstract)

Conclusions

"(i)Such organisations promote equality/reduce inequality, for these groups and the most poor, (ii) associated with lower rates of borrowing for consumption, and (iii) poor HH with social organisations in villages with social organisations spend smaller proportion of income on healthcare (do not get into debt)." (p.614-615).

"As new forms of co-dependency, emerging as a self-organising response to precarity, the social organisations discussed in this article represent a core feature of resilience: adaptive capacity. In the absence of formal social welfare and social assistance programmes and the uncertainty of rural livelihoods, such organisations are likely to play a significant role in the shaping of resilience. This is not only in terms of coping and adapting, but through facilitating stronger networking to address some of the underlying causes of vulnerability and precarity. Moreover, by developing or enhancing inter-dependency, particularly when youth involvement is strong, these organisations contribute to the strengthening of community life and social cohesion. This is particularly important at a time when, in rural areas, out-migration and changing social trends are weakening existing community life and structures. The mode through which these organisations contribute to increased resilience appears to be through

reducing inequalities by addressing social welfare needs of the more vulnerable community members. They do this by facilitating an environment which encourages adaptation and risk-taking to diversify in the face of expanding precarity.” (p.620)

Reviewer comment

This is a very detailed and careful study of different forms of community support networks in Myanmar. It shows that women, particularly FHH, and disabled people received more support and were less precarious in villages with community support networks than in villages without them.

34. Hagedoorn, L.C., Bubeck, P. et al. Preferences of vulnerable social groups for ecosystem-based adaptation to flood risk in Central Vietnam. World Development, Volume 148, December 2021, 105650. <https://doi.org/10.1016/j.worlddev.2021.105650>

Research Topic

Quantitative analysis of the differences in vulnerability to flooding as well as preferences for Eco-systems-based Adaptation (EbA) benefits across income groups and gender. Vietnam.

Methodology

To assess the levels of vulnerability and preferences for EbA benefits across income groups and gender the authors implemented a household survey, which embedded a Discrete choice experiment (DCE). The DCE is a stated preference valuation method that is used to elicit the preferences of respondents for specified changes in a good or service. The DCE methodology asks survey respondents to make repeated choices between multi-attribute descriptions of a good or service that is presented to them in choice cards. The data collected through the DCE was analyzed using random parameter logit (RPL) regressions to examine the relative influence of each attribute and level on the choices that respondents make.

Relevant findings

The comparison of vulnerability measures across gender provided a slightly nuanced picture. The study found little differences in terms of *exposure*. With regard to *sensitivity*, the study found higher levels, although not significant, for women in terms of experienced severity of floods, psychological burden and recovery rates. The difference in mobility was significant and reflects that women take on more caretaker roles in the wake of a flood. In terms of *adaptive capacity* the study found significantly lower levels of education and occupation diversity for women.

With regard to women's preferences for EbA, the results indicate a correlation with their current livelihoods. For instance, rural women had higher preferences for the *seafood* attribute, which reflects the prominent role they play in the on-shore fisheries sector. Rural women also identified *Protection from storms and floods* as their preferred attribute, which reflects their higher vulnerability to floods and storms.

Conclusions

Overall, the results indicate that EbA has the potential to reduce vulnerability of the more vulnerable groups through poverty alleviation and women's empowerment. However, whether this will occur in reality will to a large extent depend on projects' design and management. Policies must be in place to ensure that EbA are tailored and benefit those that need them the most.

Reviewer comment

In the case of EbAs, women prefer correlation with their current livelihood. Women are less able to manage and recover from floods, less mobile, and less equipped for alternative livelihoods.

35. Hattam, C., Louisa Evans, Karyn Morrissey, Tara Hooper, Kathy Young, Fazlun Khalid, Mark Bryant, Ali Thani, Lorna Slade, Chris Perry, Susanne Turrall, Dominica Williamson, Andy Hughes (2020) Building resilience in practice to support coral communities in the Western Indian Ocean, Environmental Science & Policy, Volume 106, 2020, Pages 182-190, ISSN 1462-9011, <https://doi.org/10.1016/j.envsci.2020.02.006>.

Research Topic

Using the Western Indian Ocean (WIO) as a case study, this paper combines published literature and expert opinion elicited through a multi-stakeholder workshop to assess the intended and realised social and ecological implications of strategies commonly applied in the region.

Methodology

Based on discussions with stakeholders from the WIO and evidence from the literature review resilience = as the ability to resist, recover, adapt and bounce back from any kind of pressure, but not necessarily to the same state. Ecological resilience was explored in terms of changes to ecosystem services, while any evidence of social and economic change at the individual, household and community level was considered to impact social resilience.

Relevant findings

Who benefits from resilience strategies remains unclear. Gender equity is also largely missing from the social-ecological resilience framework. Strategies to date have not sought to understand or address the question of how men and women negotiate natural resources and how they are affected by, and able to respond to, shocks in the ecosystem. Yet gender mainstreaming is recognised as central to sustainable development and environmental policy and practice. However, there is a general lack of focus on household dynamics and resource allocation, with little consideration of how decisions made within individual households, impact wider resource management.

Conclusions

Resilience policies and the strategies implemented to deliver them need to focus on ways that different groups of actors construct ideas of resilience. Engaging with organisations that tap into individual, household and community values (e.g. women's organisations, health services, faith-based organisation, customary institutions) may be an important route for supporting the more subjective and relational aspects of resilience building.

Reviewer comment

Summarises key findings across literature on resilience. Lack of attention to gender equity issues and especially gender and intra-household dynamics in negotiating access to natural resources is a key finding from the review.

36. Hill, G.; Friedman, R.; Dargusch, P. (2022) Food Systems Development: The Necessary Paradigm Shift for a Healthy and Sustainable Agrarian Transition, a Case Study from Bougainville, Papua New Guinea. Int. J. Environ. Res. Public Health 2022, 19, 4651. [https:// doi.org/10.3390/ijerph19084](https://doi.org/10.3390/ijerph19084)

Research Topic

The study aims to demonstrate how including localised gender dynamics, climate vulnerability, rapidly morphing social norms, and climate analogue environments is critical in building food systems resilience and is key to designing policies, programs, and development projects that more effectively address environmental, sociocultural, and health considerations. Papua New Guinea.

Methodology

The research applied various methods to collect and triangulate food system data from the field, including rich picturing, garden visits, and focus group interviews. Three wards were visited.

Relevant findings

Local gender norms influence agricultural and food production and the need to address them when designing food systems policies and programmes, along with other ecological and social considerations. This is key particularly in the current context of climate change. Traditional gender roles in agriculture and food systems (e.g. women's central roles in the food system and primary responsibility for all subsistence activities along with cash cropping activities and some animal-derived foods) are still very rigid, despite the fact that younger people perceive themselves to be more "gender equal". The study however does not provide any further analysis.

Conclusions

As food production is a means of nutrient acquisition, as well as a component of broader socio-cultural and ecological systems, it is vital that when designing food policy, programs, or aid projects that address food systems inadequacies in Melanesia, they are underpinned by contextual understanding. This must integrate disaster and conflict susceptibility, the impacts of global environmental change and socio-cultural factors, as well as climate suitability. The study highlights the importance to include localised gender dynamics, in the analysis of food systems, though detailed findings in this area are not provided.

Reviewer comment

Highlights resilience of gender norms in the face of climate change. Need to understand and analyse gender norms and dynamics into designing food policies etc.

37. Kamara, J.K., Berhe W. Sahle, Kingsley E. Agho, Andre M. N. Renzaho, "Governments' Policy Response to Drought in Eswatini and Lesotho: A Systematic Review of the Characteristics, Comprehensiveness, and Quality of Existing Policies to Improve Community Resilience to Drought Hazards", *Discrete Dynamics in Nature and Society*, vol. 2020, Article ID 3294614, 17 pages, 2020. <https://doi.org/10.1155/2020/3294614>

Research Topic

This review analyses the characteristics, quality, and comprehensiveness of drought-resilience policies published between 1 January 1980 and 30 June 2019 in Eswatini and Lesotho.

Methodology

The authors searched electronic bibliographic databases and key organisational websites for policy documents published between 1 January 1980 and 30 June 2019. The following key search terms were used: (Disaster * OR hazard * OR drought *) AND (resilient* OR adapt * OR coping OR adjustment * OR coherent *) AND (policy OR Policies * OR rule * OR regulation * OR guideline * OR convention * OR method * OR model * OR framework *) AND (Lesotho OR Swaziland OR Eswatini).

The eligibility of policy documents was determined following a three-stage screening process consistent with the Cochrane guidelines for screening of studies in systematic reviews.

Relevant findings

Nine policy documents suggested that the stress caused by dwindling resources due to recurrent drought aggravates gender inequality and gender-based violence, particularly sexual violence, in affected communities. In addition, women and girls are culturally excluded from control of productive resources and entitlements, which undermines their ability to cope with disaster impacts. The proposed responses to address gender-based violence and inequalities =: providing gender-sensitive guidelines and inclusive participation; accelerating socioeconomic development to redistribute wealth; promoting education attainment for women; training men and women on sexual and reproductive health; empowering women through policies that encourage female participation in political leadership and training; strengthening the capacity of the police, judiciary, and social welfare to promptly and adequately respond to gender-based violence; engaging men and women on sexual reproductive health, marriage, and inheritance laws; reviewing and repealing laws and customs that confine women to minority status; and promoting universal primary education for all children.

Conclusions

The review has found evidence to suggest that gender inequality and gender-based violence are prevalent and aggravated during times of adversity, especially during drought periods. However, none of the responses proposed to educate young boys as allies in prevention of gender inequity. Evidence suggests that engaging young boys in helping to overturn gender inequity and unfair social norms is an effective and sustainable strategy for dealing with the problems. Inequality and exposure to gender-based violence weaken household and community cohesion, as well as the ability to adapt to adverse conditions.

Reviewer comment

It confirms lack of attention to gender-transformative approaches in climate and resilience-focused policies. Highlights increase in GBV especially sexual violence, as competition over resources increases due to climate change.

38. Kandal, H. A., Yacoub, H. A., Gerkema, M. P., & Swart, J. A. A. (2019). Traditional knowledge and community resilience in Wadi Allaqi, Egypt. *Journal of Arid Environments*, 171, [103987].
<https://doi.org/10.1016/j.jaridenv.2019.05.015>

Research Topic

This study investigates the impact of the formation of Lake Nasser in the mid-1960s and the establishment of a new village in the vicinity of the lake in the early 2000s on traditional knowledge of a Bedouin community, with a focus on issues related to rangeland and resettlements. Egypt.

Methodology

This study is based on questionnaires, interviews, and literature. The research was conducted in the Wadi Allaqi area in the Southeastern Desert of Egypt between 2011 and 2015. The questionnaire and interview studies were conducted with Bedouin people settled in the Allaqi Village, as well as with people still living in tents along the shores of Lake Nasser.

Relevant findings

The knowledge base of Bedouin communities have been under pressure since the formation of Lake Nasser in the early 1960s. The emergence of overgrazing and soil degradation resulting from the formation of Lake Nasser, coupled with the impact of climate change are listed as some of the main challenges. Resettled families had to learn how to cope with these circumstances and developed a grazing strategy involving the use of *Najas* species from the lake as fodder, thereby allowing the rangelands on the shore to recover from grazing each year. However, data summarized in the study indicate that men and older people in Wadi Allaqi have more experience-based knowledge on rangeland issues, as compared to women or younger people, respectively. For instance, they had significantly more experience-based knowledge concerning "Best places for making charcoal" and "How to make charcoal". Men also had more knowledge concerning "Best plant species" and "Best grazing areas before Lake Nasser formation" with respect to sheep and goats grazing. Nevertheless, some issues (e.g., traditional medicine, handcrafts, and care for newborn babies) are the distinctive domains of women, while men are more skilled in such matters as treating camel scabies.

Conclusions

The study concludes that, despite the livelihood change experience by the Bedouin community in the process of resettlement and sedentarization they have managed to readapt their knowledge to the new agro-ecological conditions. However, men show to have more experience-based knowledge on rangeland issues, then women.

Reviewer comment

The gender dimension of knowledge is not fully explored. Knowledge is associated with gender division of labour and is experiential (important point, not learned from extensionists), also suggests importance of oral history.

39. Kassie, M., Fisher, M., Murichob G., et al. Women's empowerment boosts the gains in dietary diversity from agricultural technology adoption in rural Kenya. *Food Policy*, August 2020
<https://doi.org/10.1016/j.foodpol.2020.101957>

Research Topic

This paper assesses the moderating effect of women's empowerment on the relationship between agricultural technology adoption and women's dietary diversity. Kenya.

Focal technology: Push-pull technology (PPT) is a cropping system in which cereals such as maize are intercropped with a perennial fodder legume (*Desmodium*) that repels (pushes) stemborers and suppresses *Striga* weed species. A border of perennial (e.g., *Brachiaria* species) fodder grass also surrounds the cereal crops that attract (pulls) stemborers away from cereal plants.

We estimate the impact of women's empowerment on the women's dietary diversity score (WDDS) using the Abbreviated Women's Empowerment in Agriculture Index (A-WEAI).

Methodology

Nine out of 11 counties in western Kenya were purposively selected for the study. Within these, 60 villages and 711 farm households surveyed, comprising 361 adopters/ 350 non-adopters of PPT. 688 female respondents were interviewed to provide data on women's dietary diversity and gender-disaggregated data for computation of A-WEAI. A special module of questionnaire elicited data on seven days' food consumption to capture dietary patterns of mothers. Production module designed within the household questionnaire to capture households' PPT-adoption status.

The conceptual framework adopted presents three testable hypotheses (H) related to the study's research questions (RQ).

- RQ1 and H1: How does women's empowerment influence their dietary diversity? We hypothesize a positive association.
- RQ2 and H2: What is the effect of PPT adoption on women's dietary diversity? A positive association is hypothesized.
- RQ3 and H3: What is the moderating effect of women's empowerment on the relationship between PPT adoption and women's dietary diversity? Researchers hypothesize that women's empowerment boosts the gains in dietary diversity from PPT adoption.

Relevant findings

Hypothesis 1: results show that women's empowerment in agriculture has a positive and significant effect on WDDS for both PPT adopting and non-adopting households.

Hypothesis 2: turning to the impact of PPT adoption on dietary diversity results show a positive and significant impact regardless of women's empowerment status and the measure of dietary diversity used. For households with disempowered women, PPT adoption increases WDDS by 5%. For households with empowered women, PPT adoption increases WDDS by 10%.

Hypothesis 3: results indicate that women's empowerment boosts the WDDS impact of PPT adoption. Disempowered women can achieve higher dietary diversity if their households adopt PPT, but the magnitude of the impact vis-à-vis nutrition benefits is lower than that which empowered women can achieve. A significant and positive association was found between WDDS and women's group membership, income control, ownership of assets, and decision making on credit. By pairing women's empowerment and technology dissemination efforts, the dietary diversity gain increases by three percentage points compared to matching disempowered women with technology adoption.

Conclusions

The results suggest that women's empowerment is an important determinant of women's dietary diversity, which is unsurprising. As women's empowerment levels increase, they have greater knowledge of nutrition and health and decision-making power, income control, and time to exercise that knowledge as care givers. The positive impact of women's empowerment on the WDDS is not only relevant to women themselves but can also impact the health of their family and break the inter-generational transmission of malnutrition. The study also finds evidence that the interaction of women's empowerment with technology adoption further increases dietary diversity. The adoption of PPT has a positive nutrition effect regardless of empowerment status, but its effect is more visible in households with empowered women. This result has two policy implications. First, policies aimed at achieving higher women's dietary diversity should consider pairing technology dissemination efforts with women's empowerment interventions. Second, development partners and policymakers can also enhance disempowered women's dietary diversity through the diffusion of economic and nutrition enhancing technologies such as PPT.

Reviewer comment

Women's dietary diversity is higher when women are empowered. Technologies like PPT can strengthen these effects. It is recommended to pair technology dissemination with women's empowerment interventions.

40. Kgosikoma, K.R., and Phatsimo Cotildah Lekota (2018) Agro-pastoralists' determinants of adaptation to climate change. Department of Agricultural Economics, Education and Extension, Faculty of Agriculture, Botswana University of Agriculture and Natural Resources, Gaborone, Botswana, and International Journal of Climate Change Strategies and Management Vol. 10 No. 3, 2018

Research Topic

The purpose of this study is to analyze smallholder farmers' perceptions of climate change and its stressors, their adaptation strategies and factors that influence them. Country focus is Botswana.

Methodology

The study conducted in Kweneng district, located in the southeastern part of Botswana. Multi-stage sampling was used to obtain a representative sample from three subdistricts in the district. A structured questionnaire was used to collect data by using face-to-face interviews.

Relevant findings

The study summarises key findings related to farmers' perceptions of climate change and its stressors (e.g. recurrent drought, change of rainfall patterns and related decline in productivity) as well as the adaptation strategy most commonly used by farmers. Gender was found to be a key Determinant of adaptation to climate change. However, the only finding mentioned in the study is that female-headed households are 16 per cent more likely to adapt to climate change than male-headed households. This was partly attributed to willingness of women to change their livelihood strategy in an effort to support their families.

Conclusions

Gender, along with age, household size, poverty and lack of access to credit significantly and negatively affect adaptation to climate change. Policy recommendations focusing on gender are not provided.

Reviewer comment

Gender-related findings with regard to adaptation strategies are not further explored. The hypothesis is that women's care-giver role drives their willingness to adapt their livelihood strategies. This hypothesis is not further discussed or validated with data and evidence.

41. Khoza, S., Leon Tielman de Beer, Dewald van Niekerk & Livhuwani Nemakonde (2021) A gender-differentiated analysis of climate-smart agriculture adoption by smallholder farmers: application of the extended technology acceptance model, Gender, Technology and Development, 25:1, 1-21, DOI: 10.1080/09718524.2020.1830338

Research Topic

This research assesses the socio-psychological processes and determinants that influence climate-smart agriculture (CSA) adoption decisions among men and women smallholder-farmers in Malawi and Zambia. The study tested the applicability of socio-psychological theory in ascertaining intricate gender-differentiated behavior and attitudes shaping farmers' micro-level decision-making in CSA adoption.

Methodology

An exploratory-sequential mixed methods research design informed the collection of qualitative and quantitative data. Key informant interviews and focus group discussions were conducted in the preliminary qualitative phase that was subsequently followed by a cross-sectional survey in the quantitative phase. Mixed methods sampling was used where for the qualitative phase. In the quantitative phase, random sampling was used to select households to participate in the study.

Relevant findings

Beliefs determining CSA adoption

Perceived ease of use: As women farmers are the primary labor providers even in male-headed households, CSA adoption is likely when they perceive that the technologies will be free from additional drudgery.

Perceived usefulness: If women view CSA technology as enhancing the performance of their farming activities, they were likely to want to adopt and continue using it in the future.

Perceived climate risk: Climate risk seems more significant to men than women and results could suggest that men were more likely to share and influence each other to adopt CSA based on PCR. This could also be related to the skewed access to climate change and risk information which is more accessible for men than women.

Cognitive processes determining CSA adoption

Technology relevance: Farmers' decisions to adopt CSA or not was informed by their consideration of whether a particular CSA technology was relevant and applicable to their situations and contexts. For instance, treadle pumps were not a preferred technology for married women because the pumping routine make them too tired to fulfill their conjugal role, resulting in fights.

Tangible benefits: Farmers, men and women, place importance on the need to see tangible benefits from CSA for them to adopt.

Experience: For men, positive experience with other CSA technologies was key in adoption of new agricultural technologies. The lack of local viable markets again brings to light issues of gendered-vulnerability when the mobility of women is compared against men and underlying risk drivers, such as poverty, which could be tackled through income-generation in CSA.

Intention and actual usage behavior

Compared to men, social processes were more likely to determine CSA adoption for women.

Conclusions

The hypothesis that the socio-psychological determinants of CSA adoption are gender-differentiated holds.

Reviewer comment

The authors argue that for practitioners and policymakers there is a critical need for gender-specific behavioral change communication strategies and inclusive participatory engagement.

42. Khoza, Sizwile (2021) "Gender mainstreaming in risk reduction and resilience-building strategies: local conceptualisation of gender and masculinities in Malawi and Zambia." Disaster Prevention and Management: An International Journal (2021). <https://doi.org/10.1108/DPM-03-2021-0066>

Research Topic

This paper looks at local conceptualisations of gender and framings of men and masculinities, at a local level in Malawi and Zambia. The aim is to improve the way that gender is conceptualised and framed in work on gender mainstreaming in smallholder farming.

Methodology

Qualitative data from 70 key informants from relevant government departments, NGOs and traditional leadership, and community members. Data was captured through 16 KIs and 6 FGDs in two study sites, with an average of 9 participants per group. Study sites both have high levels of food insecurity and disaster risk, with communities vulnerable to climatic hazards.

Relevant findings

The paper asserts in its analysis of gender framing that currently, the gender lens used in development is imposed on communities from a Western perspective and there is not enough emphasis on gender relations, nor on the potential of positive masculinities to 'contribute towards gender transformation and achievement of gender-equitable resistance' (p.225). It raises questions about who constructs gender and why they do it.

"...in neglecting local perspectives, the gender lens applied in development fails to allow the communities to hold the lens and look through it themselves, drawing from their cultures, histories and norms. [...] Ignoring gender relations existing at a local level may catalyse hostility between men and women, which may arise when researchers and practitioners only rely on external fabrications of gender. For this reason, there is growing advocacy for applying African feminist theories underpinned by an understanding of the local level gender relations and how African women tackle patriarchy and disempowerment." (p.217)

Gender mainstreaming in Disaster Risk Reduction (DRR) needs to consider the colonial foundations of gender inequality as gender is not ahistorical (p.218)

The field research looks at three areas – the understanding of gender at a local level, the roles and responsibilities of institutions on gender mainstreaming, and men and masculinities at a local level. It found that the conception of gender among KIIS was informed by traditional approaches to gender and donor requirements, but that they raised questions about the current conceptualisation of gender used in projects and how effective it is.

Community members spoke of the importance of men and women working together 'to share the burden of the challenges from climate change' and raise concerns about the impact of men's alcoholism on their capacity to collaborate.

The roles and responsibilities have of institutions were understood to have been to empower women, through women-targeted approaches, but key informants also question this focus and raise the importance of encouraging women and men to work together.

The FGDs highlighted how subordinate and marginalised masculinities in the community are missed (e.g. unmarried men are not included in cookstove initiatives or FGDs), and the relationship between insecurities among subordinate men and the restriction of women's participation in training and income generation. Examples given were of men who don't want women to travel to markets in town because they fear that they will be seduced by richer urban men. The research also highlights how women were conceived as reinforcing patriarchy by preferring to elect male leaders. It considers that this may be women 'conforming to their reality' because of the norm of households being led by men.

Conclusions

Local communities targeted by gender mainstreaming need to be recognised as the main actors who shape gender and any approaches that exclude local perspectives and import external ideologies are likely to fail. (p.224)

Positive masculinities that harness men's abilities to be caring, non-violent, cooperative, and dependable will contribute towards gender transformation and achievement of gender-equitable resilience. (p.225)

...there is growing advocacy to decolonise gender in international development. Unfortunately, this is almost impossible without decolonising the institutions that implement global approaches down to the local level. (p.225)

Reviewer comment

This paper highlights the importance of locally led definitions of gender and the potential of work on masculinities and jointness (collaboration between women and men), all focal areas of GTAs in agriculture.

The highlighting of gender in relation to colonisation was also interesting, in particular the assertion on P.218 that 'colonialism entrenched hegemonic patriarchy in the family system', through process of labour migration and the creation of income earning opportunities in urban centres, and through legislation on land ownership, marriage and property inheritance. This ties in with other literature on the labour reserves.

43. Knippenberg, E., Jensen, J., Conostas, M., (2019) Quantifying household resilience with high frequency data: Temporal dynamics and methodological options. World Development, Volume 121, 2019, p. 1-15, <https://doi.org/10.1016/j.worlddev.2019.04.010>

Research Topic

Methods for measuring resilience, exploring three conceptual approaches that draw on the same data set captured in a highly food insecure population in Malawi.

Methodology

It uses a data set collected under the 'Measuring Indicators for Resilience Analysis' (MIRA) project, piloted in Malawi during a humanitarian emergency in 2016/2017. The project developed a short monthly household survey using enumerators embedded in communities to track the adverse effects of self-reported shocks and indicators of food security, over the course of one year. It was preceded by a larger baseline survey to allow researchers to investigate how time-invariant household characteristics affect the dynamics of shocks and food insecurity. A total of 576 households were included from 31 villages, in a mix of high-risk flood zones and low risk non flood zones in the Shire River watershed.

Relevant findings

P.2 The paper looks at the data set through three conceptual lenses based on literature:

- 1) A household's assessment of the persistence of shocks on their own welfare
- 2) The likelihood that a household's characteristics fall below a normative threshold of welfare.
- 3) The set of household characteristics that best predict future wellbeing.

Across all three, it evaluates the Coping Strategy Index (CSI), a composite weighted score of various strategies households engage in when faced with short-term food shortages.

The different lenses offered complementary insights but the paper highlights how different lenses have more or less relevance for different types of interventions and decisions.

With regards to gender, it was found that female headed households were impacted by illness more than male headed. This could relate to a common constraint faced by female-headed households of available family labour. In other respects, female-headed households were not shown to have lower resilience to shocks than male-headed. In fact, the female headed households were shown to be better off in the midst of a food emergency when data collection began, than male-headed (p.6, 3.3) though the reasons for this are not analysed in this paper. The trend then converges between male and female headed households over the year.

Conclusions

From abstract: Across approaches, our study consistently finds that shocks and food insecurity are very persistent, and that households living in the flood plains are more resilient. When focusing on specific shocks such as illness, the gender of the household head matters as well.

The paper demonstrates how multiple complementary approaches can identify different but overlapping factors related to resilience, painting a fuller, richer picture. It also demonstrates the empirical benefits derived from using high-frequency data sets in the study of resilience.

Reviewer comment

The focus of this paper is primarily methodological. It does not offer detailed insight into gender norms in relation to economic resilience. Its interest is more in how resilience is framed and measured. The aspects that were of interest were:

Its innovation in finding a cost-effective way of collecting monthly data, using enumerators in the community and a short survey (10-15), allowing for household level analysis.

Its highlighting of the duration of the impact of a shock as a useful measure of resilience, "A household may experience the effects of drought long after the objectively measured drought is over."

The need for subjective measures of shocks and recovery: “the household’s perception of shocks, which allows us to use a household’s own internal accounting to approximate how that shock affects a household’s wellbeing.” (p.6)

44. Lawless S, Philippa J. Cohen, Cynthia McDougall, Sangeeta Mangubhai, Andrew M. Song, Tiffany H. Morrison, 2022. Tinker, tailor or transform: Gender equality amidst social-ecological change, *Global Environmental Change*, Volume 72, 2022, 102434, ISSN 0959-3780, <https://doi.org/10.1016/j.gloenvcha.2021.102434>.

Research Topic

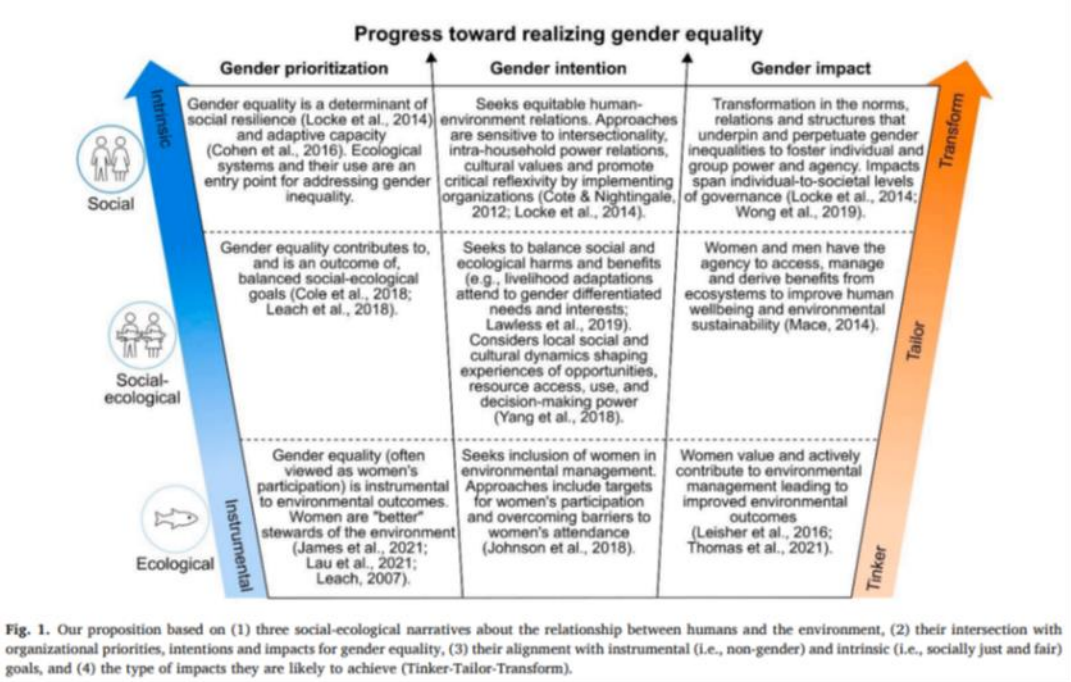
Fisheries in the Pacific - Fiji, Solomon Islands and Vanuatu.

We use the case of small-scale fisheries in the Pacific Islands region to explore the proposition that different social-ecological narratives: (1) ecological, (2) socialecological, and (3) social, shape the gender equality priorities, intentions and impacts of implementing organizations.

Methodology

Interviews with regional and national fisheries experts (n = 71) and analysis of gender commitments made within policies (n = 29) that influence small-scale fisheries. To explore these data, we developed a ‘Tinker-Tailor-Transform’ gender assessment typology (p.1).

Through examining the policies and practices that influence small-sale fisheries, we assess the efficacy of gender equality priorities, intentions and impacts. By ‘policy’ we refer to written binding and non-binding regional, national and organizational gender equality commitments made to small-scale fisheries. By ‘practice’ we refer to the actualization of commitments, for instance via extension services, development projects or investments offered by regional agencies, national governments, NGOs, the private sector, and/or civil society, henceforth, ‘implementing organizations’ (p2). Figure below p3, and on page 10, results are mapped against this.



Relevant findings

1. Implementing organizations need to question and potentially reorient their theories of change. For example, under the ecological narrative, gender approaches may be perfectly executed to increase the attendance (i.e., physical presence) of women in environmental management efforts, and this may indicate progress toward more gender-equal participation. However, without commitment to intrinsic outcomes (i.e., justice, equity or empowerment), these efforts are only likely to achieve the ecological

narratives' (limited) visions and ambitions for gender equality (refs). Reorientation toward the intrinsic value of gender equality is important because when fundamental human rights, including gender equality are secured, all people are far more able to benefit from natural resources and efforts to manage them.

2. Gender approaches need to move away from a narrow focus on reaching greater numbers of women, to multiple dimensions of empowerment (refs). This step may require reimagining what gender equality 'success' looks like and how it should be measured. To do so, well-established and tested frameworks including the Women's Empowerment in Agriculture Index (ref), the 'Reach, Benefit, Empower, Transform' framework (ref) and the gender integration continuum (ref) offer guidance. To move toward deep and sustained social change, these approaches need to challenge and reconfigure system attributes that perpetuate gender inequalities (McDougall et al., 2021). For example, efforts must also be directed across all levels of governance, and influence the informal (e.g., values, attitudes, beliefs and skills) and formal (e.g., laws and policy) domains (ref). This degree of change involves addressing inequalities in gender power relations at all levels.

3. Implementing organizations need to consciously interrogate their own discursive positions and beliefs to unsettle habituated ways of thinking and acting (refs). Our analysis revealed instances where fisheries officers came to 'realize the value' of gender work when given the opportunity to engage with external experts, trainings or undertake gender research. This form of self reflexivity (or internal process of inquiry) is an opportunity for people to see the world differently, adjust their frames and openness to gender sensitivities (ref) p.12.

Conclusions

Explored the proposition that differences in social-ecological narratives shape gender equality engagement and impact. To perform this task, we developed the 'Tinker-Tailor-Transform' gender assessment typology and used it to analyse policy and practice impacting upon small-scale fisheries in the Pacific Islands region. Respondents indicated a range of successes in progressing gender equality across ecological, social-ecological and social narratives. This progress tended to be related to the greater inclusion of women and consideration of their needs and interests. Accounts from this research, and broader evidence, suggest that this type of progress may make certain experiences of livelihoods, management and governance efforts linked to small-scale fisheries more equitable, and provide important precursors to deeper social change. However, we found most approaches and impacts to be modest relative to what we know from gender and development literature to be possible and needed. The modesty of progress and approaches was surprising given the surrounding dominant rhetoric about the importance of social change (i.e., human development) in the social-ecological and social narratives. The methodology we apply, including the 'Tinker-Tailor-Transform' typology we develop, offers a critical tool for implementing organizations to engage in more self-reflexive processes, reorient toward more intrinsic visions of gender equality, embrace more ambitious (and current) theories of change, and measure and assess progress against more robust indicators of impact (p.13).

Reviewer comment

The Tinker Tailor Transform methodology is very rich and valuable and we can consider this as one of our frameworks for understanding how implementing agencies conceptualize and work with gender unequal norms. This article raises fundamental questions about how to link gender transformation with ecological approaches (which is of great relevance to our work if we are to say that we think they are in some ways mutually supportive). The whole article would need to be read again carefully as it is not possible to bring out all their points. However, I have to say that I am not clear that they have made the point that gender transformation can help meet ecological goals. Perhaps they do not want to make this point, but it seems vital to me, and I think we need more understanding if ecological goals in themselves are only instrumentalist. How are we to support them? A keynote paper.

45. Liru, P.; Heineken, L. Building Resilience: The Gendered Effect of Climate Change on Food Security and Sovereignty in Kakamega-Kenya. Sustainability 2021, 13, 3751. <https://doi.org/10.3390/su13073751>

Research Topic

Climate change and resilience theory, Kenya.

Using resilience theory as a lens, we established that women use different pathways to mitigate the effects of climate change on their livelihoods.

Methodology

Resilience-building refers to the ability of people to cope with the stresses and shocks of climate change by anticipating, preparing for, responding to and recovering from them [30]. There are three different pathways that farmers use, namely developing their absorptive coping capacity, adaptive capacity and transformative capacity. (p.2)

1. Absorptive resilience refers to the ability of individuals to cope with the impacts of climate change on their livelihoods within their immediate means. This is reactive and often short-lived, especially if subjected to successive shocks.
2. Consequently, individuals need to develop their adaptive capacity, which refers to their ability to manage future risks and bounce back when shocks occur.
3. Last is the need to develop transformative resilience in terms of access to different resources, as well as change certain practices that enable them to move beyond the vulnerability threshold.

Relevant findings

Changes in weather patterns affect the availability of food, access to food, utilization of food, the stability of food supply and nutrition of people. All these elements of food security are gendered due to women's role in agriculture, access to resources, and the differentiated gender roles and responsibilities expected of them. (p.2)

Climate change has profound impacts on women, due to their reliance on the natural environment for survival and limitations placed on their ability to access different capitals due to patriarchal societal values. Women farmers typically lack the natural, financial and physical capital of men to augment their capacity to adapt, but as this study has shown, they do not lack agency. They use and have come to expand both their human and social capital to mitigate the effects of climate change on their livelihoods. All the three forms of resilience essential for livelihood adaptation were evident, which served to not only alleviate the impact of climate change on their livelihoods but enhance the food security and sovereignty of the rural women farmers. (p.10)

- At first, women responded to climate change through adopting various coping strategies to deal with food insecurity. However, these were merely absorptive coping strategies where they utilized their existing assets to survive, which were not sustainable and were typically over-reliant on the natural environment. These strategies did little to improve the food security or food sovereignty of women and increased their vulnerability to further climate-related shocks. However, using their indigenous knowledge of weather patterns and farming, the women incrementally adapted to the effects of climate change.
- The adaptive strategies were more proactive, where the women used intentional practices to cope with known shocks and stresses. Adaptive capacity emphasizes flexibility, and the capacity of a system or an individual to make incremental changes in terms of continuous adjustment, learning, and innovation. In terms of adaptive strategies, through the various women's groups and community organizations, women in Kakamega share knowledge and decide on what climate-resilient crops to plant based on their shared knowledge and prediction of weather patterns, as a kind of insurance against unpredictable climatic conditions. For example, they have moved towards a greater integration of crops with livestock production to enhance productivity, efficiency and sustainability. The ability to share financial and technological knowledge and other forms of human capital influences the effectiveness of this adaptation strategy. In the process, women become both the producers and practitioners of indigenous climate knowledge. While this did improve their food security and sovereignty, these practices remained over-reliant on natural resources and were influenced by broader patriarchal relations that affected their access to various capitals.
- Transformative strategies of building resilience were observed among women farmers in Kakamega, which involved using their agency to make intentional changes or arrangements to halt or lessen the risk factors of climate change, including their vulnerability, poverty and inequality that pre-dispose them to climate-related shocks. Here, women's social capital was key to building the resilience of rural women, who used their social capital to gain access to financial, natural and physical capital that enabled them to collectively deal with the impact of climate change. Women's groups linked women farmers to institutions that provide technical climate information and how to adapt to changes in weather patterns. While institutions may provide advice on climate change and how to act, the literature has proved that there are different levels of uptake. An important finding of this study was just how important the links between governmental institutions, civil organizations and communities are to enable people to cope but also challenge gender inequality and patriarchal value systems (here follow examples).

Conclusions

The study found that initially women adopt coping strategies that are reactive and not sustainable, but soon adapted their farming strategies, using their indigenous knowledge to exercise some control over both their food security and food sovereignty. Besides this, they use their human and social capital to expand their networks of support. By linking up to other organizations and gaining access to government support, they are able to challenge patriarchal relations that perpetuate poverty and inequality and bring about more transformative and sustainable responses to climate change.

Reviewer comment

So far, this is the best article of all. It demonstrates women's sharp awareness of climate change and provides significant detail (not cited above) as to women's challenges and how they are responding, and also shows (linking up to earlier papers) that some strategies create other problems, particularly their coping strategies (absorptive resilience).

Again - and this is powerful - there is a very strong message that women diversify crops, diversify their varieties, etc and are actually much more effective than men (who may plant one kind of sugar cane only) in managing the vagaries of weather etc caused by climate change. They are truly more resilient. The need to provide, process and prepare food for the family is the driving force.

Hypothesis (from me) could be that men-led farming systems, supported by almost the entire extension system and its many actors, are leading to catastrophe and destroying agro-ecosystem functionality. Whereas women, due to their marginalization, are actually forced into resilience by drawing on their indigenous knowledge, creating new ITK through their close observation of weather and agro-ecology, etc.

The article makes the point that I have also been supporting for some years, that women's agency unlocks resources, and that they need institutional support for this.

46. Lottering S.J., P. Mafongoya, R.T. Lottering. 2021. Assessing the social vulnerability of small-scale farmers to drought in uMzinga, KwaZulu-Natal, International Journal of Disaster Risk Reduction, Volume 65,2021,102568,ISSN 2212-4209,<https://doi.org/10.1016/j.ijdr.2021.102568>.

Research Topic

"The effect of climatic events such as droughts, will have severe and disproportionate impacts on poor rural communities and small-scale farmers who base their livelihoods on rain fed agriculture. These groups are considered socially vulnerable to climate change, due to various socio-economic conditions that increase their susceptibility to droughts, specifically socio-economic droughts which are associated with a failure of water resources to meet water demands leading to adverse impacts on humans and the economy of a given region. This has led to a growing emphasis on research assessing the social factors that sustain a community's vulnerability to natural hazards. The aim of this article was to assess and identify social vulnerability amongst small-scale farmers and rural communities in the uMzinga community in the KwaZulu Natal province of South Africa..." (p.1)

Methodology

"...and rural communities in the uMzinga community in the KwaZulu Natal province of South Africa using an adapted social vulnerability index (SoVI). Nine social indicators were identified using qualitative and quantitative research methods. Indicators were identified from questionnaire surveys and literature that has proven the role that certain social factors play in sustaining vulnerability."

They construct a social vulnerability index described in some detail in the article. 85% of respondents were women. They used FGDs as well as surveys. (p.1).

Relevant findings

"[49] and Rogan [50]; that reveal there are large poverty gaps between males and females in South Africa. About, 56.8% of female headed households were poor compared to 36.3% of male headed households." (p.7)

"...collection and their responsibility for providing food, places them at a greater risk when drought occurs [51]. In addition, the patriarchal nature of men in rural areas prohibits women from having any decision-making powers, which constraints their ability to adapt to changing climates. Women are generally underrepresented and have little influence over community strategies [52]. As result, policies that may be presented to a community to replace traditional crops with better suited crops for periods of droughts, may only be focused on the needs of the male's fields and not address the needs and priorities of women in their household gardens. These factors contribute significantly to the social vulnerability of women to droughts. Information obtained from (FGDs), with female small-scale farmers suggested that females are largely excluded from decision making and various development programmes, which is of great concern considering that a majority of the farmers are females."

Conclusions

Women are majority of farmers but have higher vulnerability than men. They are not part of community decision-making networks.

Reviewer comment

A good review of social vulnerability network but not very much insight into gender relations. They even talk about "gender headed households"! But this article confirms that women are more vulnerable because they are excluded from decision-making networks and development programmes.

47. Lwanga-Ntale, C. & Owino, B.O., 2020, 'Understanding vulnerability and resilience in Somalia', Jàmbá: Journal of Disaster Risk Studies 12(1), a856. <https://doi.org/10.4102/jamba.v12i1.856>

Research Topic

Explores experiences of Somalis during the 2011 and 2016 crises, examining the link between vulnerability and resilience, and the role played by international humanitarian responders in resilience building. The aim is to provide information on how different population groups responded to and managed to survive recurrent shocks; the prevailing drivers of marginalisation and exclusion; mechanisms through which these are maintained; and the role of external stakeholders.

Methodology

A review of published and grey literature preceded fieldwork and focused on risk, vulnerability and resilience during crises in the study period. Field consultations were carried out in four study sites: Kismayo Urban, Kismayo Rural, Baidoa and Beledweyne. Stratified random sampling was used to identify and select respondents and ensure inclusion of all groups (youth, older persons, women, minority clans and persons with disabilities). Key informant interviews were also carried out. Wide range of participatory research methods, including participant observation, focus group discussions (FGDs), household dialogues, livelihood analysis, well-being analysis and gender analysis.

Relevant findings

Households adopted different coping strategies depending on their resource endowments. These included social and organisational coping strategies, divesting of non-essential domestic assets; and diversification of income generation and food production strategies. Marginalisation and exclusion of minority clans, women and persons with disabilities in access to resources and participation in decision-making processes continue to undermine efforts to build household resilience in Somalia. The culture of patriarchy and gender discrimination was found to limit the participation of women in socio-economic activities. Cash transfer programmes targeting women were the major investments by INGOs in order to enable households to cope with the crises and to strengthen their resilience to future shocks.

Conclusions

Patriarchal norms hinder women's capacity to build resilience. Government and development actors need to coordinate efforts to deliver tailored services to marginalised social groups including women.

Reviewer comment

Norms continue to marginalize women and affect their earning capacity. There is a lack of effective institutional support.

48. Maharani, C. D., Moeliono, M., Wong, G. Y., Brockhaus, M., Carmenta, R., & Kallio, M. (2019). Development and equity: A gendered inquiry in a swidden landscape. *Forest Policy and Economics*, 101, 120-128. <https://doi.org/10.1016/j.forpol.2018.11.002>

Research Topic

Market-driven development is transforming swidden landscapes and having different impacts along intersections of gender, age and class. In Kapuas Hulu, West Kalimantan, Indonesia, Dayak communities practicing swidden agriculture are making choices on maintaining traditional land use systems, and engaging in rubber, oil palm and conservation (REDD+) in their livelihood strategies. Although REDD+ has been heralded as an alternative to oil palm as a sustainable development option, it is still far from full implementation. Meanwhile, oil palm has become a reality, with large scale plantations that offer job opportunities and produce new sources of prestige but create contestations around traditional land use systems. (p.120)

The article provides evidence for often negative change in women's access to assets, due to oil palm, and is concerned that REDD+ will build on the more inequitable gender relations now existing rather than challenging. It provides suggestions for gender-equitable REDD+.

Methodology

Combines sustainable livelihood framework (SLF) focus on assets with the gender asset agriculture project framework (GAAP) to understand how (larger) agricultural development interventions impact the gendered distribution of assets. Key assets studied are - land, labour, social capital. NB - (they explore) the intangible assets of social capital, which implies the capability to access and use land and labour, under the unique cultural context of the swidden Dayak community. (p.121)

This study is also historical. It compares livelihood strategies over time and especially how gendered access to the selected assets has changed over time, in four communities which traditionally practice swidden agriculture but are now moving into oil palm (particularly).

Swidden = "land use system employs a natural or improved fallow phase, which is longer than the cultivation phase of annual crops, sufficiently long to be dominated by woody vegetation, and cleared by means of fire." (p.120).

Relevant findings

Has the development of rubber and oil palm affected the gendered gap in access and rights to land? In the traditional swidden system, everyone within a lineage, regardless of gender, inherits an equal access to land as long as it is used for communal practice such as swidden. When swidden land is converted to cash cropping, access to land becomes contested. State structures, in formalizing smallholder plots, create a hierarchy that favors men, and individuals with greater access to information and more knowledgeable regarding the inheritance structures have power in land conversion and land use. They tend to be the first in claiming individual rights to communal swidden land by converting to cash crops or trees. In the process, this often left women with less access and rights.

Has the development of rubber and oil palm crops affected the gendered gap in labour? The redeployment of labour from swidden agriculture to engage with oil palm does not seem to affect men negatively, as most of their labour contributions have been substituted by agricultural inputs and technology. Such changes, however, widened the gendered access to labour for swidden. Despite the use of agricultural inputs, women's labour in swidden remains unchanged, and has even increased due to the environmental consequences of shortened fallows. This has constrained their engagement in wage labour for oil palm as casual workers, which does not provide adequate compensation and benefits. In the case of rubber, because of the nature of its production nature and proximity to longhouse, women have the freedom to manage and balance their time to both rubber and swidden production, but this has also increased the amount of work that women bear.

Has the interaction with oil palm and rubber affected gendered gap to social capital? Traditionally, men through bejalai have had wider access to more diverse social networks in comparison to women whose networks were often limited to the longhouse. For women in a remote longhouse, improved road access facilitated connections to itinerant traders. In spite of this, the gendered gap to social capital has not necessarily diminished. Our finding showed how reliance on itinerant traders and asymmetric information on price and has in fact lowered women's bargaining power. The finding also showed the widening gap of access to traditional labor network exchanges of beduruk, and its inherent social network, as the system is disrupted by men's labour redeployment for oil palm. (p.126)

Conclusions

The authors recognize the potential value of REDD+ particularly in the context of oil palm (p.126 – three potential scenarios, potential gender costs and benefits of each). This should be reviewed again as it is closely argued.

Even though REDD+ processes involve safeguards to ensure social and gender equity, these have been narrowly implemented and largely relegated as checklist exercises without a clear strategy for addressing gendered power relations (refs). Consequently, REDD+ can be easily turned into a top-down approach that supports existing socially constructed gender relations, where men typically represent the household. Information is not always shared with wives, and women's access gap to social capital (in the form of a network of information sharing) could be widened. To ensure equal access to REDD+ information, it is important to explore the different social foci where information is traditionally shared and to create opportunities and safe spaces for dissent and resistance.

Reviewer comment

A keynote paper. This paper shows how “traditional” ways of managing assets through swidden agriculture, the importance of the longhouse, etc, were significantly changed through the building of roads and the introduction of oil palm. Although women managed to secure some benefits, overall oil palm has been directed at men and men have benefited. In other words, gender relations have changed substantially. REDD+ risks being introduced and implemented on top of this increasingly gender-equitable situation. Typical REDD+ processes for addressing gender are unlikely to work because they are too narrow.

49. Maharjan, A., Ishaan Kochhar, Vishwas Sudhir Chitale, Abid Hussain, Giovanna Gioli, (2020) Understanding rural outmigration and agricultural land use change in the Gandaki Basin, Nepal, *Applied Geography*, Volume 124, 2020, 102278, ISSN 0143-6228, <https://doi.org/10.1016/j.apgeog.2020.102278>.

Research Topic

This study investigates agricultural land use change during the period 1990-2010 and empirically tests the effects of migration on these changes at the meso scale – VDC level. It also examines the differentiated effects of various types of migration and the gendered impacts of migration on agricultural land use change. Nepal.

Methodology

The research adopted a multi-disciplinary approach integrating macroscale longitudinal geospatial analysis with qualitative methods (key informant interviews and focus group discussions conducted in 2017) and quantitative econometric analysis. Census 2011 was used for econometric analysis. Such longitudinal approach helps in better understanding agricultural land use change in the context of population and environmental change. The primary data collected via qualitative methods is used to validate the relevance of the findings of the quantitative analysis in the period 2010-2017, and provides an in-depth view on local perceptions of the issues under study. The geospatial analysis was based on The LULC dataset - developed by the International Centre for Integrated Mountain Development (ICIMOD).

Relevant findings

The analysis of agricultural land use change showed a declining trend in agricultural area in Lamjung and Nuwakot districts. Among the main variables, internal migration of men and women showed a statistically significant relationship with agricultural land contraction. However, male internal migration showed the negative relationship, and internal female migration showed the positive relationship. When men migrate the remaining family members, particularly women, continue to cultivate the land. However, when women migrate, only older people are left in the household, thus leading to abandonment of agricultural work.

For instance, in Lamjung district, the development of hydropower plants has provided employment opportunities for men, while their wives have migrated to urban areas, thus abandoning agricultural land. Nonetheless, the major role played by women in agriculture clearly emerged in the gender-differentiated views they held on the performance and future of agriculture. Men tended to be more dismissive and less interested in the future of agriculture. Women's answers were far more concrete and hopeful.

Conclusions

Findings show that when men migrate internally, it leads to agricultural intensification, whereas internal migration of women leads to agricultural land abandonment, confirming the crucial role of women in agriculture. Future agricultural development in Nepal should place women at the centre of agricultural planning and policy. Efforts are necessary to promote gender-friendly technologies in agriculture and build their overall capacities to increase the productivity of available agricultural land and restore abandoned land. This is crucial particularly in the context of Nepal, where climate change, land scarcity and abandonment and increasing dependence on food imports, can jeopardize the food security situation of the country.

Reviewer comment

This paper shows the centrality of women to agriculture. When they migrate, agriculture collapses. Yet women face many challenges including suitable technologies.

50. Makate, C. Marshall Makate & Nelson Mango (2017) Sustainable agriculture practices and livelihoods in pro-poor smallholder farming systems in southern Africa, African Journal of Science, Technology, Innovation and Development, 9:3, 269-279, DOI: 10.1080/20421338.2017.1322350

Research Topic

"Climate variability and change threaten and impact negatively on biodiversity, agricultural sustainability, ecosystems, and economic and social structures - factors that are all vital for human resilience and wellbeing. To cope with these challenges, embracing sustainability in food production is therefore essential. Practising sustainable agriculture is one way of ensuring sustainability in pro-poor farming communities in low-income countries. Sustainable agricultural practices (SAPS) are those practices enabling farmers to meet current and future societal needs for food, fibre, ecosystem services and healthy lives. This study evaluates the dynamics of farm-level adoption of sustainable agriculture practices and their effects on maize productivity, crop income and food adequacy, using data from the Chinyanja Triangle (CT) in southern Africa." (p.1)

Methodology

"This study evaluates the dynamics of farm-level adoption of sustainable agriculture practices and their effects on maize productivity, crop income and food adequacy, using data from the Chinyanja Triangle in southern Africa. We apply joint estimation techniques to estimate consistently the impact of sustainable agriculture practices on maize productivity, crop income and food adequacy."

Relevant findings

"Households with membership of community groups within the CT and those that had access to agricultural information were 45.5 percentage points more likely to adopt SAPS than their counterparts who were significant at 5%. Consistent with findings of Teklewold, Kassie, and Shiferaw (2013), Isham (2002), Marenja and Barrett (2007), our results confirmed the importance of access to social capital as a determinant of technology adoption, including SAPS."

"Our results show that within our sample from the CT, female headed households were less likely to adopt SAP packages (as indicated by the SAP index). This result could be owing to the possibility that women might have less access to resources such as land, education and information on improved agricultural technologies which may lower their chances of comprehensively adopting SAPS as compared with men (Doss and Morris 2000). Our results are consistent with other findings (Doss and Morris 2000; Manda et al. 2016) on SAPS adoption." (p.6)

Conclusions

"We established that the uptake of sustainable agriculture practices significantly improves productivity, income and food adequacy. The systematic targeting of reducing gender-related imbalances, enhancing ties to boost access to social capital, diversifying revenue sources to improve wealth and facilitating access to other strategic resources can conclusively enhance the uptake of sustainable agriculture practices, thus improving livelihoods. We recommend the integration of sustainable agriculture practices into rural development policy frameworks in southern Africa."

Reviewer comment

This article confirms other data that women face a number of GBCs to adopting sustainable agriculture practices, but it does not really explain why or analyse these constraints in sufficient depth.

51. Maltitz, L.V & Bahta, Y.T. Empowerment of smallholder female livestock farmers and its potential impacts to their resilience to agricultural drought. In *AIMS Agriculture and Food*, 6(2): 603–630. May 2021. DOI: 10.3934/agrfood.2021036

Research Topic

Mixed approach that includes a survey, A-WEAI, Pearson's chi-square coefficient, and reflective dialogue to analyze the gender-specific factors that empower women and men to adapt to agricultural drought. South Africa.

Methodology

Mixed-method study, a multi-stage sampling. The 127 respondents selected from a list of female smallholder livestock farmers provided by the Northern Cape Department of Agriculture, Forestry, and Fisheries (DAFF). A questionnaire was constructed based on the A-WEAI. To calculate the Gender Parity Index (GPI), the focus was on married women or women in a permanent relationship with men who were also actively involved in the farming operation. Additionally, data were collected from single, divorced, and widowed women to determine whether there was any relationship between marital status and resilience to drought. To determine the empowerment status of the female livestock farmers who participated in the study, the A-WEAI was applied. The Pearson's chi-coefficient was used to determine any possible correlation between other factors not included in the A-WEAI and empowerment. Reflective dialogue with women carried out to gather information on specific challenges.

Relevant findings

The A-WEAI shows that 38.7% of the women and 83.6% of the men were empowered in the five domains of empowerment. The dimensions contributing most to disempowerment of female smallholder livestock farmers are excessive workload (52.97%) and access to credit decisions (17.7%). The restriction on time has spill-over effects such as lack of information (less time to access information) that in turn has a negative effect on knowledge (such as new technologies etc.), which can be used to adapt to drought and improve resilience. Lack of ownership of assets (11.3%), lack of input into productive decisions (10%), and lack of group membership (8%) also contribute to their disempowerment.

Social or cultural norms regarding gender division of labour and asset ownership have a definitive influence on women's disempowerment. No significant relationship was found between age, marital status, education level of the women and their empowerment status. Conversely a significant relationship emerged between the farming experience of the women and their empowerment status, which implied that experience added to knowledge and resilience.

During dialogues, engaged female farmers voiced their challenges and concerns, which include: stock theft, finding clean water for their animals, and lack of funding to purchase the necessary infrastructure (such as fencing). A need exists for continuous training on cattle health and herd management. An efficient gender-specific funding mechanism must be constructed to assist female smallholder livestock farmers.

Conclusions

Female smallholder livestock farmers in the Northern Cape Province of South Africa face many challenges influencing their vulnerability to agricultural drought. The majority of women livestock farmers were disempowered and less resilient to agricultural drought compared to men livestock. The main attributes of the disempowered women were time/workload, access to and decisions on credit, ownership of assets, input into productive decisions, and group membership. Livestock women farmers in dual households experienced relatively good gender parity. The study provides the following recommendations:

- Women in each community should work together to help each other in terms of time availability, such as an afterschool center managed by some of the women, with remuneration.
- The government should provide a clear policy on land-related issues so that farmers can access credit. Female farmers should be empowered to make their own decisions through training.

- Group membership must be further encouraged, and a women's farmers union can concentrate on the specific challenges of female farmers.

Reviewer comment

Shows women face wide range of gender-based constraints at all levels.

52. Mkhathshwa, N. (2017) The gendered experiences of children in childheaded households in Swaziland, *African Journal of AIDS Research*, 16:4, 365-372, DOI: 10.2989/16085906.2017.1389756

Research Topic

Investigated the gender dynamics of living in child-headed households (CHHs) in a rural area in Swaziland that experiences high levels of drought, poverty and HIV and AIDS. Using a qualitative research methodology, the study examined ways in which children in CHHs meet their daily family needs and address their vulnerabilities according to their gender, focusing on the experiences of the children.

Methodology

The study sample consisted of 10 households, with 5 boy and 5 girl-headed households from the chiefdoms within the area. A semi-structured interview guide was used to conduct interviews in the respondents' own homes. The study focused on the gendered coping strategies used by the children to sustain their household welfare. Three factors were examined: leadership, food provision and education.

Relevant findings

The findings show that birth order conferred headship or leadership to the eldest sibling irrespective of their gender. Variations in the performance of the three factors, which were influenced by the gender of the household head, were observed. Generally, the children acted in accordance with their socio-cultural norms demanded in fulfilling the role of leadership and food provision. Boy-headed households become disadvantaged because of the boys' reluctance to take tasks which would contravene traditional Swazi notions of masculinity. This was exacerbated by societal expectations of the independence of boys. Hegemonic masculinity puts boys at a disadvantage when societal expectations require them to enact their masculinity through independence, rather than by drawing on the support of their neighbours/family/social networks. However, girls conformed to traditional Swazi norms. Societal compassion with the vulnerability of girls produced sustainable social arrangements and fostered resilience in girl-headed households.

Conclusions

The study shows that boy households found it much harder to be resilient than girl households because boys are expected to show independence, to "man up" and they are not really supported within the community or by teachers. They also seek to conform to masculine norms e.g., they try to secure cooked food from neighbours rather than ingredients, etc.

Though girl households could be challenged by younger males in the household, normative expectations in the community generally expect women to be managing the household on a daily basis. So the community (neighbours, teachers etc) were much more prepared to help and support girl households including through helping them prepare fields, support education etc. Boys wanted to demonstrate independence but did not have inputs.

Reviewer comment

This study is not top quality because it suffers from a lack of critical reflection on gender roles and if they should actually change. It rather supports the idea that boys and girls should conform to gender roles but that boys need more support in this. Overall though this article is remarkably interesting because it shows that households led by girls are more resilient because they conform with normative gender expectations about how women lead and manage households. Boys are unable to provide to the same extent and receive much less support and their households are less resilient.

53. Molua, E.L., & Justine E. Ayuk (2021) Male-female sensitivity in climate- induced income insecurity: some empirical evidence from farming households in Northern Cameroon, *Development in Practice*, 31:8, 1014-1039, DOI: 10.1080/09614524.2021.1937528

Research Topic

This paper compares the exposure of female-headed and male-headed rural households to income risk induced by climate variation.

Methodology

Three hundred and twenty households and their farms were observed in three Northern provinces of Cameroon. Snowball sampling was employed to identify and select household heads for the semi-structured interviews (320 in total). The analysis is based on a cross-sectional study undertaken to gain insights into the experiences of male- and female-headed households, respectively, under climatic stressors.

Relevant findings

Coping strategies and climate risk mitigation: Coping strategies of women are impeded by a number of factors; they have less access to resources and own fewer assets that are essential for coping and mitigating climate risks. Women's rights to land, and rights of access, use, and to make decisions about land and land-based resources are often denied.

Impacts on household security: Due to climate risks, traditional food sources may become more unpredictable and scarcer. In the study areas, women face loss of income and harvests. Related increases in food prices make food more inaccessible to low-income households, in particular, to women and girls, whose health declines more than male's health in times of food shortages.

Effects of climate variation on income: When variations in rainfall and temperature increase significantly, households experience negative returns to income levels, with female-headed households experiencing stronger impacts - indicating the insufficiency of current coping strategies in cushioning the households against stronger shocks.

Conclusions

Gender inequalities in access to resources, including credit, extension services, information and technology, should be considered in developing activities designed to curb climate change effects. The omnipresence of the extension service effort and accessibility to knowledge and innovation that enables male and female smallholders to improve their productivity and strengthen local markets, would further help particularly the female-headed smallholder farmers to increase their incomes and improve food availability in the region and beyond via responsive markets. Paying particular attention to the specific needs of female-headed households and finding a means to see to this will ensure that women - burdened by unequal access to resources - adapt to the vagaries of climate and ensure livelihood security for the teeming population. Governments should be encouraged to incorporate gender perspectives into their national policies and action plans.

Reviewer comment

Perceived and actual risks on agricultural income, coming on the heels of climate variation, threaten the livelihoods of households managed by both men and women.

54. Mosso, C.; Pons, D.; Beza-Beza, C. A Long Way towards Climate Smart Agriculture: The Importance of Addressing Gender Inequity in the Agricultural Sector of Guatemala. *Land* 2022, 11, 1268. <https://doi.org/10.3390/land1108126>

Research Topic

The research assesses the extent to which Climate Smart Agriculture (CSA) strategies address gender dynamics by supporting women's access to climate information, CSA training and decision-making spaces in which resource management strategies are defined. Country focus is Guatemala.

Methodology

Research findings are based on the analysis of an online survey - 2021 among a group of 109 rural extensionists who attended three climate services for agriculture workshops. The survey assessed: (1) participants' perceptions of the access and use of climate data/information; and (2) participants' perceptions of challenges and opportunities that gender dynamics pose for the adoption of climate change adaptation and mitigation strategies in the rural sector of Guatemala's Western Highlands. Data Analysis was performed using NVivo 12 qualitative analysis software.

Relevant findings

Gendered barriers to CSA: 72.3% of the respondents considered that there were differences in the ability to access climate information between rural men and women. From this group, 94% = social norms including machismo (30.3%), lower levels of education among women (24.2%), and exclusion from training activities (18.2%). A total of 85.4% of respondents = differences in the level of participation in decision-making processes among rural men and women. Less than half of the respondents (47%) considered that there were differences in the way climate change affects women and men. From this group, 56.3% stated that women are more vulnerable to climate change than men.

Opportunities to overcome barriers 54.3% of the agricultural extensionists who responded to this survey stated that at least 51% of the farmers they advised were women. Extensionists identified training activities for both men and women on gender issues (36.8%) and implementation of more inclusive agricultural training (36.8%) as key strategies for overcoming existing barriers.

Conclusions

For CSA strategies to be sustainable and effective in the improvement of adaptation capacity and resilience they must consider context-specific gender dynamics and social values, while aiming to reduce existing gender inequity. Lack of access to information, training, and decision-making spaces among rural women were identified as general manifestations of gender inequity in the Western Highlands of Guatemala and, consequently, as barriers to effective CSA approaches. However, further exploration is needed for a better understanding of the way in which these dynamics occur in the Western Highlands of Guatemala, as well as their intersection with other factors, such as the race, ethnicity, sexual orientation. Understanding these dynamics would allow the development of effective CSA strategies that, could contribute to reducing gender inequity and increasing adaptive capacity and resilience in the context of climate change, both at local and global scales.

Reviewer comment

Although (50% approx.) extension services say they advise women gender constraints remain. Normative barriers are, in themselves, not being challenged. Some awareness that climate change affects women differently but not much detail.

55. Muchacha, M., & Mushunje, M. (2019). The gender dynamics of climate change on rural women's agro-based livelihoods and food security in rural Zimbabwe: implications for green social work, Critical and Radical Social Work, 7(1), 59-72. Retrieved Jan 16, 2023, from <https://bristoluniversitypressdigital.com/view/journals/crsw/7/1/article-p59.xml>

Research Topic

This paper explores the gender dynamics of climate change on rural women's agro-based livelihoods and food security in Zimbabwe and proffers green social work as a strategic and relevant social work tool to address the challenges. It argues that green social work offers a holistic approach, tackles structural issues, and encourages social workers to advocate for pro-poor social policies. It asserts that "building on its historic roots in the betterment and preservation of human welfare, social work is a natural advocate of environmental perseverance, sustainability, and women's rights." (p.61)

Methodology

This is a discussion paper based on desk research.

Relevant findings

"...there are two major themes that dominate the literature on the gendered dimensions of climate change: women as vulnerable or virtuous." This binary representation ignores intersecting issues of age, class, ethnicity or race; and distracts from the fact that

the issues which underlie women's vulnerability are unbalanced relations with men; gender and power inequalities, that determine access to resources, information and the availability of options and choices. (p.60)

There is a need to address structural power relations between men and women, and gender inequalities. It is imperative that social work interventions that are directed at addressing environmental crises and injustices be anchored in tackling unbalanced power relations and other intersectional factors that compound the impact of environmental crises. (p.61)

In Zimbabwe, the impacts of climate change include the escalation of droughts, floods and extreme temperatures, leading to reductions in agricultural production, food insecurity reliance on food aid, reduced income and increased poverty. As primarily a rain-fed agro-economy, drought is a particular threat. Within this context, it is estimated that about 50% of women in Zimbabwe are involved in agriculture, and approximately 35% of rural households are headed by women. (p.59)

Land reform processes only intensified women's marginalised position, by continuing "to apply inheritance and land allocation rules that discriminate against women"(p.64). Climate change increases local food prices because of food shortages (p.62).

Class and age of women affect their vulnerability – older women are particularly at risk, as well as those of lower incomes. E.g., Women with higher economic and political status, better education, and exposure, benefited more than poorer, lower-status women, from both the processes of land reform and from government-subsidised farming implements such as seeds and tools in subsequent years. (p.64)

Green social work is posited as a holistic approach which "addresses environmental crises from various fronts, exploring and integrating various spaces, such as culture, economy, politics, emotional well-being, the physical environment and spirituality." In contrast to ecological social work, with which it overlaps, green social work is distinguished by "addressing the structural factors underlying environmental crises, for example power relations, gender inequalities and poverty." (p.66)

It has an explicitly political/advocacy focused dimension, including influencing the state response to climate change, and for the "equitable distribution of resources, pro-poor and gender-sensitive social policies", and mainstreaming the active participation of women in social policymaking and decision-making processes.

Conclusions

The paper calls for social workers in Zimbabwe to adopt a green social work approach, at all institutional levels "as part of professional bodies, civil society organizations, pressure groups and social movements, and government social and agriculture ministries, as well as through being individual change agents and activists." It encourages them to facilitate community dialogue and forums to inform women about their rights and how to claim them, and to 'engage the government and policymakers to advocate for the equitable redistribution of land'. It highlights the (then) new constitution of Zimbabwe as an opportunity for political engagement and advocacy on the rights of women. It also suggests that social workers can promote environmentally friendly agricultural practices among women and their families and encourages them to explore further how intersections of age, social and economic class, and gender increase or reduce climate risks.

Reviewer comment

The concept of green social work is interesting and has some relationship with GTAs, in its emphasis on addressing the structural and relational dimensions of gender inequality, and on understanding the impacts of climate change, and social and economic inequality in an integrated way. The authors' calls to action to social workers in Zimbabwe seems ambitious (combining political advocacy, community work, agricultural advice and research), it would be interesting to know more about the sector and how feasible the combination of roles is in practice. It could be helpful to explore connecting GTAs and the green social work movement.

56. Myeni S, Wentink GJ. (2021) A gendered approach to drought-coping mechanisms: A case of the Lubombo region, Eswatini. J transdiscipl res S Afr. 2021;17(1), a744. <https://doi.org/10.4102/td.v17i1.744>

Research Topic

The purpose of this study was to determine and evaluate gender-differentiated coping mechanisms of withstanding drought used by communities in the Lubombo region of Eswatini.

Methodology

Research conducted based on a mix of qualitative and quantitative methods. In Phase 1 literature review on gendered coping mechanisms of drought in the region. FGDs carried out to assess impact of drought on communities and validate the focus of the study. In phase 2, self-completion survey questions were administered to participants in all five tinkhundla to establish the drought-coping mechanisms employed in the Lubombo region. All data were combined to determine the gendered coping mechanisms employed in the Lubombo region.

Relevant findings

Gender-differential coping mechanisms during drought reflect the resources and traditional roles associated to men and women by displaying that men have better opportunities over women. For example, 97.5% men and 92.5% women agreed that, regardless of the situation or disaster, men continue to be heads of the household and women continue to manage their domestic responsibilities. Men however felt burnt out by having to continue performing the role of breadwinner. Women on their side continue to have weak decision-making power and do not participate in any drought awareness initiatives.

Drought-coping mechanisms include migration, drought relief scheme (foreign aid), crop management, water management, sale of livestock and household activities.

- Migration is a prominent coping mechanism used by men to fulfil their societal role of being 'bread winners' and financial providers of their households. Women are forced to assume traditional male roles (30.77%) like herding the cattle and income generation. Respondents concluded that migration is necessary but undermines the livelihoods of the community.
- Drought relief schemes is given in the form of cash, food, seed and fertilizer. Men are main target for training.
- Crop management involves cultivating more water-efficient crops. Women in charge of weeding and management of the crop. Women plant nuts, cassava and sweet potatoes as substitute for maize starch. Male coping mechanisms relating to crop production include men hiring trucks and driving them to prepare the fields for production.
- Water management practised through activities such as regulating domestic water usage, regulating water to fulfil basic household activities, separating water gathered by quality, either for consumption or for cleaning and watering plants. 60% of respondents attested to the fact that water management at the household level was predominantly done by women.
- Sale of livestock allows a household to get money, food or services. Food stock and household expenses were said (60% both male and female respondents) to overwhelm families forcing them to sell assets to raise money to buy food.
- Household coping mechanisms involves mostly skipping meals to save food. Children eat first and adults share food, reducing the portions of food being eaten. Other coping mechanisms employed mostly by women include cooking outside to save electricity, switching on necessary lights at night only, collecting wood and selling it to neighbours in exchange for food and money, doing laundry for others in exchange for food, and starting back yard gardening. Many men work as casual labourers.

Conclusions

Gender relations, norms, and roles shape the opportunities and constraints experienced by people in the community. The disparities between men and women - founded in cultural beliefs and practices as women took a domestic role whilst men were tasked to be providers. This perspective directly impacts on the lines of vulnerabilities during drought, declaring women as the target population for social aid such as food packages from foreign aid. Therefore, integrating gender equality into legislation and practice remain a critical stepping stone towards building a gender-sensitive environment.

Reviewer comment

Keynote study which shows how gender norms shape women and men's specific vulnerabilities and drought-response strategies.

57. Mwongera C., K.M. Shikuku, J. Twyman, P. Läderach, E. Ampaire, P. Van Asten, S. Twomlow, L.A. Winowiecki. 2017. Climate smart agriculture rapid appraisal (CSA-RA): a tool for prioritizing context-specific climate smart agriculture technologies Agric. yst., 151 (2017), pp. 192-203 doi.org/10.1016/j.agsy.2016.05.009

Research Topic

The climate smart agriculture rapid appraisal (CSA-RA) is a mixed method approach that draws on participatory bottom-up, qualitative, and quantitative tools to assess the heterogeneity of local contexts and prioritize context-specific CSA options. Imperative if countries are to respond to the COP21 agreement and meet intended nationally determined contributions (INDCs). The CSA-RA = assesses biophysical including climatic, socio-cultural, economic and technological characteristics at the household, farm and community/regional level (p.192). Tested in Uganda and Tanzania.

Methodology

The CSA-RA combines participatory rural appraisal (PRA) and rapid rural appraisal (RRA) tools into one methodology, that disaggregates the gender dimension and includes resource mapping; climate calendars; historical calendars; cropping calendars; organization mapping; transect walks; key informant interviews; farmer interviews; and pairwise ranking matrix. It collects qualitative and quantitative data from stakeholders (farmers, local leaders, researchers, local-level agricultural experts, private sector actors, donor organizations, and policy implementers), allowing expansive analysis, triangulation and validation (p.192).

Relevant findings

The CSA-RA is a participatory, bottom-up approach (that can be applied across multiple scales and actors) used to understand CSA priorities, synergies and trade-offs important in the local context, that can then be used to inform regional, national, and international public and private sector investments. Key operational staff in development agencies and national governments who design lending and climate finance projects as well as practitioners who implement CSA, need access to these tools. The CSA-RA was developed to address and fill these gaps by identifying the key challenges in smallholder agricultural systems within different contexts and develop a participatory prioritization of a set of CSA practices that can respond to such challenges for targeted implementation. The CSA-RA is rigorous but easy to use by various stakeholders and can be adapted to fit any agricultural system. As such, the CSA-RA (i) justifies the need for context specific CSA targeting that operationalizes the CSA plans and other development strategies and (ii) provides criteria and process that guide identification of appropriate CSA options for local contexts.

Conclusions

The CSA-RA is a participatory, bottom-up tool that provides a huge amount of gender-responsive data (this is central) across the tools. The article is fundamentally about validating the tool rather than highlighting specific gender findings.

Reviewer comment

It does not inform us in great detail about gendered findings but rather highlights a valuable tool for actually getting such findings to then analyse and build on.

58. Okello, D. (2020) Gender effect of entrepreneurial orientation on dairy farming career resilience in Kenya, Cogent Food & Agriculture, 6:1, 1863565, DOI: 10.1080/23311932.2020.1863565 <https://doi.org/10.1080/23311932.2020.1863565>

Research Topic

Examine gender effect of entrepreneurial orientation on dairy farming career resilience in Kenya. Specifically, the study examined the moderating role of gender on the relationship between future orientation (FO), market orientation (MO), risk-taking orientation (RO), social orientation (SO) and entrepreneurial resilience of dairy agripreneurs in Kenya. Focus on agripreneurial resilience (AR).

Methodology

Surveyed 682 respondents: 480 male and 202 female dairy agripreneurs in Murang'a County, Kenya using a cross-sectional study design. Data was collected using semi-structured questionnaire using personal interview. Data were analysed using partial least square-structural equation modelling PLS-SEM and multi-group analysis.

Evans and Wall (2019) defined resilience as the capacity of entrepreneur to bounce back from business challenges and maintain his/her profitability. Shadbolt and Olubode-Awosola (2013) define agripreneurial resilience as the ability of dairy agripreneurs to adopt and adapt to changes in agribusiness environment, while taking advantage of opportunities presented by the changes. Agripreneurs are faced with so many obstacles and uncertain outcomes which they need to overcome in order to have a profitable venture. Hence, resiliency is an important attribute for entrepreneurs. Agripreneurial resilience has three elements, buffer, adaptive and transformability capacity (Evans & Wall, 2019). Buffer capacity is the ability of an agripreneur to main constant production while faced with shocks and disturbances in the business. Adaptive capacity is the capability of agripreneur to respond to change through change in the structure of the agrienterprise such as membership to groups without affecting the function of the farm. Finally, transformability capacity is the ability of dairy agripreneurs to engage in intra-chain upgrading such as diversification into new enterprises (refs).

Relevant findings

This study finds that:

- ... women have significantly higher risk-taking propensity which increases their AR, when compared with male agripreneurs. A plausible explanation for this is that it is due to the empowerment of women through access to and control of resources, which has increased their desire for financial independence and need for achievement. This could be attributed to women's empowerment initiatives driven by both government and non-governmental agencies, and women-driven social groups such as the merry-go-round which supports them to empower their families.
- In addition, women's inherent performance-driven personalities, the fact that women spend more time with the family and at the homestead and their being self-organized in all that they do, has given them access to productive resources.
- Access to productive resources such as land and livestock has always been biased towards men, which has restricted women from investing in agriculture. However, in the past decade, women in Kenya have been given the right to own and inherit productive assets which have triggered them to have an internal locus of control and a need for achievement which in turn triggers their risk-taking capability. This result is contrary to most studies (refs) that have found higher scores in risk-taking propensity among men. However, it contributes to the body of knowledge that indicates that dairy women agripreneurs are more risk taking if they have access to productive resources. This dispels the stereotype that women are risk averse. Another explanation is the fact that, the dairy industry requires good organization, management and time in comparison with crop farming. Women being more precise in nature and present at home as compared to men, will play significant role in dairy enterprises hence they take high risks in such investments
- Social orientation reduces the resilience of male agripreneurs. This is intuitive because compared to females, men are less socially oriented in Murang'a county, Kenya. The society has portrayed men in this county as independent and they have to struggle alone to make end meet. Therefore, the result indicates that if men perceive themselves as more socially oriented, the agripreneurial resilience decreases.

Conclusions

Results show significant gender differences across the agripreneurial orientations. The direct effects relationships indicate that future, market and risk-taking orientation of female agripreneurs had a positive and significant impact on agripreneurial resilience (AR). While, for male agripreneurs, future and market orientation had a positive and significant impact on AR; but social orientation had a negative impact on AR. Gender moderates the entrepreneurial orientation-agripreneurial career resilience relationship whereby female agripreneurs had statistically significant higher risk-taking propensity.

Reviewer comment

This is not a high-quality study but it nevertheless has some interesting and thought-provoking data. The key lesson for me is that the author associates women's higher levels of AR to their participation in courses on dairy (increasing their knowledge) and to the effect of women's empowerment training. This has had the effect of women expecting to succeed and in so doing mobilize

the resources they need. The author also suggests that women's everyday roles (HH management etc) provide them with business management skills.

59. Padmaja R, & Kavitha, K. & Pramanik, Soumitra & Duche, Vishwambhar & Singh, Y. & Whitbread, Anthony & Singh, Ramesh & Garg, Kaushal & Leder, Stephanie. (2020). Gender Transformative Impacts from Watershed Interventions: Insights from a Mixed-Methods Study in the Bundelkhand Region of India. Transactions of the ASABE (American Society of Agricultural and Biological Engineers). 63. 153-163. 10.13031/trans.13568

Research Topic

Explore gender perspectives on water security by exploring an integrated water management approach for agriculture, livestock, and human consumption. The data were generated in a watershed project to enhance drought resilience of farming through groundwater recharge and agroforestry interventions in the water-scarce Bundelkhand region of Uttar Pradesh in central India (p.153).

Methodology

Quantitative data were collected from 700 individuals in five villages (three treatment villages and two villages where watershed interventions were not implemented). In addition, 33 semi-structured interviews and eight focus group discussions were conducted to understand local gender norms at the project sites. (p.153)

Following the argument put forth by Mitra and Rao (2019) and Leder et al. (2017), rather than using gender as a proxy for women, more nuanced perspectives that conceptualize gender as socially constructed relationships, shaped by intersecting ties of ethnicity, caste, class, age, marital status, religion, and other forms of social difference, are adopted in this analysis. (p.154)

Relevant findings

Women are underrepresented as intervention receivers and are formally disadvantaged in participation scheme management and/or access to scheme services, which results in a lack of equal access to benefits.

Men and women were not aware of and knowledgeable about national policies, acts, regulations, and goals that prioritize equitable access to resources, participation, and benefits for men and women in the three villages where the watershed intervention project was implemented.

Women and men in the three intervention villages scored low on all three components of the GILIT tool: (1) access to scheme resources; (2) access to scheme membership, leadership opportunities, and decision-making; and (3) access to scheme benefits = the scheme approach showed little or no sensitivity to gender equality.

The watershed intervention project aimed to ensure equal benefits for both men and women in terms of access to water. Strong inherent gender and social norms did not allow women to participate or share the benefits (p.158)

- Many programs sustained at national / local levels not disclosed to women by local governing body (i.e., Gram Panchayat).
- No women participated in elections in the study villages.
- Decisions about water allocation are made by male community members. Patriarchal norms limit women's participation in these discussions and decisions. Except during the rainy season, especially years with good rainfall, women are most affected by the decrease in water or non-availability of water/ women are responsible for securing water for domestic use/ livestock rearing (plus get firewood and fodder). Despite the watershed program, women have to fetch water with hand pumps or open wells that are often 1 or 2 km distant. Women and children older than 10 years are involved in fetching water. Check dams are usually constructed far from village, which makes accessibility to women unfeasible. The watershed intervention project made women aware that check dams increase the groundwater level, and thereby open wells have sufficient water to meet their domestic needs.
- Women have low access to information about the watershed interventions; do not have enough leisure time to watch television or radio to increase their general access to information and do not own mobile phones, but they use the mobile devices of other household members, usually men.

- In farming, men make most of the decisions, with little or no input from women.
- Women were usually not provided with information on training, and/or the training was not held at times or locations that allowed women to participate.
- Women never attended any other meetings in the village.
- No support services related to agricultural product collection, sorting, or marketing were provided through the watershed interventions
- An informal advisory role was played by the scientists and scientific officers involved in the watershed project.

Gender division of labour = women's domestic and time-intensive tasks such as sowing, weeding, and harvesting. Men engage in technology and market-related tasks, which require them to leave the domestic space and engage with wider social networks. (p.159)

Compared to women from the upper caste, women from the scheduled castes have greater mobility, less strict norms, can move freely to purchase inputs, such as fertilizer. Social norms sometimes oppressive and rigid for women belonging to the upper caste/ social structure itself works to keep the hierarchy in place. Based on women's need for money, marriage, and education, teak saplings were provided to 80 families with small daughters (up to 5 years old) to be planted on bunds in the fields (Leder, 2017). These teak trees would be sold upon maturity (20 years later) to pay for marriage dowries for the girls. The project's slogan was *Ladhko ko khet, ladhkiyo ko medh, har modh par, sau sagaun ka pedh* ("Land to boys, bund to girls, on every bund, 100 teak trees"). The slogan means that land ownership is for boys, while girls can be invested in through a higher dowry, which promises a husband of higher value and through this, women's possible upward social and economic mobility. Two important insights emerge from this understanding. This indicates that even if interventions are targeted to help women according to their stated need for economic support, these interventions can lead to reinforcing gender dependence, despite being well intended.

Conclusions

For gender transformative results via an integrated watershed management approach, it is important to make technical knowledge available to women. Because functional literacy allows community members and leaders to act collectively and harness the benefits of watershed interventions, efforts must be undertaken to achieve higher functional literacy for women through training, e.g., participatory gender training (Leder et al., 2016). Enhanced awareness of women's equal rights and deliberate efforts promoting women's agency are critical for sustainable development of watersheds. Involving women in building up the social capital will enable women and young people to challenge the existing gender and cultural norms while reducing possible conflicts in the community and households. Watershed intervention projects require gender transformative approaches across all social levels, from the household and community to the watershed project team. New common watershed guidelines (Wani et al., 2015) provide resources and policy support to address issues of gender equality and the status of vulnerable groups. This study provides initial answers to a larger question that requires further research: How to combine economic and technical interventions with social and institutional interventions to challenge social norms that reinforce inequality in order to make interventions more inclusive, thereby empowering and enhancing the overall well-being of men and women, and boys and girls. (p.161)

Reviewer comment

This is a moderately high-quality article. I cited at length from findings because this provides strong "on the ground" data to show how systematic, and systemic, gender norms (caste norms are not really explored) come together to exclude women from learning about, contributing to, and benefiting from watershed projects like this which really impact upon their daily lives, including in sourcing water for livestock.

60. Patel, S.K., Agrawal, G., Mathew, B., Patel, S., Mohanty, B., & Singh, A. (2019). Climate change and women in South Asia: a review and future policy implications. World Journal of Science, Technology and Sustainable Development.

Research Topic

South Asian region is a focal point owing to its vulnerabilities to climate-sensitive diseases, dependence on climate-sensitive livelihoods, projected levels of crop decline in the region, and high rates of poverty and malnutrition. Women are particularly vulnerable to climate change, and this affects women disproportionately during different extreme events.

Methodology

Literature review.

Relevant findings

Social norms play a significant role in the way women react to disasters (p.151):

- Mobility is typically limited - studies indicate that women are not able to swim during disasters (MacDonald, 2005; UNDP, 2013). The impact of climate-induced migration of men is such that in the event of a disaster in Bangladesh, women do not go to emergency shelters as they do not have men to accompany them (UN Women, 2015). Women are among the vulnerable sections during disasters in Bangladesh, as they are mainly dependent on their husbands for decision making and have less knowledge about cyclone warnings, making it difficult for them to evacuate in time (Kabir et al., 2016). Women in Pakistan are more severely affected during disasters, as their mobility outside the village is restricted and they become dependent on men for their survival during disasters
- Even when women survive a natural hazard, they encounter violence and sexual harassments (Ahmad, 2012). Another study shows that violence against women is aggravated during disasters in Sri Lanka owing to the violence that already exists even prior to disasters (Fisher, 2010).
- Women who migrate across borders from Nepal and Bangladesh to India because of climate change impact are vulnerable to trafficking. They are promised by agents to find work as maids but are instead forced to work in brothels (Action Aid, 2016).
- In the long-term following disasters, women in South Asia face hurdles owing to existing laws and norms that inhibit their ability to adapt to climate change impacts. In Bangladesh, women's vulnerability is aggravated owing to poor access to courts and inheritance laws, because they lose access to property after their father or husbands are killed in disasters. This burdened with the loss of assets and limited capacity to get loans (Ahmad, 2012).
- Wife beating becomes common during the period of rehabilitation (Bari, 1998). Poor rural women in Afghanistan find it difficult to access financial services, hindering their capacity for employment opportunities or adjusting to climate change impacts (UNEP, 2009).

P.152:

- Disasters lead to increase in cases of depression and suicides among women in the Maldives due to displacement and destitution. Women become susceptible to psychological problems due to collapse of social support networks (ADB, UN and World Bank, 2005). In a study conducted 18 months after an earthquake in Pakistan, a higher percentage of women had post-traumatic stress disorders than men (Naeema et al., 2011)

Women in South Asia adopt strategies to cope with climate change impacts:

- Women made portable mud stoves in Bangladesh for use in the future. Productive assets such as ploughs, fishing nets and other valuables were kept under the soil by women to protect them from cyclones. Women also built elevated platforms for family members with disabilities (Dankelman et al., 2008).
- In Sri Lanka, the workload of women increased in the event of deforestation, as they had to spend additional time on searching fuelwood. They coped with this issue by reducing the frequency of trips and increasing the load in addition to involving men in the activity (Awumbila and Momsen, 1995).

- Some of the coping strategies adopted by women in drought-affected states in India included skipping meals, whereas few others had two instead of three meals per day. Women from landless labour families in Marathwada reduced the quantity of food consumed (Kermode et al., 2007). In Pakistan, women skipped meals to cope with floods and there was a rise in the use of communal latrines and open defecation (WFP, 2010).
- In the wake of migration of men, women's income from productive labour has been useful for the survival of families in Bangladesh; however, occupational hazards are a cause of concern. Further, working outside is considered inappropriate for women, leading to disputes in the family (Pouliotte et al., 2009).

Conclusions

- Alternate livelihood options need to be worked upon by provision of resources for employment activities that have scope to develop in specific areas. Self-help groups can be formed to encourage livelihood generation among women. Women also need to be provided access to loans and credit facilities so that they may be able to take up employment opportunities. (p.161)
- Since women usually have indigenous knowledge of managing the environment and adapting to climate change and climate-induced extreme events, they need to be engaged in local-level strategies for combating the effects of extreme events.
- At a broader level, steps need to be taken to ensure women's participation at the community level and in decision making at all levels of the government. This can be enabled by formal discussions between disaster management committee of the government and the community. Through a participatory approach, women need to be engaged in deliberations and strategies for coping with and adapting to climate change and extreme events affecting the area (Ahmad, 2012; Bryan and Behrman, 2013).
- The governments in India, Afghanistan, Nepal and Pakistan need to improve infrastructure for sanitation, ensuring that people have access to drinking water so as to build resilience against climate-sensitive diseases.
- Afghanistan, Bangladesh, India and Pakistan need to make stringent efforts to improve indicators on MMR, anaemia and climate-sensitive diseases. This can be done if the governments take proactive steps to improve access to education and health services of women and take required steps that can steadily bring a change in social norms that place women at disadvantageous position as compared to men.
- Climate change adaption of different South Asian countries government administrators and policy makers have been reactive rather than proactive, and mainly focussed in strengthening the DRM systems and procedures. This approach has resulted in accumulated casualties and economic losses that were significantly higher than usual. Policy makers in South Asia region should begin to take a more proactive approach on climate change and related risks and should revise the policies in the socio-cultural context. It is imperative that governments should make concerted efforts to mainstream gender and climate risk management into their planning and development processes through both structural and non-structural activities designed to mitigate the overall climate change risk.
- Local agencies must update information on a regular basis after each extreme event induced by climate change, in accordance with international standards. Accurate and comprehensive loss assessment practices must be implemented. The governments must also allow the data to be archived in geo-referenced and open-source databases that are accessible to stakeholders and that can be of use for women-centric strategic planning on climate change.

Reviewer comment

This is a great review of the literature and makes key points. Recommendations are valuable and point again at the lack of gender in institutions.

61. Pike, I.L., Hiltonb, C., Österle, M., Olungah, O., (2018), Low-intensity violence and the social determinants of adolescent health among three East African pastoralist communities. Social science & medicine (1982), 202, 117-127. <https://doi.org/10.1016/j.socscimed.2018.01.022>

Research Topic

The social determinants of adolescent health in three nomadic pastoralist communities from northern Kenya, looking at one community directly affected and one less affected by intercommunity violence, in each location.

Methodology

Physical and psychological health was assessed, through anthropometric measures (physical) and a self-reported questionnaire (psychosocial). This included a longitudinal panel study from 2008-2011, randomly sampling at least 30 households in six sites for a total of 215 households. Each household was visited four times over two years.

They conducted a participatory wealth ranking exercise with five to six elder men in each community. The indicators and levels of wealth were taken from this exercise.

Relevant findings

Climate change is impacting the livelihoods of pastoralist communities in Kenya, along with intercommunity violence: (p.118) "Over the past twenty years pastoralist communities in northern Kenya have experienced four multi-year droughts, a profound erosion of access to grazing areas (Galaty, 2014; Galvin, 2009; Hobbs et al., 2008), and perhaps the most dramatic shift has been the widespread incursion of small arms (Mkutu, 2007). All of these experiences are set against a backdrop of noteworthy socioeconomic marginalization with poor integration into the national infrastructure in Kenya ... Wealth in pastoralist communities can be highly variable, but also perilous, with one severe drought or livestock raid threatening to push families into poverty."

The study found some differences in adolescent health outcomes for female and male adolescents. Adolescent girls were to some extent protected from food insecurity because of their physical proximity to the kitchen, due to their role in domestic tasks:

"Wealth appears to play a less obvious role in predicting girls' and young women's nutritional well-being. Girls and young women are vital to household production including spending considerable time helping with cooking, childcare, collecting firewood, and other important tasks. Their proximity to homesteads and kitchens may help buffer against the relationship between lower nutritional status and lower wealth...The cultural tradition of ensuring adolescent girls are given food preferentially as they reach marriageable age may also explain part of the limited variation in nutrition by wealth." (p.124)

Girls go to school in lower numbers than boys, and on the whole, attendance at school among pastoralist communities is low. For boys, attendance at school has a positive correlation with health indicators such as height and BMI. For girls this correlation is not there. The authors explain that this may be because of boys' activity level when at home (helping with herding and away from home) as well as girls' proximity to the kitchen at home.

Family and household experiences are considered central to understanding the links between adversity and health. The paper quotes another study looking at the nutritional impacts of variations of wealth between individual wives in polygamous households. "The author suggests that even small differences in women's income generating skills or activities can influence the nutritional outcomes of their children even when they live in the same compound." The authors of the current study suggest that in the pastoralist context, for girls in particular, household level variation is an important predictor of health, rather than broader measures of wealth within a compound or community. (p.125)

Conclusions

The circumstances of individual households were shown to be key in determining the psychosocial health outcomes for adolescents. This includes the variations between co-wives' households within the polygamous compound settings that are typical of pastoralist communities:

"How households, including individual wives within polygynous households, navigate food security and the response to violence plays a formative role in predicting the emotional well-being and future opportunities for young people."

Reviewer comment

This paper was interesting in its focus on adolescent wellbeing, suggesting that the age of 10-14 is particularly vulnerable for boys, when they are no longer regarded as young children but do not yet have the resources to be independent. From a gender perspective, it was interesting to note how traditional roles gave adolescent girls some protection from a health perspective, and that being of 'marriageable age' meant that they were given nutritional preference in households. It was also interesting to consider the significance of women's roles within polygamous compounds, and how outcomes can vary according to the co-wife's income and food security.

62. Po, Y.T., J., Hickey, G.M., (2018) Local institutions and smallholder women's access to land resources in semi-arid Kenya. Land Use Policy, Elsevier, vol. 76(C), pages 252-263. DOI: 10.1016/j.landusepol.2018.03.055

Research Topic

How smallholder women access land resources under institutions in semi-arid Kenya, following a major constitutional reform no land succession passed in 2010.

Methodology

In depth KIs (77), 12 FGDs (134) and 8 community meetings (363 participants) in Makueni County, Kenya. Conceptually, the authors build on the social relations approach to gendered resource allocation (Kabeer, 1994; Kabeer and Subrahmanian, 1996 Kevane, 2012, Meinzen-Dick et al., 1997), access theory (Ribot and Peluso, 2003), and social-ecological resilience thinking (Adger, 2006 Carpenter et al., 2001 Cote and Nightingale, 2012).

Relevant findings

The land succession act was found to be gender neutral on balance:

"The institutional analysis based on the social relations approach demonstrated how a gender-aware but gender-neutral constitutional clause in land succession presents a major step, however one of many, toward addressing gender inequality in Kenya's land tenure system. Our findings also revealed how existing structural causes and local norms on gender relations within a household can undermine national policy objectives." (p.259, Section 5.)

Even though women are aware of their rights, they sometimes don't want to act on this awareness e.g., to pursue legal land titles, as they don't want to jeopardise family ties with parents or spouse.

Relational mechanisms for land resource access, through kinship and community were shown to be important. Women described sharing money/resources with husband to show deference and building trust and reciprocity with parents in law to protect their access should they become widows. Through building relationships with neighbours, women can rent land locally if needed. (4.3)

"Relational mechanisms also contribute other preparatory, recovery and adaptive function of resilience. Un-married daughters and divorcees have comparatively fewer mechanisms of access to land resources."

Land ownership customs are complex, and land has significance beyond being a resource for agricultural production:

"Beyond agricultural assets, land access embodies a new settlement, a place of retirement, and a place to 'rest the bones' after death." (4.1.7)

Divorcees can be buried on her husband's land after death; this strengthens children's claim to the land.

Gender norms are changing but women's empowerment through financial independence can still be perceived as a threat to family unity:

"The dichotomous view of larger household assets, such as land resources and cattle being controlled by men, and smaller household assets such as poultry and micro-credit loans controlled by women appear to factor into the belief that women's empowerment through financial independence can lead to disintegration of family unity." (4.2) At the same time some parents recognised the benefits of gender equality, e.g., daughters sending more money home than sons, and expressed ambitions for their daughters to be educated and buy their own land.

Conclusions

From abstract: "Although some women are aware of their rights to inherit and own land, Kamba (majority ethnic group in Makueni county) women were generally reluctant to claim land resources through customary institutions and/or land registration processes. This stemmed from a desire to maintain gender dynamics within the household and to maintain their current relational access to land and other livelihood resources."

“Women, as daughters, sisters, wives, mothers, grandmothers, divorcee, and widows, were found to face vastly different pressures in land resource access. They reported using relational access mechanisms to cope with, and adapt to, land resource constraints. When combined with rights-based mechanisms of access, women could better secure future generations’ land resource access, especially in cases of skipped-generational households.”

The authors suggest that there is an emerging need to examine resource access by rural women from various class, education and social-economic levels, moving beyond the categorization of women based on familiar relationships, as this research has done.

Reviewer comment

This is a detailed and fascinating analysis of the interplay between different institutions (formal, customary, relational) in relation to women’s land access, showing how movement/change at one institutional level translates in practice at another. The categorisation of women by familial relationships (daughter, sister, widow etc), is helpful. The research demonstrates the importance of the relational dimension – in particular the insight into how women prioritise relational access to land even when legal access is technically on offer, showing how education on rights is not sufficient if action is not taken to support change in gender norms and relations more broadly.

63. Poudel, Shobha & Funakawa, Shinya & Shinjo, Hitoshi & Mishra, Bhogendra. (2020). Understanding households’ livelihood vulnerability to climate change in the Lamjung district of Nepal. Environment, Development and Sustainability. 22. 10.1007/s10668-019-00566-3.

Research Topic

This paper explored two vulnerability assessment indices, livelihood vulnerability index (LVI) and IPCC vulnerability index by surveying 150 households from three village development committees (VDCs) in Lamjung district, Nepal. Research questions: What are the key contributing factors of vulnerability to climate change for the mountainous households? Who are the most vulnerable households on the basis of location, well-being status, and gender of the household head?

Methodology

Data collected through a mixed method involving quantitative and qualitative approaches. The LVI was used to understand the contribution of demographic, social, and physical factors to climate change vulnerability. The IPCC vulnerability index was used to assess exposure, sensitivity, and adaptive capacity. The study was based on the following research questions:

1. Is there relationship between location of the households and exposure, sensitivity, and adaptive capacity?
2. Is there relationship between well-being status of the households and exposure, sensitivity, and adaptive capacity?
3. Is there relationship between gender of the household head and exposure, sensitivity and adaptive capacity?

Relevant findings

LVI analysis showed that climate change vulnerability varied according to the gender of the household head. The percentages of male- and female-headed households are 62% (n=93) and 38% (n=57), respectively. The analysis of vulnerability according to household head was relevant in the context of rural mountainous areas of Nepal, as the number of female-headed household (FHH) was increasing because of migration of male members to other places and countries for better opportunities.

The analysis shows that female-headed households had slightly higher exposure (0.395) and sensitivity (0.296), as well as lower adaptive capacity (0.549) in comparison with male-headed households, although FHHs were over-represented among well-off households. FHHs reported higher numbers of natural disasters such as landslides and food shortage. Similarly, a higher percentage of FHH (66.66%) reported increasing number of insects/health problem, higher food-insufficient months (4.1 months), a higher percentage of households (3.5%) with food accessibility decreased. Females are traditionally bound to household work, but in recent years, they have been facing increased burdens of household work along with agricultural activities due to the absence of their male counterparts. Female-headed households also had a higher percentage of landlessness (89.47%) and lower percentage (22.81%) of involvement in saving and credit cooperatives, which contributed to reduce their adaptive capacity.

Conclusions

The findings have an implication for Nepal's climate change policies and programs, such as national adaptation programs of action (NAPA) 2010, local adaptation programs of action (LAPA) 2011, and climate change policy 2011, which ignored the concept of differential vulnerability in terms of analysis and adaptation planning. The findings demonstrate the importance to consider the gender of the household head, alongside other determinants, in vulnerability analysis and adaptation planning. Both NAPA and LAPA are oriented toward addressing climate hazards but overlook the underlying causes of vulnerability and the process to identify and address the most vulnerable populations. A micro-level analysis of vulnerability may help in designing more effective adaptation plans at the local level.

Reviewer comment

Clearly shows differential abilities of female and male headed households to handle climate change due to underlying vulnerabilities.

64. Purcell, S. W., A. Tagliafico, B. R. Cullis, and B. J. Gogel. 2021. Socioeconomic impacts of resource diversification from small-scale fishery development. *Ecology and Society* 26(1):14.
<https://doi.org/10.5751/ES-12183-260114>

Research Topic

The predicted future shortfall in seafood production from tropical small-scale fisheries demands support to help diversify income streams and food production for coastal communities. Livelihood diversification can comprise the enhancement or addition of components to existing fisheries, yet the likely socioeconomic impacts are unclear. With a long history of non-deleterious introductions, the marine snail "trochus" (*Rochia nilotica*) was introduced to Samoa from 2003 to 2006 to offer a new artisanal fishery resource (pages aren't clear).

Methodology

15 years later, surveyed 303 fishers - structured questionnaires / mixed effects models to evaluate how fishery has contributed to fisher well-being and factors influencing socioeconomic impacts.

Relevant findings

This is from the literature review, the material selected for conclusion summarizes the overall finding clearly.

- Gender inequality in fisheries income is widely known (refs), yet the underlying causes are diverse. Compared to male fishers, women could be disadvantaged by being restricted to just gleaning, processing, and selling activities (Santos 2015, Furkon et al. 2019), being restricted to intertidal areas (Siar 2003, de la Torre-Castro et al. 2017), having poorer access to boats (Gerrard and Kleiber 2019), being given lower prices by traders (Purcell et al. 2016), and even being excluded in some fisheries, thus obtaining less catch (Chapman 1987, Hauzer et al. 2013).
- Women might invest less time in fishing because of their involvement in child raising, farming, caring for elders, and housekeeping (Harper et al. 2013, de la Torre-Castro et al. 2017, Biswas 2018).
- Gendered disparities in fishery income might also be attributable to the proclivity of women to retain part of their catch for subsistence consumption (Harper et al. 2013, Gillett and Tauati 2018).
- In certain cultures, men are held as income earners and so might sell a comparably larger part of their catch (Kronen et al. 2007, Fröcklin et al. 2014). Consequently, while prices and landings for harvested seafood might be gender equitable, women fishers might gain lower weekly income (de la Torre-Castro et al. 2017, de la Torre-Castro 2019).
- Framing poverty in Pacific Island fisheries on income alone might be inadequate because fisheries contribute substantially to local diets (Pollard and Abbott 2004, Charlton et al. 2016).
- Fishery catches and incomes strongly influence fisher well-being. The concept of well-being includes their economic yield, food supplies, employment, and psychological and physical health (Coulthard et al. 2011, Weeratunge et al. 2014). Satisfaction with

one's livelihood source is one, of many, useful measures of wellbeing, derived directly from socioeconomic surveys (Coulthard et al. 2015, Purcell et al. 2016).

- Fishery dissatisfaction can impact on fishers' mental and physical health (Coulthard et al. 2015, Pollnac and Poggie 2006), and undermine the relationships between fishers and management institutions (Trimble and Johnson 2013). An integration of objective (e.g., food or income) and subjective well-being measures (e.g., satisfaction) thus offers a more holistic perspective on impacts within social-ecological systems (Weeratunge et al. 2014).

Conclusions

Most fishers consumed part of their catch and both fisherwomen and fishermen shared harvests informally within communities, thereby bolstering resilience of the socioecological systems at the community level. More than one-quarter of fishers sold part of their catch, and the new earnings represented 17% of their gross income from all sources. Fishing incomes were gender equitable and influenced by fishing frequency and capital assets (boats). Most fishers were satisfied with income from the relatively new fishery and improved income was reported by a majority of fishers, especially those younger and less experienced. Additional money from the fishery was most often spent on food, church tithing, and school fees. This relatively new fishery fostered positive well-being outcomes that were gender inclusive. Extrapolations of annual incomes across the fishery reveal a rapid return on investment from foreign aid funded development. The study reveals that certain coastal artisanal fisheries can be gender equitable and that benefits are likely underestimated because of subsistence consumption and informal distribution networks. Diversifying the marine resources accessible to small-scale fishers offers a promising strategy to support coastal livelihoods and strengthen resilience of social-ecological systems (p.13).

Reviewer comment

This is a keynote article. It provides a useful summary of gender-based constraints to women in fisheries (see findings above) and the conclusions are important. They refer to the fact that fishers give away an important proportion of the snail catch (approx. 30% to 50%), consume themselves, sell, etc. Women as well as men fish and consume and benefit from good community relations. This paper shows that we are not measuring enough aspects of well-being if we rely purely on economic measures.

65. Randriamparany, ST., Mahefosa Randrianalijaona, T. (2022). The Vulnerability of Antandroy women to droughts in Ambovombe Androy (Madagascar). International Journal of Disaster Risk Reduction. Volume 72. <https://doi.org/10.1016/j.ijdr.2022.102821>

Research Topic

This paper looks at the factors behind Antandroy women's vulnerability to the risks of droughts.

Methodology

Documents on gender, vulnerability, droughts and the Androy region were collected and analysed. Then the authors conducted interviews and a survey to understand the Antandroy's living conditions. 100 households were selected for the survey in 10 villages, where 14,259 households live.

Relevant findings

One third of surveyed households are headed by women. Elsewhere in the paper (p.8, 4.5), there is reference to drought as a possible cause of out-migration by men to seek employment, though this is not explored in detail. (p.6, section 4.2)

16% of women in union have one or more co-wives. The paper refers to 'competition with co-wives' as a factor which increases women's susceptibility to drought impacts but this is not explored further. (p.6, section 4.2)

Surveyed women perform 64.5% of household chores, limiting their time for income-generation. With regards to income generating activities, female labor is concentrated in agriculture (91.2% versus the national average of 72.2%), making them particularly susceptible to the impacts of drought. (p.6, section 4.3)

Low levels of female education limit their work opportunities outside agriculture (Section 4.1 and 5).

Women traditionally have a role in fetching water which is time consuming during periods of drought. Survey respondents reported spending more than 3 hours per day on fetching water, and this is not guaranteed to be from a safe source. (p.8, section 4.4)

Early marriage leads to early pregnancies and women in the region have a higher number of children due to cultural norms and lack of access to education and birth control, which places additional financial burden on the household and also exposes women to healthcare challenges associated with low levels of maternal care. (p.8 & 9, section 4.6)

Women are physically affected by the impact of drought - which leads to dehydration, exposure to diseases from non-drinkable water, and weight loss. (The paper does not say how men are affected). (p.9, section 4.6, c)

Conclusions

(From abstract) Several factors lead to women's high degree of vulnerability, such as the particular climate of Androy, household's size, duties, difficulties faced by the agricultural production, early marriage, migration and low educational levels. To contribute to the reduction of their vulnerability to droughts, Atandroy women should reinforce their adaptability by developing income-generating activities to diversified sectors, using varied seed and protecting the environment, with the definitive eradication of slash-and-burn cultivation and bushfires.

Reviewer comment

The analysis within the paper is quite topline. It lists constraints facing women and makes links in a general way between these constraints and their vulnerability to the impacts of climate change. It is still of value as an example of the difficulties women have to respond to the impact of climate change, given the normative constraints of early marriage, high birth rate, low literacy levels, and high dependence on subsistence agriculture. Beyond this it points to interesting topics though these are not explored in detail - such as the feminisation of agriculture due to climate-driven outmigration by men, and the dynamics of polygamous households in situations of resource constraint such as drought. The reference to physical impact of drought on women's health and physical strength is interesting as an aspect of the relationship between climate change and economic resilience.

66. Rios, A.M.P., Manuel Francisco Díaz Baca, Karen Johanna Enciso Valencia, Natalia Triana Ángel & Stefan Burkart (2022): Gender inequalities in the Colombian cattle sector: an econometric analysis, Development in Practice, DOI: 10.1080/09614524.2022.2098256

Research Topic

This document analyses gender inequalities in the Colombian cattle sector using census information on the cattle activity, disaggregated at the territorial level.

Methodology

Different sources of information on cattle production, sociodemographic contexts, agricultural credit, and poverty were gathered from the 2014 Household Census to develop the descriptive analysis and the estimation of the econometric model. The econometric model developed through a linear regression models that relate gender with cattle production at the municipal level.

Relevant findings

The results show differences in the participation of men and women in multiple aspects of cattle ranching. Men dominate cattle farming, and women's participation is lower. This may originate from the difficulties women face in inheriting family businesses and land, given the gender biases in the generational transfer that favour first-born sons over daughters. Likewise, women have less weight in productive decision-making, indicating that family structures and gender roles limit the time they can dedicate to decision-making and associativity. Likewise, gender roles can impede women leadership by assigning work to women in private spaces (such as caring for the home and animals) and to men in public spaces (such as the sale and marketing of products). The differences also extend to education, access to credit, and farm size. The aforementioned factors can affect the production of women-led UPAs. The estimates of the econometric model reveal a negative relationship between cattle production and women's participation at the municipal level. Municipalities with more women-led UPAs tend to have lower cattle inventories and milk production. Cultural differences and gender roles hinder women's access to cattle, considering that they play a specific role

in production unrelated to administration and commercialisation, but focused on care and cleaning. Time constraints caused by housework also limited access to information/technical assistance can also impair their ability to maintain and increase cattle inventories.

Conclusions

Gender gaps in agriculture, and particularly in the cattle sector, have the potential to affect national agricultural production in Colombia and with it the well-being of rural families that depend on it. This highlights the importance of policy decisions that feature a gender approach. Through an active intervention of the State, particularly through education programmes, credits for rural women, awareness creation of gender roles, and land tenure/inheritance regulations, a first step can be taken towards reducing gender gaps.

Reviewer comment

The article is not focused on climate.

67. Rosen, J.G., Mulenga, D., Phiri, L. et al. "Burnt by the scorching sun": climate-induced livelihood transformations, reproductive health, and fertility trajectories in drought-affected communities of Zambia. *BMC Public Health* 21, 1501 (2021). <https://doi.org/10.1186/s12889-021-11560-8>

Research Topic

Climate-induced disruptions like drought can destabilize household and community livelihoods, particularly in low- and middle-income countries. This qualitative study explores the impact of severe and prolonged droughts on gendered livelihood transitions, women's social and financial wellbeing, and sexual and reproductive health (SRH) outcomes in two Zambian provinces (p.1).

Methodology

In September 2020, in-depth interviews (n = 20) and focus group discussions (n = 16) with 165 adult women and men in five drought-affected districts, as well as key informant interviews (n = 16) with civic leaders and healthcare providers, were conducted. A team-based thematic analysis approach, guided by the Framework Method, was used to code transcript text segments, facilitating identification and interpretation of salient thematic patterns (p.1).

Given the potential synergistic impacts of this growing climate crisis and gender-based inequities on the SRH outcomes of women and girls, there is an urgent need to interrogate the intersection of climate change, gender, and SRH in Zambia, more closely. The economic impacts of climate volatility on rural communities, whose livelihoods are inextricably linked to agriculture and farming, are well-documented in sub-Saharan Africa [26–28]. "Drought, specifically, threatens agricultural productivity, resulting in heightened food insecurity and diminished household incomes; these processes can catalyze other downstream risks, like early marriage and transactional sex, associated with poverty." While disruptions to essential health services are anticipated with extreme weather events [29, 30], the impact of severe and persistent climate irregularities, like drought, on provision of routine SRH services (i.e., contraception, family planning) is unknown (p.2).

Limited infrastructure and support for climate-responsive agricultural practices have also rendered these districts (ie the study districts) particularly susceptible to poorer crop yields in times of drought. Fewer than half (45%) and 40% of Zambian farmers do not use fertilizer on their fields and plant hybrid maize seeds, respectively, rendering agricultural outputs particularly vulnerable to rainfall anomalies [34]. (p.3)

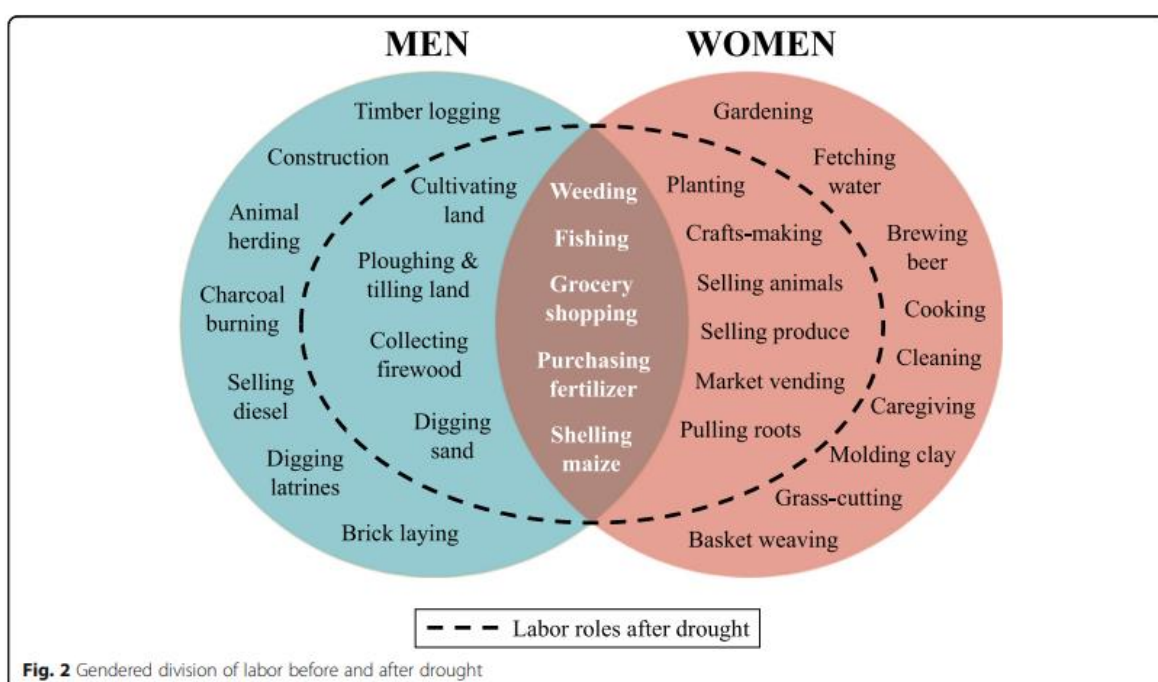
Relevant findings

For most participants, climate change was manmade and attributed to human interactions with the environment, specifically deforestation. Men, in particular, narrated a cyclical process whereby reduced economic prospects attributed to drought drove men to industries (i.e., timber logging, charcoal production) contributing to deforestation, which subsequently worsened climate change processes. (p.5)

A few participants, primarily women in rural districts, discussed spiritual or celestial origins to changing rainfall patterns in their communities. (p.5)

Economic strife and food insecurity propelled men and women to pursue alternative sources of employment to generate household income lost to drought. Prior to drought, participants described structured, patterned divisions of labor by gender. Men primarily held jobs involving manual labor (e.g., farming, construction, fishing), while women divided their time between micro-entrepreneurial activities and household obligations, like cooking, cleaning, and child-rearing. Even in dual breadwinning households, where men and women participated equally in non-household labor, employment opportunities (and, therefore, potential earnings) were highly gendered, with men managing small-scale businesses (e.g., planting and tilling farms) and delegating women to sell commodities, like produce and fish, generated from micro-enterprises (n.b. citation indicates that people believe the gender division of labour must be kept to). (p.6)

As drought virtually eliminated viable employment opportunities in small-scale agriculture, demand for alternative income sources outstretched the available supply. While employment opportunities were available in burgeoning industries like timber logging and charcoal burning for men and beer brewing for women, some of these opportunities were seen as temporary or seasonal, offering little in terms of employment security or stable income. Participant narratives suggested that drought blurred a historically gendered division of employment roles. A shrinking labor market forced men and women to prioritize potential earnings over workforce preferences, pushing women into jobs with heightened manual labor demands and men into labor sectors traditionally dominated by women (see Fig. 2). (p.6)



- Some women perceived out-migrating men to have abandoned their families by permanently resettling in other communities, leaving their spouses to become the sole caretakers of the household and family. In these circumstances, children remained in their mother's care, constraining employment prospects for women who suddenly absorbed more caregiving responsibilities. For these reasons, participants unanimously explained that female-headed households were among the most vulnerable to hardships in times of drought (p.7)
- Marital relationships were challenged in times of drought, particularly when disagreements around household purchases could not be reconciled. Tightened household incomes required more joint decision-making in even small household purchases; routine purchases, like clothing and children's toys, were seen as luxuries in the context of drought. As women became increasingly involved in household budgeting and financial decision-making, they reported more frequent quarrels with their partners about household purchases, particularly when disagreements arose regarding men's discretionary use of income. (p.8)
- Many women were overburdened by a perceived imbalance in household and income-generating responsibilities, feeling especially unsupported by sporadically employed or unemployed male partners. Participants explained that reduced employment prospects in agriculture were to blame for high rates of male unemployment, which many women suggested was the driver of truant behavior and alcohol use among men in their communities.

- While drought affected all facets of life, the financial and social repercussions of drought, from increased breadwinning demands to heightened food insecurity, disproportionately impacted women. An immediate consequence of diminishing household incomes is the inability to afford school fees, which resulted in growing rates of school attrition during droughts. Even in circumstances where families could afford school fees or could rely on other relatives to offset these financial burdens, some caregivers withdrew their children from school, so they could support their families financially through labor.
- Children, particularly girls, from the most vulnerable households were sent to other communities for domestic work. Some women acknowledged that the financial insecurity propagated by drought forced girls into early marriages (n.b. parents also get bridewealth).
- Drought-induced financial insecurity prompted women, in some cases, to exchange or sell sex to support themselves and their families. Some participants explained how women would trade sex for food, money, or farming inputs. Female-headed households were perceived as particularly vulnerable, as many lacked the appropriate farming tools and inputs to support their own farms, which are traditionally owned and maintained by men.

Conclusions

Across districts, participants emphasized the toll drought had taken on their livelihoods and communities, leaving farming households with reduced income and food, with many turning to alternative income sources. Female-headed households were perceived as particularly vulnerable to drought, as women's breadwinning and caregiving responsibilities increased, especially in households where women's partners out-migrated in search of employment prospects. As household incomes declined, women and girls' vulnerabilities increased: young children increasingly entered the workforce, and young girls were married when families could not afford school fees and struggled to support them financially. With less income due to drought, many participants could not afford travel to health facilities or would resort to purchasing health commodities, including family planning, from private retail pharmacies when unavailable from government facilities. Most participants described changes in fertility intentions motivated by drought: women, in particular, expressed desires for smaller families, fearing drought would constrain their capacity to support larger families. While participants cited some ongoing activities in their communities to support climate change adaptation, most acknowledged current interventions were insufficient (p.1).

Reviewer comment

A keynote article because though it focuses primarily on how health outcomes are associated with climate change and failing agriculture, it provides extremely rich gendered data on the climate x impact on economic resilience of women and FHH. Superb.

68. Sen, D., (2018) "Fempreneurs or organic tea farmers? Entrepreneurialism, resilience and alternative agriculture in Darjeeling, India", *Journal of Political Ecology* 25(1), p.732-747.
doi: <https://doi.org/10.2458/v25i1.22386>

Research Topic

Women organic tea farmers build economic resilience through dual enactments as "organic farmers" and as "entrepreneurs." In substantiating both, women question the limited optics through which Fair Trade type sustainability ventures measure their work for a tea cooperative, as well poorly recognizing their entrepreneurial work in their households and community. Women are deeply aware of the politics of Fair Trade where their productive and reproductive labor is appropriated through the labor of organics, where women not only produce the organic green leaf tea but also produce narratives of Fair Trade's success in its certification and gender audits. Thus, to understand what sustains the new wave of "sustainable agriculture" in the global South, we must explore the intersections of organic farming practices with emerging discourses and practices of gendered entrepreneurialism in organic farming communities.

Methodology

Anthropological observation over many years.

a) women in Darjeeling's poorest villages have developed a very different understanding of what they term "riks" (a variation of the English word risks) than microcredit-promoting development NGOs, when the latter discuss 'risky economic behavior'.

b) these different interpretations of credit and risk reveal the situated nature of economic practices and highlight the deep connection between inter- and intra-household power inequalities. The latter, women believe, puts them at danger as they engage in borrowing and repaying money. Fair Trade, with its promise of empowerment, is unable to mitigate these issues.

c) women navigate these strictures by demonstrating creative collective social agency (including through establishing Makhmali SHG) that in turn has had important, unintended consequences for them and community resilience (p.738)

Relevant findings

In assessing the success of alternative/organic agricultural practices in "certified communities" one must take into account the intense community level performances and practice that relate to, but go beyond, agriculture. Such attention demands that we widen our aperture when "seeing" women farmers and realize the pressures they face in doing the labor of organics and Fair Trade. This includes their work in producing organic tea, hours spent in training to pass certification of internal control measures, attending Fair Trade training, and taking cultural and financial risks to become entrepreneurs. In a situation of intense precarity where sales of green organic tea leaves to plantations only brings income for seven or eight months a year, these community level performances and entrepreneurialism are a way for women farmers to make visible the labor that goes into sustaining their households - while continuing their commitment to organics and building economic resilience for their families. These performances at the community level in order to gain respect as female entrepreneurs, as well as for the transnational audience waiting to bring them into the folds of global regulatory regimes, remind us that the path of sustainable agriculture requires this "dual enactment" of performance work to thrive. While transnational justice and sustainability regimes want to empower women farmers, this "dual enactment" of performance remains concealed. (p.746)

Conclusions

"My objective in this article is to highlight how women organic farmers build resilience through dual enactment as "organic farmers" and "entrepreneurs." In substantiating both positions, women question the limited optics through which Fair Trade-type ventures measure their work for the cooperative, as well as the misrecognition of their entrepreneurial work in their households and communities. Women are deeply aware of the politics of Fair Trade where their productive and reproductive labor is appropriated through the labor of organics where women not only produce the organic green leaf tea but also produce narratives of Fair Trade's success in certification's gender audits. Their images and narratives are used to build virtualities (West 2012; see also Cornwall and Edwards 2014; Cornwall, Harrison and Whitehead 2007) which elide the everyday battles over resource and recognition at the heart of these organic tea-producing communities. In the context of these deep fissures around resource access, women's situated knowledge of tea production and associated skills are denied in a patriarchal setup by a context-specific vector of shame (see also Schroeder 1999). However, such disciplining practices encounter women's critical manoeuvres around the denial of their labor. Women farmers often use the word "riks" [a variation of the English word risk] to draw attention to the dangers of their specific entrepreneurial ventures in addition to tea production. Entrepreneurial performances and practices affect their reputation and household dynamics, but women hold onto them for the survival of their households and their own aspirations" (p.734)

Reviewer comment

This article is very different. It is basically an account of the author's field work and reflections and not academic. It, however, goes extremely deeply into how women's work as entrepreneurs was continually contested within the community and within the household, and how women fought back. An interesting observation is that so long as earning an income was considered to be part of a housewife's work then it was okay. But as time moved on and the Self-Help Group became about loans, women's reputations were severely threatened and criticised. This may be (reviewer thought) a reason why many organisations try to maintain a welfare dimension, which is often criticised as inefficient, because it maintains the image of women as being primarily carers etc.

69. Serote, B.; Mokgehle, S.; Du Plooy, C.; Mpandeli, S.; Nhamo, L.; Senyolo, G. (2021) Factors Influencing the Adoption of Climate-Smart Irrigation Technologies for Sustainable Crop Productivity by Smallholder Farmers in Arid Areas of South Africa. *Agriculture* 2021, 11, 1222. <https://doi.org/10.3390/agriculture11121222>

Research Topic

This study identified the factors influencing smallholder farmers' decision to adopt Climate Smart Irrigation Technologies (CSIT) in the Limpopo Province of South Africa.

Methodology

Data were collected through the completion of a face-to-face structured questionnaire by 100 smallholder farmers selected through convenience and purposive sampling. A probit regression and OLS model were used to identify factors that influence the smallholder farmer's decision to adopt CSIT and the level of adoption.

Relevant findings

Males and females were represented equally in the Capricorn district. While in the Vhembe district, a large proportion of the sampled farmers were males. However, males have a higher likelihood of increasing the level of adoption. This implies that the probability of males adopting CSIT increases by 68% compared to their female counterparts. This is often a result of dominant cultural biases where males still have exclusive rights to make farm decisions regarding short-term and long-term adjustments. In contrast, women are often faced with social, cultural and financial barriers that limit them from accessing the required resources for CSIT adoption [30,64].

Conclusions

The study shows that gender differences influence the adoption of CSIT. Recommendation -stakeholders in agricultural sector, such as extension officers, should continue educating smallholder farmers about CSIT. This can be done by organizing workshops and farmers' days. In order to fast track the adoption of the CSIT amongst SHF, there is a need to provide capacity building, technology transfer demonstrations, and strengthen knowledge dissemination as part of the science-policy interface.

Reviewer comment

Specific recommendations on gender are not addressed in the conclusions. It demonstrates however the importance of norms privileging male decision-making around adoption, and lack of attention to alleviating constraints facing women.

70. Singh, R.K., Kumar, A., Singh, A. et al..(2020) Evidence that cultural food practices of Adi women in Arunachal Pradesh, India, improve social-ecological resilience: insights for Sustainable Development Goals. *Ecol Process* 9, 29 (2020). <https://doi.org/10.1186/s13717-020-00232-x>

Research Topic

Sustainable use of biocultural diversity can help achieve Sustainable Development Goals (SDGs) in many ways. The tribal communities of bio-culturally rich Northeastern India rely heavily on local food resources and knowledge for livelihood security. Focus is on analysing remote and less remote (transitional) communities, the focus is entirely on women's Indigenous Technical Knowledge (ITK).

Methodology

To understand (i) access and use patterns for biocultural plants and animal species from diverse ecosystems and (ii) species diversity and conservation strategies deployed by Adis. We employed a novel participatory approach called "recipe contest" to mobilize Adi women from 15 randomly selected remote and transitional villages. The winners of the contest were then randomly sampled as well. Data collected through interviews, focus group discussions and transect walks from 75 women, were analyzed using qualitative and quantitative methods.

Tribal communities living in remote locations of India, particularly of mountainous ecosystems differ from those of more settled and manipulated landscapes in terms of available resources and socio-cultural and political resource base (Joshi 1989) (Fig. 1). These economically marginal communities, unable to access external resources, have developed ecosystem-based knowledge systems (refs), to survive and maintain biocultural diversity (Toledo 1999). Over time, they have also evolved culturally and ecologically responsive strategies for adapting to multiple stressors (refs). Identifying these strategies can provide meaningful insights for policymakers attempting to develop place-based approaches to land and resource use, and meeting targets of SDGs, especially for biocultural hot-spots (refs). Based on insights from Toledo (1999), Maffi (2010), and Davidson-Hunt et al. (2012), we conceptualized Adi biocultural resources as “a plant-animal-cultural continuum contributing to resilience and sustainable livelihoods”. (p.2)

Relevant findings

Adi women, particularly those living in more remote communities, possess a rich knowledge about nutritionally important local food species and their traditional products. They rely heavily on local plant and animal species accessed from diverse land use systems and conserved using proven traditional practices. (p.16)

- 14 different plant species were used as food, 33 as both food and ethnomedicine, and 13 species were one of the major means to generating income.
- 17 animal species also play crucial roles in food and sustaining cultural identity of the Adis.
- The knowledge on integrated strategies applied by Adi women for accessing and using food resources was found to improve biocultural diversity and social-ecological resilience.

Women respondents from traditional and transitional villages differed significantly with each other for use of plant and animal species for food, ethnomedicine, and income generation, with the former using significantly more number of species.

Women from traditional villages had more knowledge on species conservation practices than their transitional counterparts. Home gardens were the most important source of diverse food plants, followed by jhum lands. Moang community forests and home gardens displayed more species diversity than other land use systems.

Changing lifestyles and land use patterns have accelerated the erosion of biocultural knowledge; particularly in and around transitional villages. Women living in transitional social systems closer to urban centers knew less about traditional ways of cultivating, harvesting, and using local food resources from diverse ecosystems, and used fewer of these foods (cf Turner and Turner 2004; Singh et al. 2010). As a result, their resilience to health and environmental risks is expected to magnify from the erosion of biocultural diversity and the shift away from nutritionally rich local foods to marketed foods, which are often high in unhealthy fats and sugars (Turner and Clifton 2009). Irrigated and mono-cropping systems and implementation of PDS (public distribution system) in India (Kothari 1997), along with emergence of new food stocking and distribution patterns (Behera et al. 2016) with the easy availability of commercial and marketed foods, offer seemingly more efficient alternatives for the people living in or nearby towns, who previously used primarily local and uncultivated species (Kothari 1997). The detrimental effects of new and changed globalized diets are reflected not only in the poorer health of individuals, but also in the farming systems and management of local ecosystems (Parks et al. 2005; Nautiyal et al. 2007) (p.16).

Conclusions

- Promotion of reciprocal learning between formal and informal knowledge systems can enhance the adaptive capacity of local communities to cope with emerging social-ecological stressors and provide new avenues to science-policy establishments for improving outcomes vis-à-vis SDGs.
- Emphasis is on promoting local/regional dialects as means for rapid diffusion of biocultural knowledge, especially in areas where social-ecological changes are taking a toll natural capital and livelihoods. For example, creation of “Village Knowledge Banks” in collaboration of science-policy institutions, while retaining local identities and ownership of resources, can provide localized education and learning platforms, particularly in areas rich in biocultural resources but facing threats.
- Relevant provisions in government policies and programs [e.g., mid-day meal and public distribution system] for promoting nutritional local food. Consequent increase in demand for locally produced nutritious food can motivate communities for strengthening species conservation measures as well as increasing livelihood opportunities.

- Special empowerment drives need to be launched for the capacity building of disadvantaged women groups (e.g., Adi women in our case) through Indian policies on promotion of micro and small industries, food processing and value addition, skill development, and eco-tourism such that they are not only better equipped for livelihood security in face of looming threats but may also contribute to achieving some targets of SDGs and biodiversity conservation policies.
- Coordinated planning between agriculture (including horticulture), forestry, land development corporations, and housing authorities for reversing and arresting the adverse impacts of land use changes and erosion in biocultural diversity. Aligning local conservation practices with flagship national programs (e.g., Sustaining Himalayan Ecosystems) can specifically help achieve the SDGs targets of “Zero Hunger” and “Life on the Land”.

Reviewer comment

A detailed article. Main point is that ITK is still strongly present in remote and transitional communities but is being rapidly eroded in transitional communities with costs to biodiversity (because ITK has management strategies as well) and also to human health because a lot of convenience food is now being consumed.

71. Singh R.K., Zander, K.K., Kumar, S., Singh, A., Sheoran, P., Kumar, A., Hussain, S.M., Riba, T., Rallen, O., Lego, Y.J., Padung, E., Garnett, S.T. (2017) Perceptions of climate variability and livelihood adaptations relating to gender and wealth among the Adi community of the Eastern Indian Himalayas. *Applied Geography*, Volume 86, 2017, Pages 41-52, <https://doi.org/10.1016/j.apgeog.2017.06.018>

Research Topic

The perception of climate variability, and adaptation strategies, of women and men Adi farmers in Arunachal Pradesh (India), the Indian state identified as most vulnerable to climate change.

Methodology

Interviews and FGDs with 65 male and 71 female Adi farmers, combined with observation and transect walks. Three authors lived for extended periods in the villages which facilitated informal interactions. More detail on the sampling and data collection methods is on p.43, section 2.4.

The conceptual framework referred to Chambers (1989) livelihood adaptation approach and considered the idea that perceptions of male and female farmers may vary according to their differing decision making roles and respective assigned responsibility, and the relative ability to access internal and external resources in order to adapt to climate variability. (p.43, 2.3)

Comparison is made between individual adaptation strategies of poor and wealthy men and women farmers, before and after the year 2000, used as a benchmark year, because most of the initial changes such as introduction of crop varieties, establishment of research and technology agencies and development of new markets, took place after this year. (p.44, 2.4.2)

Relevant findings

“Compared to female respondents, male respondents were on average better educated, had more contact with extension agencies, were more likely to have had exposure to training in how to manage agriculture in the face of stressors, and had better access to distant markets. Women on the other hand, were more likely to be connected strongly to networks of indigenous institutions within which they could gain access to required resources and were more likely to depend on common property resources”. (p.44,3)

“Poorer farmers (both men and women) still rely on local plants, insects and animals to predict weather upon which they made decisions about agricultural activities”. (p.45, 3.3.1)

The diversity of adaptation strategies between wealthy and poor farmers, is much higher after the year 2000 than before, indicating that wealthy farmers are able to pursue livelihood strategies in response to climate change that are not available to poorer farmers. The differences in livelihood options for women and men also become more pronounced after the year 2000, though the greater difference is to do with wealth rather than gender. (p.48, 3.4)

"Making handicrafts, accessing forest-based resources (food and medicinal plants), fishing, hunting and rearing pigs and poultry (using a diverse mix of local and exotic breeds) were the key diversification adaptation strategies among poorer women (and poorer men to a lesser extent)." (p.48, 4.2)

"Wealthier farmers, on the other hand, have both diversified and intensified their production..." (p.49, 4.2)

"Poorer farmers, particularly women, are likely to become even more vulnerable because so far their adaptation strategies have been depending on rainfed systems and forest resources, both increasingly impacted by climate variability." (p.50, 4.3)

Conclusions

The perception of climate change is shaped by livelihood activities and differs according to gender and income level. Women are more sensitive to the changes because they perceive that their livelihoods are more impacted.

However, the support that is available for agricultural intensification and other adaptation measures, responds to the needs of wealthier male farmers. The paper calls for policy changes to tailor support to poorer women and men, rather than only to wealthier farmers. E.g., P.50 'the advice needs to be relevant to cropping in jhum on hillsides, not rich river valleys.'

It also acknowledges that the traditional knowledge on which poorer farmers depend is likely to be challenged as climate changes.

Reviewer comment

The research is methodologically and conceptually robust and very interesting, in particular its analysis of how the perceptions of climate variability change according to men's and women's roles and income levels. Its main conclusion is that income level is as significant as gender in determining sensitivity to climate change and available adaptation strategies.

72. Singh P, Tammy Tabe, Tess Martin (2022) The role of women in community resilience to climate change: A case study of an Indigenous Fijian community. Women's Studies International Forum, Volume 90, 2022, 102550, ISSN 0277-5395, <https://doi.org/10.1016/j.wsif.2021.102550>.

Research Topic

Climate change is projected to have wide-ranging effects on natural resources, environment, agriculture and fisheries. The various impacts of climate change pose significant risks to Pacific Island communities and people's livelihoods. A community consists of different groups of people and requires contributions from all members to build community resilience to the impacts of climate change. The Indigenous Fijian (iTaukei) women have skills, knowledge, and unique local and traditional experience that can increase the resilience of the community; however, their voice is not often heard in a patriarchal community. An understanding of gender roles in a community is essential for adaptation to climate change. Fijian communities exhibit elaborate social support system and traditional practices that demonstrate high level of dynamism and institutional support in the community's response to risks.

Methodology

Semi-structured interviews, focus groups, and participant observation.

Relevant findings

Economic resilience in the village = alternative income sources that provide for a diverse economy. This diversity was dependent on the ecosystem services in the surrounding natural environment. Given the village's vulnerability to flooding, the existing livelihood sources of fishing and farming were not considered adequate to recover from a possible huge financial loss following natural disasters. Women indicated their understanding of the need to manage ecosystems for the economic development of the village, which is largely dependent on natural resources.

Financial stability of a community is one of the key indicators of economic resilience. A majority stated that flooding is frequent and causes the highest economic loss in the form of damage to properties and assets in the community. Flooding also destroys agricultural crops = major source of food but also a source of income for the community. Men = inclement weather events such

as floods and cyclones reduce number of fishing days = loss of fishing equipment and damages to fishing boats and coastal infrastructure. These events increase expenditure and affect normal income sources. (p.7)

Women recognize importance of being financially stable to be able to support their family, - assist their husbands in their work or engage in income generating activities of their own – e.g. sugarcane crushing season, some women assisted their husbands in the field to harvest sugarcane. Women also collected and sold freshwater and marine animals e.g. shellfish, beche-de-mer, crabs, and prawns, and women have established a good market for seaweed. Economic resilience and location are closely related (closeness to resources they depend on eg fish etc) (p.8)

The women stressed the importance of mangrove fishing - great value to their income activities. Other marketing opportunities inc. sale of tiger prawns and freshwater clams (mussels). Women have minimal involvement in offshore and deep-sea fishing as this is considered to be men's domain. (p.7)

Young men seek wage employment while the village women contribute towards households' social and economic status by diversifying their income streams through activities such mud-flat gleaning, market gardening, crafting, grocery and bakery sales. Women are mainly involved in selling of freshwater and marine food products. Harper et al. (2013) state that in Fiji the role of women in fisheries is changing from subsistence to commercial, with surplus being sold at the local markets. This additional income is viewed as a form of insurance for periods of scarcity, which are likely to arise in the case of a natural disaster.

This shift from subsistence-oriented to market-oriented production has created a new dimension for women, who are able to save money for use during difficult times.

Loss and damages associated with natural hazards can be emotionally devastating particularly for women who are most often on the receiving end of a man's anger. Women emphasized the importance of being psychologically and financially stable to cope with the consequences of natural hazards.

Men think that women should manage household tasks, including caring for the children and the elderly. Women are well-aware of this fact and are often in a quandary over engaging in paid work or staying home. Nevertheless, women in Votua are contributing towards their village economy.

Loss of assets, destruction of crops, damages to the infrastructure, demands of the recovery and reconstruction, and disruption of basic utilities leads to psychological distress especially among the men, who are usually the household head. Such situations create an adverse social environment, where most often, it is the women who bear the brunt of men's frustration and anger. The majority of the women engaged in the individual interviews indicated that when women are emotionally disturbed, it becomes difficult to focus on disaster recovery.

The church is a very important source of support and community capital.

Kinship and local networks are described as key attributes of *social resilience* that contributed to community disaster preparedness, response and recovery, and adaptive measures. Social relationships are firmly based on mutual support and trust. Mutual assistance is the most prevalent aspect of community social capital, which in the context of this study implies social relations that facilitate collective action.

Evident during the period of the field study when a significant flood affected the area. The social support systems, such as the ones created by the women during this period of flooding, were observed to be strong factors contributing to community resilience.

Further examples of the strong social relationships were noted when women used their networks to share ideas on marketing strategies, and skills such as sewing, quilting, crafting, and gleaning of freshwater and coastal animals to diversify income sources. This study confirmed the findings from previous studies that social support structure is a key contributing factor to community resilience to climate change (refs). The concept of vanua with its key elements of natural resources, place, culture and people was identified as great source of social resilience. It helps to keep the villagers connected to each other. This was also reflected by Ryle (2016).

Conclusions

Building women's adaptive capacity relative to climate change will require working with men to change the societal norms and values associated with gender roles since men are the key decision makers. Although, changing the mind-set of a patriarchal community is a challenge, it can be achieved gradually by increasing the number of community outreach programs that require

equal contributions from both men and women. Traditional leaders also have significant influence over local governance policies and programs as do the religious community leaders such as church pastors who bring a powerful voice to issues of importance. Without their support, it would be difficult to change the mind-set of people who traditionally think that women should not be involved in decision-making processes. The study also highlighted that women have profound knowledge of the availability and distribution of aquatic resources, particularly freshwater, inshore and mangrove species. They understand the impacts of climate change on these resources, as well as the need for their conservation for a sustainable future. Thus, upcoming research on women's local and traditional knowledge is warranted as it has the potential to provide information that is relevant for informing decision making on natural resource management.

Reviewer comment

This article focuses on community resilience– the main conclusion is that The Indigenous Fijian (iTaukei) women have skills, knowledge, and unique local and traditional experience that can increase the resilience of the community, however, their voice is not often heard in a patriarchal community. This article, as several others, highlights the importance of social capital and social resilience to women' economic (and general) resilience. Church is important as are other networks. Women's deep knowledge is not captured in village decision-making bodies because women aren't supposed to speak.

73. Sonko, E., Florkowski, W.J., Agodzo, S., Antwi-Agyei, P. (2020), Subsistence farmer knowledge of strategies alleviating food insecurity in the context of climate change in the lower river region of the Gambia. Food Security: The Science, Sociology and Economics of Food Production and Access to Food, Springer; The International Society for Plant Pathology, vol. 12(3), pages 607-624, June. 10.1007/s12571-020-01024-z

Research Topic

This paper examines subsistence farmer knowledge of strategies to cope with food insecurity and progressive climate change in the Lower River Region of the Gambia.

Methodology

Survey data collected in 18 villages in the region. The villages were randomly selected from six districts. In each village, 10 farming households were selected from lists of rice and maize farmers provided by extension agents and village heads. The region is sparsely populated and identified as particularly vulnerable to climate change impacts.

The questionnaire assessed the awareness of rural households to respond to food insecurity by choosing a strategy given weather extremes and gradual climate change. More detail on sampling and data collection is on (p.612, 3.2.1.)

Relevant findings

The research looked at respondents' knowledge of 11 strategies, of which nine are directly related to farm practices, and two to insufficient food crop production (temporary and long-term migration/sending remittances to the farm). The paper asserts that knowledge of a strategy is a pre-requisite for implementation, though it does not look at constraints to implementation beyond awareness. It looks at a range of factors which influence farmer strategy knowledge, including formal education, secondary household activities, gender, government support, farm size and income.

It is interesting that with the Gambia having a young population (average age estimated at 43.2 in 2013), the average age of respondents was 34 years old.

Women appear to have a 11.5% higher probability of knowing about crop diversification as a strategy, compared with men. 'It is likely associated with women commonly cultivating a wider variety of crops, including vegetables, many of which mature within a period of time that is shorter than in the case of major grains. (p.616, 4.2.9). Female respondents also are 24% more likely than men to know about integrated pest management though the reasons for this are not given. (p.613, 4.2.1)

Higher education levels were shown to have a correlation with permanent migration as a strategy, but not temporary. (p.616. 4.2.10 and 4.2.11).

Income influenced respondent's knowledge of crop rotation, alley farming, contour farming and IPM, with farmers with higher incomes leading to greater awareness of these strategies. Farmers who practice agroforestry were more aware of crop diversification, changing the planting date, contour farming and zero tillage. (p.620)

Conclusions

It is important for decision-makers to look at the experiences of remote and poor communities when addressing food insecurity. (Introduction)

The paper draws general conclusions beyond this about the importance of increasing access to formal education for women and men. (p.608)

Reviewer comment

The analysis on gender in this paper is not conclusive. The points summarised in the discussion are somewhat overstated and, in some places, contradict what is stated in the body of the paper. E.g., the assertion that being female increases the probability of knowing about contour farming (p.620, 5) is at odds with the conclusion on p.614, 4.2.6 that men are more likely to know about this strategy. The awareness of women of long-term migration/remittances is also only 1% greater than men's though is emphasised in the discussion on p.620.

Overall, the paper lacks an integrated analysis, which would link and relate different elements of an intersectional lens, and/or explore of the reasons behind the differences in awareness of adaptation strategies, between farmers of different income levels, genders, farm sizes etc. It also does not look beyond awareness of strategies to examine other opportunities/barriers.

74. Tran, Van Thanh, An-Vo, Duc-Anh, Mushtaq, Shahbaz and Cockfield, Geoff. (2022). "Nuanced assessment of livelihood resilience through the intersectional lens of gender and ethnicity: evidence from small-scale farming communities in the upland regions of Vietnam." *Journal of Rural Studies*. 92, pp. 68-78. <https://doi.org/10.1016/j.jrurstud.2022.03.011>

Research Topic

The intersection of gender and ethnicity in relation to individual and household livelihood resilience, in the context of ethnic minorities in the Northwest Mountainous regions of Vietnam.

Methodology

The study used the Household Livelihood Resilience approach (HLRA) which assumes that livelihood resilience is built upon five capital assets: human, financial, physical, social and natural. It gathered survey data on 'subjective' resilience, mapping responses onto these five assets, which it categorised as a bottom up rather than top-down approach. (Lists of household indicators for each capital asset are in Table 1, 3.2.) 240 surveys with household heads were carried out through face-to-face interviews, in the Phu Yen district, one of the five most impoverished districts in the NW mountainous region, where four primary ethnic groups are present: Thai and Muong at low altitude and Dao and Hmong at middle and high elevations.

Relevant findings

Women participants' responses indicated that their overall household level of resilience is lower than men's, largely driven by differences in human and natural capitals. In particular for human, capital, male respondents were on average more educated, and women reported that their household ability to practice livelihoods were more affected by personal health problems (77% of women respondents compared with 46% men). (4.1)

Regarding natural capital, men reported a greater crop diversity on their farms and a higher level of farm equipment ownership than women. {...} In addition, Vietnamese men often have better connections with local governments. These connections could bring more opportunities for men to interact with government officials in the diversification process of crops to cope with a changing climate.

However, ethnicity was found overall to cause more significant variations across all five capitals. The study concludes that ethnicity has a relatively more substantial role than gender in governing the five livelihood categories in the studied groups. (4.3)

The study looks at particularities of gender as a factor within different ethnic groups and finds variations – for example, Muong women score much more highly than Muong men on social capital, while men in other ethnic groups score higher than women in the same group. Hmong women had particularly low levels of financial, physical and natural capitals.

Conclusions

The authors mention some limitations of their approach – one is that it is a snapshot of a moment in time and is not a dynamic measure of how livelihood resilience is changing. Also, the indicators capture the ‘what’ but not the ‘why’ of each households’ capital assets. The authors also recognise that they have not given resilience indicators differentiated weighting, and that the indicators do not capture all essential variables, omitting for example indigenous knowledge. (5.3)

...this paper illustrates the importance of analysing perceived livelihood resilience by integrating gender with other factors (e.g., ethnicity) supporting the idea that livelihood resilience is tightly linked to individual knowledge, culture, attitudes and perceptions. The paper calls for targeted policy interventions for the most disadvantaged groups. ...it is critical to consider women and ethnic minority groups to increase their household livelihood resilience and overall national poverty alleviation. (6)

75. Yanda, P.Z., Mabhuye, E.B. & Mwajombe, A. (2021) Linking Coastal and Marine Resources Endowments and Climate Change Resilience of Tanzania Coastal Communities. *Environmental Management* (2021). <https://doi.org/10.1007/s00267-021-01553-z>

Research Topic

The inter-linkages between natural resources endowments and climate change resilience of communities along the coastal region of Tanzania. The study examined the role of coastal and marine resources in climate change adaptation of coastal communities and how the interrelationships are influenced by gender, education, bylaws and membership in social networks.

Methodology

The study was conducted in the Rufiji District in the coastal area of Tanzania, in two villages from two wards selected in consultation with district authorities. The research used FGDs, KIs and a household questionnaire to gather quantitative information. FGDs were held with community members with long personal experiences with the coastal areas and those who were knowledgeable about the socio aspects of coastal communities. In addition, 77 respondents were interviewed through the household questionnaire.

Relevant findings

Conceptual framework: “literature shows that non-climate factors such as socio-cultural, poverty vs. wellbeing, location, age, gender, education, political etc, have the potential to aggravate or relieve a livelihood from climate stressors.” The paper outlines the ways that the coastal resources support local communities, and the ways in which climate change is impacting these resources. It looks at the degree to which the communities depend on coastal resources for their livelihoods. The analysis of gender within this is very brief. Two main points are made:

Adaptation to Coastal Communities, Gender: Historically, in all fishing communities, men are predominantly doing fishing activities, so they have greater access to the benefits from this. Women are mainly engaged with mat weaving, subsistence agriculture and food vending. “In this study, male headed household have a relatively high level of resilience to the impact of climate change, including famine, hunger and flooding, which are regularly reported to have been occurring in both Mchungu and Kivinja A villages.” Gender norms are changing in some areas – e.g. it was noted that men as well as women are now engaged with fetching water.

Social networks: more women than men engage with social networks, which can increase resilience, by “providing material resources such as loans when a member is considered to be in a hard time, and humanitarian needs such as providing food to disabled families.”

Conclusions

Coastal resources are key to livelihoods in the study areas. Climate change is impacting these resources. E.g., fish stocks can be reduced, and indirectly, agriculture can be affected – e.g. excessive flooding can cause lack of crops. The paper does not examine

the gender dimensions of these beyond concluding that male-headed households have a greater diversity of income sources and so are more resilient.

Reviewer comment

The gender dimensions of the analysis are limited. It is of interest in that it looks at coastal and marine resources and details how these are being affected and how this in turn affects communities and livelihoods.

76. Wei Zhang, Marlène Elias, Ruth Meinzen-Dick, Kimberly Swallow, Carlos Calvo-Hernandez & Ephraim Nkonya (2021): Soil health and gender: why and how to identify the linkages, International Journal of Agricultural Sustainability, DOI: 10.1080/14735903.2021.190657

Research Topic

Many technical options for soil improvement or restoration sit 'on the shelf', there are many sociocultural, institutional, economic, and policy barriers hindering their adoption at large scale. Gender inequality is deeply embedded in and reinforces these barriers and represents a wicked problem requiring context - and culture-specific understandings and approaches.

Methodology

Literature review

Relevant findings

Literature review

Addressing soil erosion and build-up of SOM require longer-term investments such as terracing, bunding, planting pits, fallowing, and tree planting. Gender analysis of differences in knowledge and perceptions about these practices, and preferences (especially time preferences) can help to identify reasons for gender differences in adoption.

Are extension services providing women with adequate information? Are the tree species being promoted primarily for timber (which men tend to control), or consistent with women's preferences, such as for fruit trees?

Time constraints.

Examination of gender differences in capitals and assets = constraints to adoption. Terracing, bunding and check dams are often labour-intensive and may require equipment or cash outlays, both for initial investment and recurring maintenance. Where women have limited control over the necessary labour, physical, or financial capital, they may be unable to adopt these practices on their plots.

Political capital can contribute: in Ghana, Goldstein and Udry (2008) show that women and men with lower political standing were less likely to leave land fallow, which can restore soil health, but also risks having the land appropriated by others.

Intra-household distribution of rights affects management practices. Women with use rights to the land may not have sufficient management rights to plant trees, and if they do not have the fructis rights they would not share in the benefits.

Understanding these relationships can inform programming. In Kenya, World Agroforestry (2020) used the Integrating Gender and Nutrition within Agricultural Extension Services toolkit when designing their initiative, which invites both men and women to training, and encourages 'try out' of planting pits and different tree species so that women and men can see how much labour is required and what benefits result in the short and long run (addressing trade-offs). Where women lack land rights that would enable them to plant trees for soil enhancement, they have also promoted biomass transfer practices, adding nutrient-rich *Tithonia diversifolia* leaves harvested from farm boundaries or nearby areas (Place et al., 2007).

Gender-equitable land reforms increase adoption, investment, as well as gender equity. Ali et al. (2014) find that land registration in Rwanda increased investment in bunds, terraces, and check dams, particularly among female headed households. Similarly, in Ethiopia, the first stage land certification increased investment in terracing and bunding, and knowledge of land rights

increased tree planting and soil conservation, particularly for female headed households (Deininger et al., 2008a; Quisumbing & Kumar, 2014).that secure women's land rights offer a transformational approach.

Conclusions

This study reviews relevant gender literature and proposes a conceptual framework to help illuminate important gender considerations for soil health management. The framework highlights how a range of separate and joint assets held by men and women in households, and the intrahousehold distribution of use, management, fructus, and alienation rights shape the management practices that contribute to, or undermine, soil health. These considerations are essential for identifying gender-based constraints, opportunities, and unintended consequences in promoting soil management technologies.

Reviewer comment

This study flags up women's differential abilities, due to a wide range of factors, to invest in soil health. This has inevitable implications, not discussed here, for women's economic resilience in food systems.

77. Wrigley-Asante, C., Owusu, K., Egyir, I.S., Mboya Owiyo, T., (2017) Gender dimensions of climate change adaptation practices: the experiences of smallholder crop farmers in the transition zone of Ghana. *Geographical Review*, 38:2, 126-139, DOI: 10.1080/19376812.2017.1340168

Research Topic

The gender dimensions of climate change adaptation strategies among smallholder crop farmers in the Forest-Savannah Transition Zone of mid-Ghana, an ecotone between the tropical forests to the southwest and the semi-arid areas of the north. Specifically, the paper aims to

1. Examine the gendered perceptions of climate change and its effect at the household level
2. Assess the different adaptation strategies of men and women and their implications for household food security
3. Discuss how adaptation strategies influence the decision-making role of women in particular, at the household level

Methodology

Interviews using semi-structured questionnaires with 612 farmers (328 women and 284 men). In addition, information and household and institutional levels was gathered using desk reviews of literature, individual surveys and 10 FGDs, with an average of 14 people per group (two mixed, four with men and four with women only.) Respondents were all 40 years+ so as to allow them to compare weather changes in the last five years with the last two decades.

This paper is part of a broader study in two administrative municipalities, Ejura-Sekyedumase in the Ashanti region, and Wenchii Municipality in the Brong-Ahafo region. These districts were purposively selected by a USAID funded project to implement the climate action for agricultural development program. Three communities were then purposively selected from each municipality.

Relevant findings

Women and men shared similar perceptions of what constitutes climate change. This may be because the men and women interviewed were all farm operators and decision makers. (p.131)

Men were slightly more likely than women to report that their production had increased over the last five years. In the Ashanti region, women were also more likely than men to report food shortages during the period April-July/August. In both communities, more women (65.2%) than men (59.4%) reported experiencing decreases in trading in agricultural goods. (p.132)

More men (83.3%) than women (69.1%) used chemical fertilizer as an adaptation strategy. The authors attribute this to men's ability to access agricultural inputs, due to their control over financial resources and access to credit. (p.133)

There were differences in the crops planted by women and men as part of the intensification of mixed cropping. Women across all communities focused on traditional crops, that can support household food security, such as pepper, garden eggs, groundnut, beans, yam and cassava. In one village, the authors report that women preferred the traditional crops as they were not so sure of

the benefits of new crops such as watermelon. This reinforces the view that women were not likely to take risks as household food security is a priority to them (p.134)

Another example of gender difference in adaptation is given on P.134/5, where men in Awisa (Brong-Ahafo) had diversified into cashew production because it is drought resistant, while women lacked the land to do this and were engaged in short-term crops such as garden eggs and okro, as well as seeking work as factory hands in a cashew processing factory.

Off farm income options for women were described as petty trading in household durables and consumables. Men, particularly young men, emigrated to Libya in search of work. The study indicates that women earning income off-farm is one factor contributing to their increased participation in household decision-making. (p.135-136) Women's engagement in petty trading has positively influenced their socioeconomic status and in turn influenced their decision-making role at the household level. However, decisions about land, the primary asset, are still the preserve of men.

Conclusions

Male and female farmers knew and could describe the nature of changing environmental conditions. Farmers have adopted both on and off farm strategies. Men have more access to some of the on-farm strategies such as use of chemical fertilizer and drought-resistant crops such as cashew (because of their access to land to grow them on); women focused more on traditional food crops to ensure household food security. Women resorted to petty trading in agricultural and consumable goods. The paper calls for institutions that support climate change adaption initiatives at the local level take gender differences into consideration and provide support to men who are affected by crop losses, as well as support to women's empowerment, and support on entrepreneurship and microcredit for both women and men.

Reviewer comment

The findings in the paper are not new and are quite top line - possibly because it focuses on gender only and no other categories of social difference. The distinction between the current situation and before the last five years is also not very clearly drawn - which may be because the response of women and men to climate change has been an intensification of their existing activities, with women focusing on food crop production and trade, and men on cash crop production. The main constraints that the paper point to which affect women's economic resilience in the face of climate change, are to do with control over land as a primary asset, control over income and access to credit. The paper also mentions the impact of the out-migration of men but does not explore this in detail.

Anne Rietveld, Scientist, a.rietveld@cgiar.org

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INITIATIVE ON
Gender Equality

Anne Rietveld, Scientist, a.rietveld@cgiar.org

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INITIATIVE ON
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